

AMPK in skeletal muscle function and metabolism

Rasmus Kjøbsted,^{*,1} Janne R. Hingst,^{*} Joachim Fentz,^{*} Marc Foretz,^{†,‡,§} Maria-Nieves Sanz,^{¶,||}
Christian Pehmøller,[#] Michael Shum,^{**,††} André Marette,^{**,††} Remi Mounier,^{‡‡} Jonas T. Treebak,^{§§}
Jørgen F. P. Wojtaszewski,^{*} Benoit Viollet,^{†,‡,§} and Louise Lantier^{¶,||,|||,2}

^{*}Section of Molecular Physiology, Department of Nutrition, Exercise, and Sports, Faculty of Science, and ^{§§}Section of Integrative Physiology, Novo Nordisk Foundation Center for Basic Metabolic Research, Faculty of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark; [†]INSERM, Unité 1016, Institut Cochin, Paris, France; [‡]Centre National de la Recherche Scientifique (CNRS), Unité Mixte de Recherche (UMR) 8104, Paris, France; [§]Université Paris Descartes, Sorbonne Paris Cité, Paris, France; [¶]Department of Cardiovascular Surgery, Inselspital, Bern University Hospital, and ^{||}Department of Biomedical Research, University of Bern, Bern, Switzerland; [#]Internal Medicine Research Unit, Pfizer Global Research and Development, Cambridge, Massachusetts, USA; ^{**}Axe Cardiologie, Quebec Heart and Lung Research Institute, and ^{††}Institute for Nutrition and Functional Foods, Laval University, Québec, Canada; ^{‡‡}Institute NeuroMyoGène, Université Claude Bernard Lyon 1, INSERM Unité 1217, CNRS UMR, Villeurbanne, France; and ^{¶¶}Department of Molecular Physiology and Biophysics and ^{|||}Mouse Metabolic Phenotyping Center, Vanderbilt University, Nashville, Tennessee, USA

ABSTRACT: Skeletal muscle possesses a remarkable ability to adapt to various physiologic conditions. AMPK is a sensor of intracellular energy status that maintains energy stores by fine-tuning anabolic and catabolic pathways. AMPK's role as an energy sensor is particularly critical in tissues displaying highly changeable energy turnover. Due to the drastic changes in energy demand that occur between the resting and exercising state, skeletal muscle is one such tissue. Here, we review the complex regulation of AMPK in skeletal muscle and its consequences on metabolism (e.g., substrate uptake, oxidation, and storage as well as mitochondrial function of skeletal muscle fibers). We focus on the role of AMPK in skeletal muscle during exercise and in exercise recovery. We also address adaptations to exercise training, including skeletal muscle plasticity, highlighting novel concepts and future perspectives that need to be investigated. Furthermore, we discuss the possible role of AMPK as a therapeutic target as well as different AMPK activators and their potential for future drug development.—Kjøbsted, R., Hingst, J. R., Fentz, J., Foretz, M., Sanz, M.-N., Pehmøller, C., Shum, M., Marette, A., Mounier, R., Treebak, J. T., Wojtaszewski, J. F. P., Viollet, B., Lantier, L. AMPK in skeletal muscle function and metabolism. *FASEB J.* 32, 1741–1777 (2018). www.fasebj.org

KEY WORDS: exercise • mitochondria • glucose uptake • plasticity • diabetes

One fundamental function of skeletal muscle is to generate mechanical force to support body posture and to facilitate a wide variety of movements. Besides this role in body motility, skeletal muscle has been shown to be important for regulating whole-body metabolism. Skeletal muscle demonstrates high malleability and can adapt

its contractile composition and metabolic properties in response to a number of physiologic conditions, including exercise. Such adaptations are reflected by changes in muscle size, fiber type distribution, contractile velocity, force production, and endurance capacity, being the result of the functional demands of the contractile activity (1, 2).

ABBREVIATIONS: 4E-BP1, 4E binding protein 1; ACC2, acetyl-CoA carboxylase 2; AICAR, 5-amino-1-β-D-ribofuranosyl-imidazole-4-carboxamide; BDNF, brain-derived neurotrophic factor; C2, 5-(5-hydroxyl-isoxazol-3-yl)-furan-2-phosphonic acid; CaMK, Ca²⁺/calmodulin-dependent protein kinase; CaMKKβ, Ca²⁺/calmodulin-dependent protein kinase kinase β; CD36, cluster of differentiation 36; DMD, Duchenne muscular dystrophy; ERR, estrogen-related receptor; FOXO3a, Forkhead box protein O3a; GBD, glycogen-binding domain; β-GPA, β-guanidinopropionic acid; GS, glycogen synthase; HDAC5, class II histone deacetylase 5; HFD, high-fat diet; KD, kinase domain; KO, knockout; LIF, leukemia inhibitory factor; LKB1, liver kinase B1; MAFbx, muscle atrophy F-box; mdKO, muscle-specific double knockout; MEF2, myocyte enhancer factor 2; mTORC1, mammalian target of rapamycin complex 1; MuRF1, muscle RING finger 1; MuSC, muscle stem cell; NAM, nicotinamide; NAMPT, nicotinamide phosphoribosyltransferase; NR4A, nuclear hormone receptor 4A; PDH, pyruvate dehydrogenase; PGC-1α, peroxisome proliferator-activated receptor γ coactivator 1α; PKD, protein kinase D; PLIN2, perilipin 2; ROS, reactive oxygen species; SIRT, NAD-dependent sirtuin; SNARK, sucrose nonfermenting AMPK-related kinase; TSC, tuberous sclerosis complex; ULK1, uncoordinated 51-like kinase 1; WADA, World Anti-Doping Agency; WT, wild type; ZMP, 5-aminoimidazole-4-carboxamide ribonucleotide

¹ Correspondence: Department of Nutrition, Exercise and Sports, Section of Molecular Physiology, Faculty of Science, University of Copenhagen, Universitetsparken 13, DK-2100 Copenhagen, Denmark. E-mail: rasmus.kjobsted@nexs.ku.dk

² Correspondence: Department of Molecular Physiology and Biophysics, Vanderbilt University, 823 Light Hall, 2215 Garland Ave., Nashville, TN 37232, USA. E-mail: louise.lantier@vanderbilt.edu

This is an Open Access article distributed under the terms of the Creative Commons Attribution 4.0 International (CC BY 4.0) (<http://creativecommons.org/licenses/by/4.0/>) which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

doi: 10.1096/fj.201700442R

This plasticity may involve short- and long-term mechanisms, leading to changes in protein abundance and activity (1–3). These changes are mediated by activation and repression of specific intracellular signaling events that govern effectors involved in metabolic pathways and transcription/translation processes of exercise-responsive genes (4). The intracellular signaling mechanisms that modify skeletal muscle function in response to exercise are regulated by perturbations in muscle cell homeostasis, including alterations in tissue perfusion, oxygen tension, redox state, calcium (Ca^{2+}) dynamics, and ATP turnover (5). Evidence suggests that ATP turnover in skeletal muscle may increase by >100-fold in response to exercise (6). Keeping cellular ATP concentrations fairly constant during such conditions represents a major challenge to the cell and highlights the vast dynamics of muscle energy metabolism. Because skeletal muscle ATP consumption increases during exercise, intracellular AMP concentrations may accumulate as a result of the adenylate kinase reaction. This increases cellular AMP/ATP and ADP/ATP ratios, leading to activation of AMPK (7). This kinase is considered a central sensor of intracellular energy status and maintains energy stores by regulating anabolic and catabolic pathways, thereby ensuring a balance between energy supply and demand (8). In skeletal muscle, acute pharmacological activation of AMPK has been shown to promote glucose transport and fatty acid oxidation (9) while suppressing glycogen synthase activity and protein synthesis (10, 11). In addition, chronic activation of AMPK reduces markers of skeletal muscle fragility (12) and enhances muscle fiber oxidative capacity by stimulating mitochondrial biogenesis (13–15). These events are initiated by AMPK downstream phosphorylation of key metabolic enzymes as well as transcription factors that modulate cellular metabolism in order to handle both current and future metabolic challenges. Several excellent reviews have examined the role of AMPK in regulating skeletal muscle function and metabolism (16–49). Therefore, this review addresses novel concepts and future perspectives of AMPK in skeletal muscle that need to be experimentally validated and tested.

AMPK STRUCTURE AND EXPRESSION

AMPK is a heterotrimeric protein complex that consists of a catalytic subunit (α) and 2 regulatory subunits (β and γ), of which several isoforms have been found ($\alpha 1$, $\alpha 2$, $\beta 1$, $\beta 2$, $\gamma 1$, $\gamma 2$, and $\gamma 3$) (50) (Fig. 1). The α subunit contains the kinase domain, activity of which is highly dependent on the reversible phosphorylation of α -Thr172 (51–53). The β subunit acts as a scaffold for binding the α and γ subunits (54) and contains a glycogen-binding domain (GBD) that likely targets the heterotrimeric complex to glycogen particles (55, 56). The γ subunit functions as a sensor of intracellular energy status through its direct binding of adenosine nucleotides (57). Besides these well-established functions, all 3 subunits contain different domains or posttranslational modifications that may locate AMPK to distinct subcellular compartments. In the α subunit, a nuclear export sequence has been found (58), and myristoylation of the β subunit has been suggested to facilitate

AMPK translocation to perinuclear speckles and mitochondrial membranes (59–61). The γ subunit isoforms differ in their N-terminal extensions, which also appear to determine AMPK localization (62–64). Thus, in skeletal muscle fibers, the $\gamma 1$ isoform is localized to the z disk, whereas the $\gamma 3$ isoform is found in the nucleus and along the z disk and I-band in a pattern that closely resembles the T-tubule/sarcoplasmic reticulum structures (64). The functional role of AMPK in different subcellular locations has not received much attention, but recent findings indicate that it may represent another way of regulating AMPK activity. For example, the unspecific AMPK activator PT-1 appears to only activate $\gamma 1$ complexes in mouse skeletal muscle, whereas it displays no isoform selectivity in HEK293 cells stably expressing each of the 3 γ isoforms (65). Because PT-1 activates AMPK by inhibiting the mitochondrial respiratory chain (65), this may signify an important role of the $\gamma 1$ -associated complexes in monitoring ATP synthesis, and the highly contraction-responsive $\gamma 3$ -containing complex may serve as the major sensor of ATP consumption in skeletal muscle. On the other hand, the PT-1 concentration previously used to stimulate isolated skeletal muscle (65) may have been insufficient to activate $\gamma 3$ -containing complexes. Recent evidence from cell-based studies also indicates that AMPK subunit composition influences sensitivity to AMP, which likely contributes to the specialized functions of AMPK heterotrimeric subtypes (57, 66–68).

The 7 different AMPK subunit isoforms give rise to 12 heterotrimeric combinations that seem to be expressed in a tissue-specific manner. Thus, in skeletal muscle preparations from human and mouse, all subunit isoforms have been detected, but only a subset of possible heterotrimeric complexes seems to exist (69, 70). In human skeletal muscle (vastus lateralis), 3 different complexes have been described ($\alpha 2\beta 2\gamma 1$, $\alpha 2\beta 2\gamma 3$, and $\alpha 1\beta 2\gamma 1$) (69), whereas 5 complexes have been identified in mouse skeletal muscle ($\alpha 2\beta 2\gamma 1$, $\alpha 2\beta 2\gamma 3$, $\alpha 2\beta 1\gamma 1$, $\alpha 1\beta 2\gamma 1$, and $\alpha 1\beta 1\gamma 1$) (70) (Table 1). In addition, some AMPK subunits are expressed in a fiber type-dependent manner (71), which may explain the relative distribution of complexes between different muscles (70). Currently, it is not known why mouse skeletal muscle appears to express two additional complexes ($\beta 1$ -associated) compared with human skeletal muscle. Based on findings in muscle-specific AMPK α and AMPK β double-knockout (KO) mice (AMPK α mdKO and AMPK $\beta 1\beta 2$ M-KO mice, respectively) (72–74), the majority of $\beta 1$ -associated complexes detected in a crude sample of skeletal muscle seem to derive from nonmuscle tissue (e.g., connective tissue, neuronal cells, adipocytes, endothelial cells, etc.). This is in line with the notion that the $\alpha 1\beta 1\gamma 1$ complex is the most ubiquitously expressed complex of AMPK (68, 75). Interestingly, the $\beta 1$ subunit is also detected in sample preparations of human skeletal muscle, but, in light of findings from coimmunoprecipitations of AMPK $\alpha 2$, $\alpha 1$, $\gamma 1$, and $\gamma 3$, it does not seem to engage in stable complex formation or contribute to any measurable AMPK activity (69, 76). Although it is generally thought that all 3 AMPK subunits must be present to form a stable complex, it has been demonstrated in cell models that stable $\beta 1\gamma 1$ complexes can form in the absence of catalytic subunits (77). Whether formation of $\beta\gamma$ heterodimer

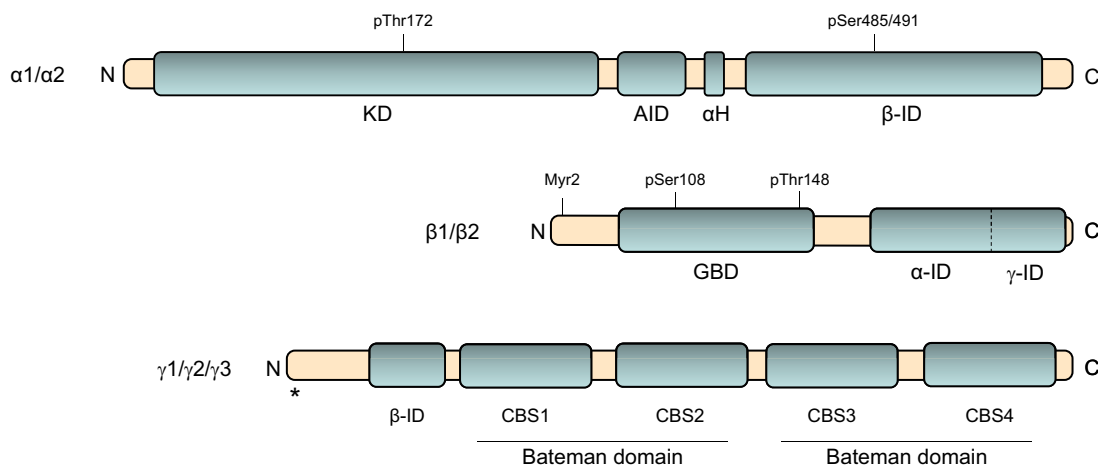


Figure 1. Structure of mammalian AMPK subunits. AMPK is a heterotrimeric protein consisting of 1 catalytic subunit (α subunit) and 2 regulatory subunits (β and γ subunit). The α subunit contains a kinase domain (KD), the activity of which relies on the phosphorylation of Thr172 by upstream AMPK kinases. The KD is followed by an autoinhibitory domain (AID) and an α -hook (α H), which seem to be important for AMP-regulated catalytic activity. At the α -C terminus, a β -interacting domain (β -ID) has been detected that binds to the C-terminal domain of the β subunit. Phosphorylation of Ser485/491 in the β -ID during insulin stimulation has been suggested to regulate kinase activity. The β subunit is subjected to myristoylation at the N terminus, which enhances phosphorylation of α -Thr172 by AMP/ADP and facilitates AMPK translocation to specific intracellular compartments. At the center, the β subunit contains a GBD that causes AMPK to bind to glycogen particles. Within the GBD, 2 phosphorylation sites have been found (Ser108 and Thr148) that seem to regulate binding capacity to glycogen particles as well as kinase activity. An α - and γ -interaction domain (α -ID, γ -ID) is located at the β -C terminus that acts as a scaffold keeping the heterotrimeric complex together. The γ -subunit contains 4 cystathionine- β -synthase (CBS) domains. These occur in tandem pairs, also known as Bateman domains, and are involved in adenosine nucleotide binding. A β -ID is located close to the γ -N terminus. An asterisk denotes that the 3 γ isoforms contain different N-terminal extensions.

complexes also occurs in mature skeletal muscle may be derived from observations in two muscle-specific AMPK double-KO mouse models. Thus, in the AMPK β 1 β 2M-KO model it seems evident that expression of β 2 protein is restricted to the myocytes (74). Interestingly, significant amounts of β 2 protein have been detected in a skeletal muscle sample preparation from the AMPK α mdKO mouse model (73), which may suggest the formation of stable $\beta\gamma$ complexes in mature skeletal muscle, assuming that single unbound subunits of AMPK are targeted for degradation. This may also be inferred from the AMPK β 1 β 2M-KO mouse model, which does not express α 2 protein in skeletal muscle (74), indicating that the β subunit is vital for maintaining AMPK α muscle protein expression. Collectively, these observations may suggest that the AMPK heterodimer ($\beta\gamma$) exists in skeletal muscle tissue and raises the possibility of a regulatory mechanism facilitating the association of catalytic subunits

with regulatory complexes. Alternatively, and somewhat speculatively, other proteins may bind to the regulatory heterodimer complex to regulate their activity or cellular localization. In this context, 12 protein kinases related to AMPK α 1 and AMPK α 2 have been detected in the human kinome (78). These are known as AMPK-related kinases, and, with a single exception, these are activated by upstream kinase liver kinase B1 (LKB1) (79). Although green fluorescent protein-transporter associated with antigen processing-tagged versions of these kinases do not appear to bind AMPK β and γ subunits (80), the sucrose nonfermenting AMPK-related kinase (SNARK/NUAK2) is activated in skeletal muscle by 5-amino-1- β -D-ribofuranosyl-imidazole-4-carboxamide (AICAR), contraction, and exercise (81, 82), indicating that SNARK activity is regulated similarly to AMPK. Is it possible that AMPK $\beta\gamma$ subunits form heterotrimeric complexes with SNARK, facilitating its regulation by adenine

TABLE 1. Relative distribution and basal activity of AMPK heterotrimeric complexes detected in human and mouse skeletal muscle

Trimer complex	<i>Homo sapiens</i> vastus lateralis		<i>Mus musculus</i>			
			Extensor digitorum longus		Soleus	
	Relative expression	Relative basal activity	Relative expression	Relative basal activity	Relative expression	Relative basal activity
α 2 β 2 γ 1	~65%	~30%	~70%	~50%	~60%	~35%
α 2 β 2 γ 3	~20%	<5%	~20%	~20%	<2%	<2%
α 1 β 2 γ 1	~15%	~65%	<5%	~25%	~20%	~50%
α 2 β 1 γ 1	N.D.	N.D.	<3%	<5%	~10%	<15%
α 1 β 1 γ 1	N.D.	N.D.	<2%		<8%	

The composition of AMPK heterotrimeric complexes was estimated from immunoprecipitation experiments in extensor digitorum longus and soleus from C57BL/6 mice as well as human male vastus lateralis. Values adapted from references 70, 76, and 487. N.D., nondetectable.

nucleotides? If so, it could be anticipated that AMPK β -deficient skeletal muscle exhibits a phenotype different from that of AMPK α -deficient skeletal muscle. Indeed, in skeletal muscle several phenotypic differences have been observed between AMPK α mdKO and AMPK β 1 β 2M-KO mice, including muscle mass, ATP levels, mitochondrial DNA and structure, citrate synthase activity, and peroxisome proliferator-activated receptor γ coactivator 1 α (PGC-1 α) mRNA levels (72, 74). In light of these observations, it has recently been reported that SNARK may be involved in the maintenance of muscle mass with age (83). Assuming that the potential binding of SNARK to AMPK β subunits induces an increase in SNARK activity and/or protects SNARK from degradation and that the association between SNARK and AMPK β is enhanced in AMPK α -deprived muscle, this could explain the increase in muscle mass observed in skeletal muscle deprived of AMPK α subunits (72).

REGULATION AND ACTIVATION OF AMPK IN SKELETAL MUSCLE

During skeletal muscle contraction, the adenylate energy charge in muscle is decreased depending on the duration and intensity of exercise (84, 85). As a result, the intracellular AMP/ATP and ADP/ATP ratios increase, which leads to activation of AMPK (7). AMPK activation occurs in two steps: stimulatory allosteric binding of AMP within the γ subunit and covalent activation through reversible phosphorylation on Thr172 in the catalytic α subunit (Fig. 2). AMPK activity is stimulated by AMP and ADP and inhibited by ATP binding to the two regulatory Bateman domains of the γ subunit. This competitive binding means that increases in cellular AMP/ATP and ADP/ATP ratios stimulate AMPK allosterically (86–90). The allosteric stimulation has a moderate effect on AMPK activity (<10-fold) (91). More importantly, binding of

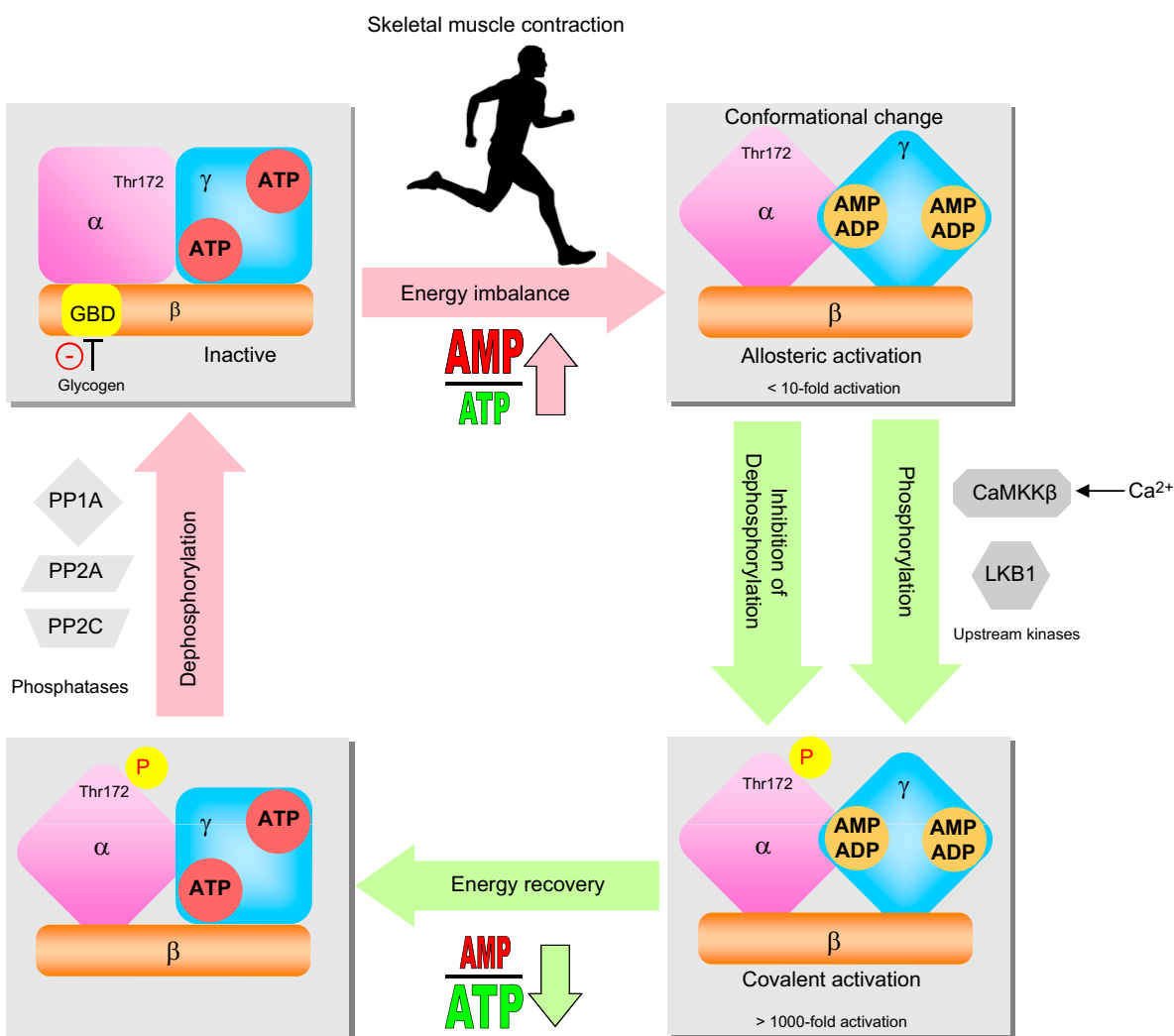


Figure 2. Regulation of AMPK in skeletal muscle during contractile activity. Exercise induces an energy imbalance in muscle, which leads to a rise in intracellular AMP and ADP concentrations. Binding of ADP and AMP at the Bateman domains of the γ subunit causes a conformational change that activates AMPK by up to 10-fold *via* an allosteric mechanism. This conformational change also triggers Thr172 phosphorylation of the α catalytic subunit by the upstream LKB1 and protects against dephosphorylation by protein phosphatases, increasing the activity 100-fold. Together the allosteric effect and α -Thr172 phosphorylation lead to a >1000-fold activation. AMPK is also activated by a rise in the intracellular Ca^{2+} concentration through α -Thr172 phosphorylation catalyzed by CaMKK β . After exercise and energy repletion, AMPK is converted back to an inactive form by dephosphorylation catalyzed by protein phosphatases (PP1A, PP2A, and PP2C) and undergoes inhibition by glycogen *via* binding to the GBD of the β subunit.

AMP and/or ADP to the γ subunit induces conformational changes that promote phosphorylation of α -Thr172 (67, 90) and permit protection against dephosphorylation by protein phosphatases PP1, PP2A, and PP2C (41, 92–95). The combined effect of allosteric activation and phosphorylation on α -Thr172 induces a >1000-fold increase in AMPK activity (91). In skeletal muscle, LKB1 is the primary upstream kinase responsible for the phosphorylation of α 2-containing AMPK complexes in response to contraction and pharmacological AMPK activators (96–99). To a lesser extent, Ca^{2+} /calmodulin-dependent protein kinase kinase β (CaMKK β) likely phosphorylates and activates AMPK α 1 complexes during long-term low-intensity exercise/contraction (100, 101) (Fig. 3). LKB1 appears constitutively active (102, 103), whereas CaMKK β activates AMPK upon an increase in intracellular Ca^{2+} concentrations, even in the absence of adenine nucleotide content imbalance (104, 105). In addition, glycogen has been shown to influence AMPK activity through its interaction with the β subunit. The β subunit contains a GBD that causes AMPK complexes to associate with glycogen particles in cell-free systems and cultured cells, and this association inhibits AMPK activity (56, 106). This inhibition by glycogen seems to affect mainly α 2-containing AMPK complexes (107). Furthermore, *in vivo* studies report an inverse relationship between muscle glycogen content and AMPK activation in rodents (108, 109) and humans (107), although this inhibition

by glycogen *in vivo* is not consistently found (110). Interestingly, the β subunit is autophosphorylated at Ser108 and Thr148, which seems to regulate AMPK activity and its binding capacity to glycogen, respectively (59, 111). In addition, findings suggest that insulin reduces AMPK activity in rat skeletal muscle likely through Akt-mediated phosphorylation of Ser485/491 on the α 1/ α 2 subunit (112).

The activation of AMPK in rodent skeletal muscle during exercise was initially described by Winder and Hardie (113), and the first reports of AMPK activation during exercise in human skeletal muscle were published some years later (114–116). Several studies have been carried out to determine the activation profile of the different AMPK complexes during exercise bouts depending on intensity and duration. Typically, AMPK activation is observed only at exercise intensities of a minimum of 60% $\text{VO}_{2\text{peak}}$ (107, 116–119). However, low-intensity exercise at 30–40% of $\text{VO}_{2\text{peak}}$ but performed until exhaustion also activates AMPK in skeletal muscle (119). Moreover, AMPK activation seems dependent on exercise duration (115, 118, 119), although not all studies report this (120). During exercise at high intensity and moderate duration, α 2-containing AMPK complexes are predominantly activated (115–118), whereas activity of AMPK α 1 complexes have been found to increase, stay unchanged, or decrease in an intensity- and duration-dependent manner (76, 101, 110, 117). Specifically, in human vastus lateralis muscle

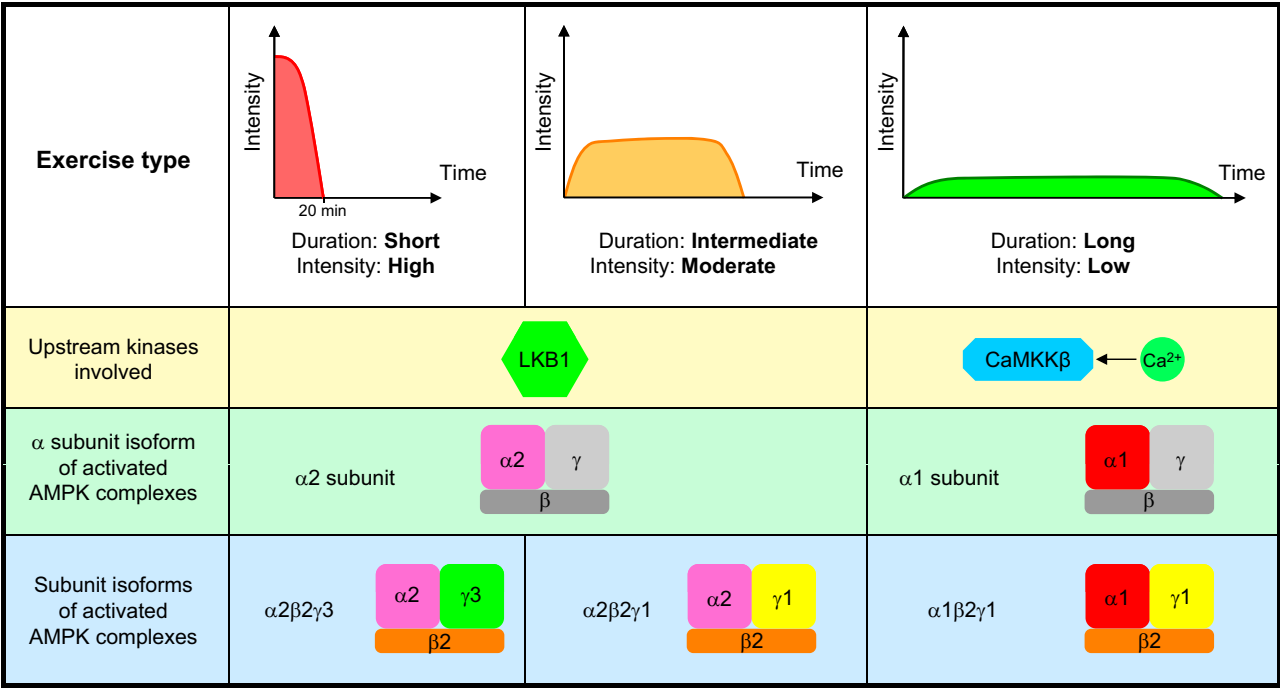


Figure 3. Activation of different AMPK complexes in skeletal muscle is dependent on exercise intensity and duration. In skeletal muscle, LKB1 is the major upstream kinase responsible for the phosphorylation of α 2-containing AMPK complexes in response to high/moderate-intensity and short/limited-duration exercise. In contrast, CaMKK β phosphorylates and activates α 1-containing AMPK complexes during long-term exercise at a low intensity. In human vastus lateralis muscle, AMPK activation is restricted to α 2 β 2 γ 3 heterotrimers during short (up to 20 min) and intense exercise, whereas the α 2 β 2 γ 1 and α 1 β 2 γ 1 complexes appear unchanged or even show decreased activation. When exercise is prolonged, α 2 β 2 γ 1 heterotrimers are activated. During lower-intensity exercise of longer duration, α 2 β 2 γ 1 and α 1 β 2 γ 1 complexes are moderately activated. Thus, in skeletal muscle each AMPK heterotrimer combination is regulated in a distinct manner during contraction depending on exercise intensity and duration, which causes a differential functional response.

during a short (up to 20 min) and intense exercise bout, AMPK activation is restricted to γ 3-containing complexes (α 2 β 2 γ 3), whereas activity of the other complexes (α 2 β 2 γ 1 and α 1 β 2 γ 1) appears unchanged or even decreased (76). When exercise is prolonged, α 2 β 2 γ 1 heterotrimers are activated (101). During exercise at lower intensities and of longer duration, a moderate increase in α 2 β 2 γ 1 activity and a weak increase in α 1 β 2 γ 1 activity have been observed (101). Thus, in skeletal muscle the different AMPK heterotrimer complexes are regulated in a distinct manner during contraction depending on exercise intensity and duration, which may cause different functional responses. How this differential activation of the specific AMPK complexes is accomplished during exercise needs further investigation. One hypothesis could be that it may in part reflect differences in AMP/ADP sensitivity of the complexes (121). Another possibility could be that the findings relate to regulation and/or recruitment of other cells types (*e.g.*, endothelial cells, adipocytes, and macrophages) within the muscle tissue during exercise. Whatever underlies the differential regulation of AMPK trimer activity in skeletal muscle during exercise, it emphasizes the complexity of these observations, which should be accounted for when evaluating downstream signaling of AMPK. Regarding this finding, distinct phosphorylation signatures of downstream targets have been demonstrated for γ 1- and γ 3-containing complexes in human skeletal muscle in response to exercise (122), supporting differential functions of the various AMPK heterotrimeric complexes. However, studies investigating the importance of the different complexes for various functions in skeletal muscle are somewhat limited. Yet, it is generally known that activating mutations in the AMPK γ 1 and AMPK γ 3 subunits promote glycogen storage in skeletal (and cardiac) muscle (123–127). Furthermore, the AMPK α 2 β 2 γ 3 complex, which is predominantly expressed in skeletal muscle, seems important for AMPK-mediated glucose transport, induction of mitochondrial biogenesis, and improvement in muscle insulin sensitivity (125, 128, 129).

Evidence indicates that AMPK activity in skeletal muscle is regulated differently depending on training status. Thus, several studies have shown that activation of skeletal muscle AMPK by acute exercise, performed at the same absolute and relative intensity, is reduced after a period of exercise training (110, 130–133). This may suggest that the working muscle has an increased ability to maintain energy homeostasis after a period of exercise training or that exercise training diminishes the effect of acute exercise on muscle energy stress, as previously indicated (134). Although these explanations seem plausible, other factors may contribute to the lower exercise-induced activation of AMPK in skeletal muscle after exercise training. One such factor could be elevated glycogen levels in trained muscle that may prevent sufficient activation of AMPK in response to acute exercise. Another factor could be the striking decrease in AMPK γ 3 protein content (10–60%) observed in skeletal muscle after exercise training (69, 130, 131, 135). Because the AMPK α 2 β 2 γ 3 complex is the major complex activated in skeletal muscle during exercise, it would indeed be expected that

the exercise-induced activation of AMPK is affected by a reduction in AMPK γ 3 protein content after exercise training. Although genetic evidence is lacking, activation of the PPAR β pathway is likely the mechanism responsible for lowering the content of AMPK γ 3 protein in skeletal muscle (136).

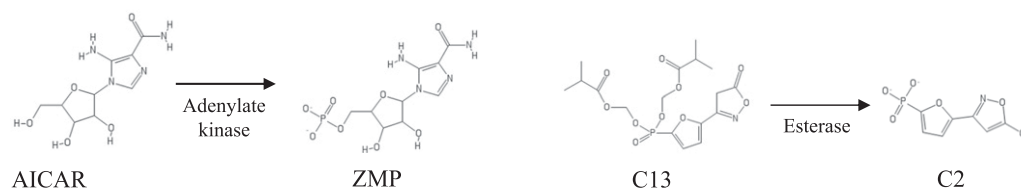
AMPK ACTIVATORS

Given its pivotal role in metabolism, activation of AMPK has long been thought of as a putative therapeutic target for metabolic disorders such as type 2 diabetes (137). Over the years, a myriad of AMPK activators have emerged in the literature (138). Nevertheless, there are no clinically attractive direct activators of AMPK available to date. This underscores the inherent complexity of activating the AMPK complex. The activators described in the literature can largely be divided into 3 subgroups based on their mechanisms of activation: AMP analogs, compounds that perturb the cellular AMP/ATP ratio, and allosteric activators (Fig. 4).

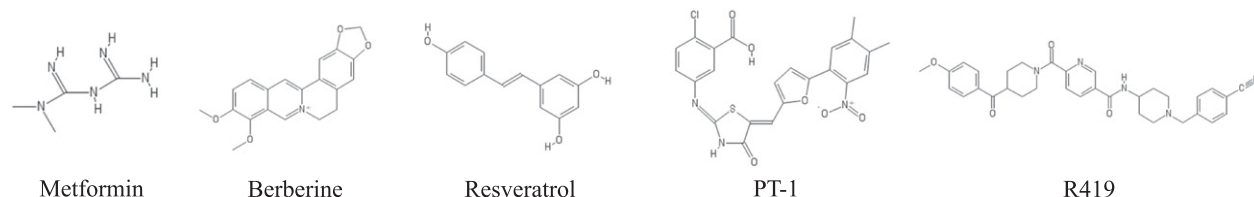
AMP analogs

The first and, to date, most studied compound in the category of AMP analogs is AICAR (139). AICAR is a prodrug and an adenosine analog that, once transported into the cell by adenosine transporters, is phosphorylated by adenosine kinase into 5-aminoimidazole-4-carboxamide ribonucleotide (ZMP) (140). ZMP is an AMP analog that binds to the γ subunits of AMPK at the same sites as AMP and elicits the same effects of activation (141, 142). However, the potency of ZMP for AMPK activation is $\sim$ 20-fold lower compared with that of AMP (139). In addition, ZMP regulates several AMP-sensitive enzymes (143). Although useful as a biologic research tool, the combination of poor selectivity, low potency, and inadequate bioavailability (144) makes AICAR an unattractive clinical candidate. ZMP is an intermediate in purine metabolism, and, although ZMP rapidly accumulates intracellularly in most primary cell types, some immortalized cell lines have high rates of purine synthesis, and hence ZMP does not accumulate (138). This could offer an explanation as to why the effects of AICAR vary depending on cell type. In skeletal muscle, AICAR appears to induce comparable activation of α 1- and α 2-containing AMPK complexes and increases glucose uptake in an AMPK α 2 β 2 γ 3 complex-dependent manner (125, 145, 146). Another AMP analog that has been shown to activate AMPK is 5-(5-hydroxyl-isoxazol-3-yl)-furan-2-phosphonic acid (C2) (147). C2 is a derivative of the prodrug compound-13 and activates AMPK with an EC₅₀ in the low nanomolar range, which is $\sim$ 1000-fold lower than AMP (147). C2 is a partial agonist of α 2-containing complexes and selectively protects α 1 complexes against dephosphorylation (148). Despite being an AMP analog, C2 does not appear to modulate other AMP-sensitive enzymes, nor does it antagonize AMP on these enzymes (148). Interestingly, recent crystal structures reveal two C2 binding sites in the γ subunit distinct from the

A AMP analogs



B Compounds that perturb cellular AMP/ATP ratios



C Allosteric activators

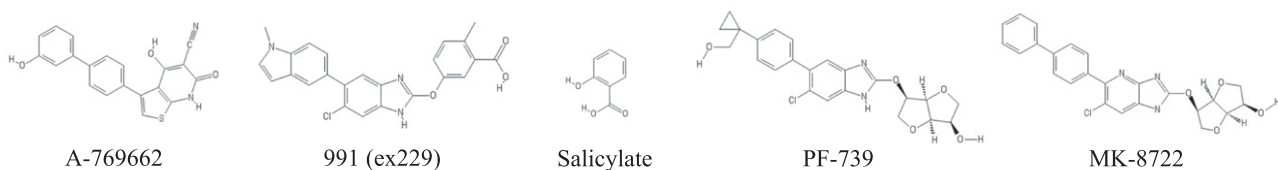


Figure 4. Chemical structure of different AMPK-activating compounds. A) Prodrugs that are converted to AMP analogs by enzymes once transported into the cell. B) Natural and synthetic compounds that increase intracellular AMP concentrations by disrupting ATP generation in mitochondria. C) Direct activators that bind in a pocket between the kinase domain on the α subunit and the carbohydrate-binding module on the β subunit (ADaM site).

nucleotide sites (149), questioning whether C2 should be classified as an allosteric AMPK activator.

Compounds that perturb the cellular AMP/ATP ratio

Because AMPK is activated by cellular increases in the AMP/ATP ratio, any mechanism that disrupts ATP generation will increase cellular AMP levels and ultimately activate AMPK. This complicates the search for attractive and specific AMPK activators because any compound that exerts an energetic stress on a cell is likely to activate AMPK. Hence, monitoring cellular stress and toxicity irrespective of selective AMPK activation is key when examining activators. The largest known classes of AMPK activators that work by this mechanism are inhibitors of mitochondrial ATP generation, such as metformin and berberine (mitochondrial respiratory chain complex I) (150, 151) and resveratrol (mitochondrial ATP synthase) (152, 153). Early studies showed that metformin acutely stimulates glucose uptake in skeletal myotubes (154). However, in mouse skeletal muscle, acute metformin treatment does not seem to alter glucose uptake despite modest activation of AMPK (155). Instead, chronic metformin treatment appears to enhance insulin-stimulated glucose uptake in an AMPK-dependent manner (155). Another compound in this class includes PT-1, which inhibits mitochondrial respiration independently of ATP synthase and appears to selectively activate γ 1-containing

complexes in skeletal muscle. Interestingly, this does not lead to increased glucose uptake in skeletal muscle (65). It will be useful to investigate how an indirect activator that acts by perturbing the cellular AMP/ATP ratio can activate only AMPK γ 1-containing complexes. As mentioned previously, localization of the specific AMPK complexes may play a role in this selectivity. More than 100 less-characterized AMPK activators have emerged in the literature throughout the years (138). A substantial portion of these are natural plant products, and, although the mechanism of activation is unknown for the majority of these products (138), most naturally derived AMPK activators with known mechanisms, such as metformin, resveratrol, and berberine, are indirect activators. Thus, it can be speculated that a considerable fraction of these plant-derived activators belong to this subgroup of indirect AMPK activators.

Allosteric activators

Of the large pool of AMPK activators in the literature, very few have been validated as direct allosteric activators with mechanisms of activation distinct from that of AMP. Of the compounds validated, even fewer have been tested in skeletal muscle. This section focuses on the compounds that have been explored in skeletal muscle. The first of such activators to be described was A-769662 (156). A-769662 binds in a pocket between the kinase domain on the α subunit and the carbohydrate-binding module on the β

subunit (157) and does not alter the cellular AMP/ATP ratio (156). A-769662 activates AMPK both allosterically and by inhibiting dephosphorylation of AMPK on α -Thr172 (158). The seminal study on A-769662 showed that administration to ob/ob mice markedly lowered plasma glucose levels (156). Interestingly, although A-769662 activates β 1-containing AMPK complexes in mouse skeletal muscle lysates, Treebak *et al.* (70) showed that A-769662-induced glucose uptake was dependent on mouse strain and appeared to occur through a wortmannin-sensitive mechanism. The majority of β 1 in skeletal muscle lysates likely originate from nonmuscle cells, and hence the effect of β 1 agonists in skeletal muscle tissue is questionable. Another recently identified allosteric activator, compound 991 (also referred to as ex229), binds to the same site as A-769662 but displays 5- to 10-fold higher potency (159). *In vitro* experiments have demonstrated that compound 991 binds 10 times stronger to AMPK β 1-containing complexes than to complexes containing β 2 (159). Nevertheless, this compound activates both β 1- and β 2-containing complexes in mouse skeletal muscle and increases glucose uptake in myotubes by an AMPK-dependent mechanism (160). Two new AMPK activators described recently, MK-8722 and PF-739, demonstrate tremendous advances in developing selective AMPK activators with desirable potency and bioavailability (161, 162). Similar to 991 and A769662, these compounds display higher affinity toward β 1, but have the ability to activate all AMPK complexes. These compounds robustly activate AMPK in skeletal muscle with EC50s in the nanomolar range. They increase skeletal muscle glucose uptake and lower blood glucose in rodents and nonhuman primates (161, 162). Reiterating the absence/low importance of β 1 complexes in skeletal muscle, the β 1-selective activator PF249 fails to activate AMPK in mouse skeletal muscle (161). The pharmacokinetic and pharmacodynamic relationships of these two novel compounds make them very attractive tools to study the effects of AMPK activation *in vivo*. Although the allosteric activators described in this review so far are synthetic compounds, another allosteric AMPK activator is salicylate, a natural plant product from willow bark and a derivative of acetyl salicylic acid (aspirin). Salicylate binds at the same site as A-769662 and predominantly activates β 1-containing complexes (163). Interestingly, in rat skeletal muscle, salicylate activates α 1- and α 2-containing AMPK complexes to a comparable extent at 5 mM. This concentration of salicylate also induces a substantial increase in glucose uptake (164). It appears that most known allosteric activators of AMPK to date, despite showing affinity toward β 1-containing complexes, can be regarded as pan activators in skeletal muscle.

The AMPK field has long sought after useful pharmacological tools to selectively activate AMPK. Although the traditional tools (*i.e.*, AMP analogs and compounds that alter the cellular AMP/ATP ratio) have provided vital insight into the biologic function of AMPK, their modes of activation make them unselective. Hence, the emergence of potent AMPK-selective small-molecule activators like compound 991, PF-739, and MK-8722 (159, 161, 162) hold high promise for the future. From a skeletal muscle-centric

view, a new generation of isoform selective activators is highly warranted. Specifically, selective activation of γ 3-containing complexes could have major implications not only for the understanding of skeletal muscle AMPK, but potentially for AMPK activation as a therapeutic strategy. So far, the strategy for small-molecule activation of AMPK has been to use a drug-binding pocket located between the α -subunit kinase domain and the β -subunit carbohydrate-binding module [also known as the allosteric drug and metabolite (ADaM) site]. Although this has proven effective for β -selective AMPK activation, this strategy is not viable for γ -selective activation. Currently, there are no crystal structures available of γ 3-containing AMPK complexes, and hence we do not know whether there are suitable drug-binding pockets in these complexes. A potential strategy is to take advantage of one or more of the 4 AMP-binding sites on the γ subunit (165), an approach partly used by the AMPK activator compound-13. However, compound-13 directly competes with AMP (148) and as such directly competes with the biologic regulation of AMPK activity. Furthermore, it is uncertain whether such an approach permits γ 3 selectivity over other γ isoforms. A recent publication on compound-13 revealed that this compound selectively activates γ 1- and γ 2-containing AMPK complexes (149), suggesting that there are indeed structural differences between the γ -subunits. Such differences may form the basis for γ 3-selective AMPK activators in the future. Regardless, further understanding of the γ subunits in the complex are warranted to design the next generation of AMPK activators. As an alternative to customized small-molecule AMPK activators, it may be speculated that the drug-binding pockets of AMPK is a site for endogenous AMPK ligands, although these have not been identified. Therefore, it may be worth investigating the presence of physiologic hormones/proteins that activate AMPK *via* its drug-binding pockets. This may permit the generation of highly selective and viable direct AMPK agonists for therapeutic use.

AMPK REGULATION BY MYOKINES

Exercise has proven to be effective for the prevention and treatment of metabolic disorders such as obesity, type 2 diabetes, and cardiovascular disease (166). This protective effect has been linked to the renewed interpretation of skeletal muscle as an endocrine organ (167). Several reports have shown that skeletal muscle produces and releases a variety of substances, called myokines, during exercise and contraction (168, 169). These myokines act as autocrine, paracrine, or endocrine factors and may mediate the beneficial effects on metabolic and physiologic responses to exercise in the skeletal muscle itself and in distant organs such as heart, lungs, brain, adipose tissue, and liver (167).

IL-6 is the most extensively studied myokine. Originally considered a proinflammatory cytokine because its plasma concentration is often elevated in patients with type 2 diabetes (170), it was later reintroduced as a therapeutic myokine because it was discovered that IL-6 is produced by contracting skeletal muscle and is released in

high amounts into the circulation to exert its endocrine function (171–173). Additionally, IL-6 functions as an important autocrine factor in skeletal muscle through the maintenance of skeletal muscle glucose homeostasis during prolonged exercise and up-regulation of its own production (171–173). The relationship between AMPK and IL-6 has been extensively examined since MacDonald *et al.* (174) found a correlation between IL-6 release from muscle and tissue AMPK α 2 activity after 60 min of exercise in humans. However, it remains controversial whether AMPK is involved in the secretion of IL-6 by skeletal muscle during exercise or, on the contrary, whether AMPK activity is modulated by the autocrine myokine action. Findings based on *in vitro* AICAR treatment have generated conflicting results. Some studies have documented a positive effect of AICAR on the myokine expression in C2C12 muscle cells (173) and depletion of IL-6 vesicles in skeletal muscle fibers (175), whereas other studies have shown a reduction in the IL-6 expression in myocytes (176) and its release by isolated skeletal muscle (177). Glund *et al.* (177) also demonstrated that the basal IL-6 release from oxidative muscle was elevated in AMPK α 2 kinase dead and AMPK α 1-KO mice models, indicating that IL-6 regulates its own release *via* AMPK in resting skeletal muscles. To complete the broad studies made using different transgenic mouse models of AMPK, it was recently demonstrated that AMPK α mdKO mice display elevated expression of IL-6 mRNA in gastrocnemius muscle, which might be due to mild skeletal muscle damage (72). On the other hand, IL-6 activates AMPK *in vitro* (178, 179) and *in vivo* (179, 180), as demonstrated in IL-6-KO mice, which exhibit decreased AMPK activity in skeletal muscle. Taken together, current observations do not provide a clear picture on the possible relationship between IL-6 and AMPK in skeletal muscle, and further research is needed to validate its manifestation. Although IL-6 is greatly implicated in mediating the health benefits of exercise and muscle regeneration after injury (181), higher systemic levels of the cytokine have been shown to induce muscle proteolysis (182). To the best of our knowledge, no involvement of AMPK in this detrimental effect has been suggested. Additionally, it has been recently proposed that the cytokine exhibits a dual pro- or anti-inflammatory effect depending on its downstream molecular pathway (183).

Other myokines produced and released by skeletal muscle are IL-8, IL-15, irisin, brain-derived neurotrophic factor (BDNF), and leukemia inhibitory factor (LIF) (184–188). These myokines have been less studied, and there are only a few studies linking them to AMPK. Lihn *et al.* (176) demonstrated that *in vitro* AICAR stimulation reduces IL-8 expression similar to IL-6 in human skeletal muscle cells. More attention has been given to the possible relationship between AMPK and IL-15. As such, IL-15 protein content has been found to be lower in skeletal muscle of AMPK α 2 kinase dead mice in both control and high-fat diet (HFD) groups (189). In addition, AMPK β 1 β 2M-KO mice exhibit lower IL-15 levels in muscle and plasma (190). The same transgenic mouse model also displays reduced plasma irisin levels in the basal state (191). Moreover, it has been shown that irisin induces phosphorylation and activation of AMPK *in vitro*

(192). Similarly, BDNF can act in an autocrine/paracrine manner to boost fat oxidation in skeletal muscle during contraction *via* AMPK (187). Finally, skeletal muscle LIF mRNA expression increases immediately after exercise and declines gradually during recovery (188). However, a definite link to AMPK has yet to be established. Interestingly, LIF has been shown to increase glucose uptake in skeletal muscle, although this does not seem to require AMPK (193).

Although it is well established that skeletal muscle expresses a variety of myokines, it is hard to believe that AMPK is the only mediator of their release. Considering the energy-preserving nature of AMPK, it may be speculated that skeletal muscle AMPK is involved in releasing myokines that facilitate restoration of myocellular energy homeostasis. If so, myokines such as IL-6, IL-15, BDNF, and LIF are possible candidates based on their promoting effects on glucose uptake and fatty acid oxidation in skeletal muscle as well as lipolysis in adipocytes (178, 187, 193–195).

AMPK IN THE CONTROL OF MUSCLE GLUCOSE UPTAKE AND FATTY ACID OXIDATION

The first evidence linking AMPK to regulation of glucose uptake and fatty acid oxidation in skeletal muscle was provided in 1997 by Merrill *et al.* (9), who observed that AICAR not only increased AMPK activity in rat skeletal muscle but also enhanced muscle glucose uptake and fatty acid oxidation (9). Later observations in the AMPK α 2 kinase dead mouse model provided genetic evidence to support a role of AMPK in regulating muscle glucose uptake but not fatty acid oxidation in response to AICAR stimulation (196, 197). The effect of AICAR on muscle glucose uptake was subsequently shown to be dependent on the α 2 β 2 γ 3 complex (125, 145, 146). In contrast to AICAR, evidence linking AMPK to increased muscle glucose uptake and fatty acid oxidation during exercise and contraction remains inconclusive. Thus, the AMPK β 1 β 2M-KO mouse model shows increased reliance on fat oxidation during treadmill exercise (74), whereas the AMPK α mdKO and LKB1-KO mouse models display decreased fat oxidation during treadmill exercise at the same relative intensity (73, 198). Moreover, exercise-induced muscle glucose uptake is impaired in AMPK β 1 β 2M-KO mice but remains intact in AMPK α mdKO and LKB1-KO mice (73, 74, 198). Findings from several AMPK-deficient mouse models also suggest that *ex vivo* contraction-stimulated glucose uptake is intact (72, 125, 146, 199, 200) or only partially decreased (74, 145, 196, 201, 202) in skeletal muscle. Similarly, rates of fatty acid oxidation are not impaired in skeletal muscle from different AMPK-deficient mouse models during *ex vivo* contraction, although an overall reduction of fatty acid oxidation may be observed in skeletal muscle of AMPK-deficient mice compared with wild-type (WT) mice (73, 146, 197, 198). However, this reduction is associated with lower cluster of differentiation 36 (CD36) and/or fatty acid binding protein content in skeletal muscle of the AMPK-deficient mouse models (73, 198), which are thought to be important

regulators of fatty acid uptake and oxidation (203). Although pharmacological activators of AMPK increase glucose uptake and fatty acid oxidation in skeletal muscle by an AMPK-dependent mechanism, AMPK is not or is only partially necessary for regulating these processes during muscle contractile activity. The reason for this discrepancy is not clear but may relate to redundant signaling during exercise or to the finding that AMPK regulates muscle glucose uptake and fatty acid oxidation during physiologic stimuli other than exercise that activates the complex in skeletal muscle (e.g., food deprivation and hypoxia). Regarding the former, the AMPK-related kinase SNARK is activated in skeletal muscle in response to exercise, and knockout of SNARK is associated with impaired exercise-stimulated glucose uptake in skeletal muscle (81). Thus, it could be speculated that SNARK may compensate for the lack of skeletal muscle AMPK, thereby masking the effect of AMPK in regulating muscle glucose uptake during exercise. Measurements of SNARK activity in skeletal muscle depleted of AMPK activity could reveal insight into this notion.

To unravel a potential role of AMPK in regulating muscle glucose uptake and fatty acid oxidation, studies investigating downstream signaling of AMPK may provide clues. It has been suggested that AMPK enhances glucose uptake by stimulating GLUT4 translocation to the muscle surface membrane (196, 204). Although not fully described, AMPK-mediated glucose uptake may involve phosphorylation of downstream target TBC1D1, which has been proposed to regulate GLUT4 translocation (205). Hence, several studies using different AMPK-deficient models have shown that several phosphorylation sites on TBC1D1 are regulated by AMPK in skeletal muscle during *ex vivo* contraction (73, 74, 122, 206–208). Furthermore, phosphorylation of TBC1D1 increases in human skeletal muscle during exercise and has been found to correlate with AMPK heterotrimer-specific activity (71, 122, 207, 209). Taken together, these observations suggest a functional relationship between TBC1D1 and AMPK in regulating exercise-induced muscle glucose uptake by redistributing GLUT4 from the cell interior to the cell surface membrane. In support of this, glycolytic skeletal muscle from various TBC1D1-deficient mouse models exhibits impairments in exercise-, contraction-, and AICAR-stimulated glucose uptake (210–213). However, these mouse models display reduced levels of GLUT4 protein in skeletal muscle, and therefore they are not ideal for assessing the possible regulatory effects of AMPK on glucose uptake. In contrast, mouse tibialis anterior muscle overexpressing TBC1D1 mutated at 4 predicted AMPK phosphorylation sites (Ser231, Ser499, Ser660, and Ser700) expresses normal levels of GLUT4, although contraction-stimulated glucose uptake is reduced by ~35% (208). Although reductions in glucose uptake after contraction may seem low in muscle overexpressing a mutated form of TBC1D1, only a portion of tibialis anterior muscle overexpresses TBC1D1 due to the transfection methodology used (i.e., electroporation) (214). Thus, improvements in transfection efficacy could potentially induce a further reduction in glucose uptake during contraction, highlighting the likely importance of TBC1D1 in regulating

AMPK-mediated glucose uptake. Interestingly, contraction- and exercise-induced glucose uptake is not compromised in skeletal muscle from TBC1D1 Ser231Ala KI mice (215, 216), suggesting that TBC1D1 needs to be phosphorylated at multiple sites by AMPK or modified otherwise to exert its effect on glucose uptake.

It is generally believed that AMPK regulates fatty acid oxidation in skeletal muscle through phosphorylation of acetyl-CoA carboxylase 2 (ACC2) on Ser212. This is supported by findings in ACC2 Ser212Ala KI mice in which AICAR fails to increase fatty acid oxidation in skeletal muscle (217). Phosphorylation of ACC2 Ser212 does not appear to play a role in regulating fatty acid oxidation during *ex vivo* contraction and exercise (218), providing evidence to support that AMPK-ACC2-independent pathways regulate fatty acid oxidation during muscle contractile activity. In addition, AMPK does not seem essential for increasing FAT/CD36 translocation and fatty acid uptake during *ex vivo* contraction (219), which may limit fatty acid oxidation during low to moderate contraction intensities (220). Considering these findings, the importance of AMPK in regulating muscle fatty acid oxidation during exercise appears sparse given the genetic evidence obtained from various transgenic mouse models. A role of AMPK in regulating muscle fatty acid oxidation in exercise recovery has been suggested based on reduced ACC activity and malonyl-CoA levels and enhanced fatty acid oxidation after a single bout of treadmill exercise (221). This is also in line with the notion that, although phosphorylation of AMPK returns to baseline quite quickly after exercise in skeletal muscle, downstream effects on substrates like ACC Ser212 are increased, which may enhance fatty acid oxidation early in the postexercise period (209, 222). This suggests that AMPK may regulate muscle fatty acid oxidation after exercise rather than during exercise.

AMPK AND PROTEIN METABOLISM

Mammalian target of rapamycin complex 1 (mTORC1) is a well-known cell nutrient and energy sensor, as demonstrated by its ability to sense amino acids and regulate protein translation as well as autophagy, two major processes involved in cellular protein turnover (223, 224). mTORC1 regulates muscle mass development by controlling protein translation initiation *via* its two major downstream targets: the p70 ribosomal S6K and the eukaryotic initiation factor 4E binding protein 1 (4E-BP1) (225). mTORC1 activity is regulated by AMPK through phosphorylation of tuberous sclerosis complex 2 (TSC2) at Thr1227 or Ser1345, which improves the ability of TSC2 to inhibit mTOR activity (226). Alternatively, AMPK can directly phosphorylate mTOR Thr2446 to inactivate the complex, which prevents Akt-mediated phosphorylation of mTOR Ser2448, resulting in inhibition of translation initiation (227, 228). In addition, AMPK phosphorylates Raptor on Ser722 and Ser792, resulting in mTORC1 inhibition (229). Hence, increased energy demand dampens protein synthesis through a balance between the AMPK and mTOR

signaling pathways (**Fig. 5**). Interestingly, rats treated with AICAR have reduced skeletal muscle protein synthesis, which is associated with down-regulation of the Akt-mTOR pathway and its downstream targets S6K and 4E-BP1 (11). Independently of mTORC1, AMPK may also inhibit translation elongation through direct phosphorylation of eEF2k (230).

Although AMPK activation inhibits mTORC1 to regulate protein synthesis, it has recently been shown that suppression of S6K activity promotes AMPK signaling (231). Indeed, the interplay between the AMPK and mTORC1 signaling pathway is tightly regulated by several upstream signals but also by several negative feedback loops operated by different kinases, such as S6K on insulin receptor substrate and AMPK (231–236). As such, AMPK α mdKO mice exhibit increased muscle mass with normal body weight compared with WT littermates, indicating a role of AMPK for controlling muscle growth (72). On the other hand, AMPK β 1 β 2M-KO mice have normal skeletal/cardiac muscle weights as compared with their WT littermates (74), which may suggest that alternative or compensatory mechanisms might compensate for AMPK deficiency in AMPK β 1 β 2M-KO mice. Although still to be investigated, AMPK activity could also be important for the maintenance of muscle protein turnover and growth in conditions of high energy demand such as exercise, caloric restriction, muscle regeneration, and repair (74).

AMPK is also known for regulating muscle protein degradation, which serves as a source of amino acids that can be used for energy production by various other organs. The ATP-dependent ubiquitin–proteasome system and the autophagy–lysosomal pathway are the major pathways for mediating protein degradation. The former involves a cascade of enzymatic reactions that tag substrate proteins with ubiquitin chains for degradation by the 26S proteasome. This system involves the activity of E3 ubiquitin ligases, which confer substrate specificity for ubiquitination and degradation. Two major E3 ligases have been described to be important for muscle atrophy: atrogin-1/muscle atrophy F-box (MAFbx) and muscle RING finger 1 (MuRF1) (237, 238). AMPK has been shown to phosphorylate forkhead box protein O3a (FOXO3a) at multiple sites, including Ser413 and Ser588 (239), which seems to control the transcription of the E3 ligases MAFbx and MuRF1, promoting muscle protein degradation (72, 240, 241) (**Fig. 5**). During starvation-induced atrophy, FOXO3a also regulates the transcription of several proteins involved in the autophagic machinery (242, 243).

Autophagy lysosomal-dependent protein degradation is an evolutionarily conserved recycling pathway for maintaining cell metabolism and protein turnover. There are three different types of autophagy. In macroautophagy, proteins, organelles, and cytosolic material are sequestered in an autophagosome to be degraded after the autophagosome fuses with lysosomes. In endosomal

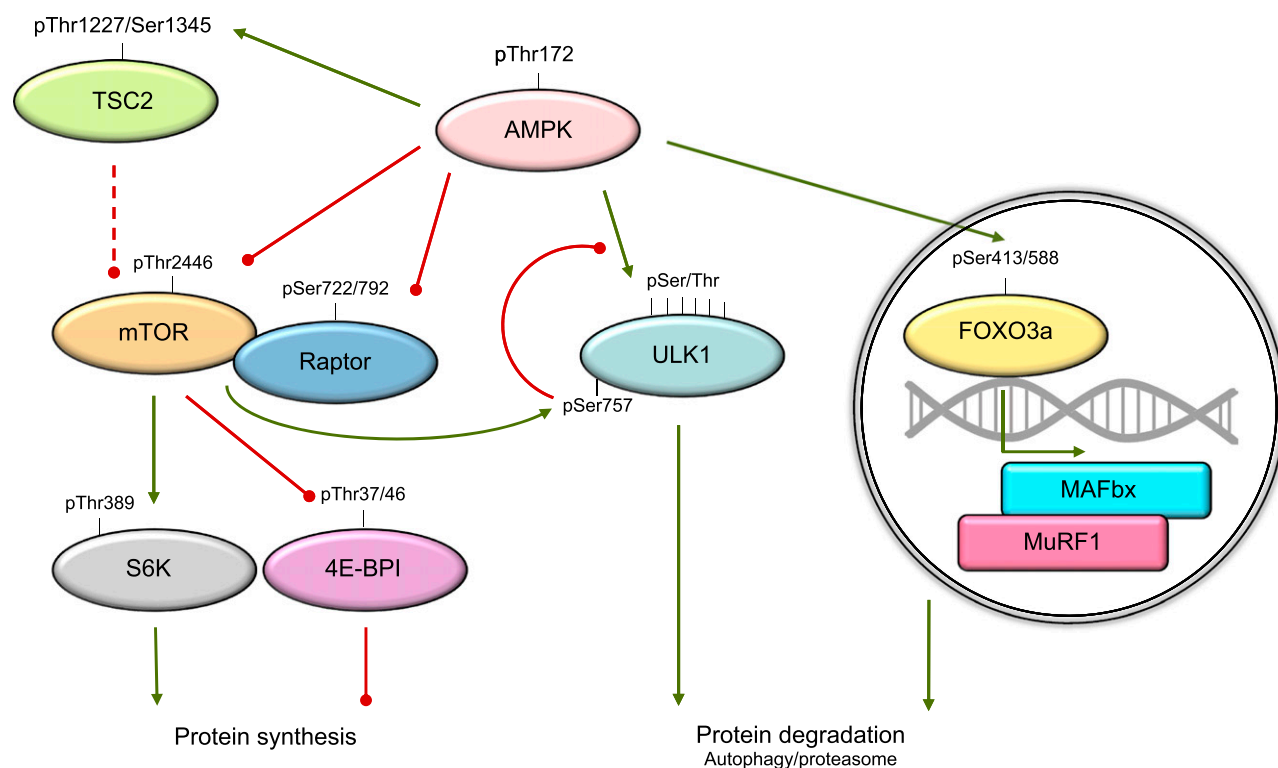


Figure 5. Role of AMPK in protein metabolism. Activation of AMPK increases phosphorylation of regulatory associated protein mTOR (Raptor) and mTORC1 itself as well as TSC2, an indirect inhibitor of mTORC1. This decreases the activity of S6 kinase (S6K) and inhibits the release of binding partner eukaryotic translation factor 4E (eIF4E) from eukaryotic translation initiation factor 4E-BP1, thereby diminishing protein synthesis. Besides its effect on protein synthesis, AMPK promotes autophagy and proteasomal-mediated protein degradation by phosphorylating ULK1 (Ser317/467/555/637/777 and Thr575) and FOXO3a, respectively. When phosphorylated, FOXO3a increases the expression of E3 ligases MAFbx and MuRF1. Green lines, promotive; red lines, inhibitory; dashed line, indirect. Adapted from Sanchez *et al.* (42).

microautophagy, portions of cytoplasm are taken up into the lumen of the lysosome to be degraded. The third mechanism is chaperone-mediated autophagy, where substrate proteins are recognized by chaperone proteins, which facilitate lysosomal membrane crossing through a translocation complex (244). Macroautophagy is robustly activated in response to nutrient deprivation and inhibited by nutrient-sensor proteins such as mTORC1 (245–248). In the last decades, several studies have identified AMPK as a main regulator of autophagy in response to food deprivation, exercise, and stressors. AMPK directly phosphorylates multiple sites in uncoordinated 51-like kinase 1 (ULK1) (Ser317, Ser467, Ser555, Thr575, Ser637, and Ser777) and promotes ULK1 function in autophagy (249–252). These observations, in addition to studies in ULK1 knockout cells, demonstrate the functional importance of ULK1 phosphorylation by AMPK for induction of autophagy. mTORC1 also regulates ULK1 and autophagy by directly phosphorylating ULK1 on Ser757, which disrupt AMPK–ULK1 interaction. These results suggest that AMPK and mTORC1 have opposite roles on autophagy regulation depending on the cellular energy status (Fig. 5). However, recent studies in humans and mice suggest that AMPK alone may not be sufficient to activate autophagy in response to exercise and food deprivation in skeletal muscle (253, 254). Autophagy-specific gene proteins (Atgs) are essential mediators of autophagy based on their role in controlling the formation of the autophagosome. Activation of the autophagic enzyme machinery (*e.g.*, Atg5, Atg7, and nutrient-deprivation autophagy factor-1) is required for skeletal muscle remodeling and proper mitochondrial function (255–259); thus, skeletal muscle autophagy is an essential process required for maintaining muscle function and whole-body energy metabolism (260). Although the role of AMPK on chaperone-mediated autophagy has not been extensively studied, recent evidence from skeletal muscle suggests that AMPK may be implicated in the regulation of chaperone-mediated autophagy to degrade specific proteins. Indeed, AMPK phosphorylates perilipin 2 (PLIN2), which increases PLIN2 degradation (261). However, the physiologic role of AMPK regulating PLIN2 protein abundance and chaperone-mediated autophagy in skeletal muscle remains to be investigated in detail. During healthy conditions, skeletal muscle is required to supply gluconeogenic amino acids (such as alanine) for the maintenance of blood glucose levels during prolonged food deprivation (262). It was recently demonstrated in mice that the glucose–alanine cycle is regulated by AMPK-mediated autophagy in skeletal muscle, which increases alanine production for hepatic gluconeogenesis to avoid hypoglycemia during food deprivation (263). Taken together, these observations suggest that AMPK acts as a regulator of protein turnover in skeletal muscle.

Different stressors, such as starvation, exercise, and hypoxia, strongly perturb the function of mitochondria and lead to the production of large amounts of reactive oxygen species (ROS), which are harmful to proteins, lipids, and DNA and impair cellular ATP production. Thus, damaged mitochondria must be either repaired or removed. Recent evidence suggests that macroautophagy

can be organelle specific in order to remove dysfunctional organelles such as mitochondria (mitophagy) (264). In atrophying muscle, the mitochondrial network is severely remodeled after food deprivation or denervation, and autophagy *via* Bnip3 contributes to mitochondrial remodeling (265). In 2011, Egan *et al.* (249) identified AMPK as a regulator of mitophagy connecting energy sensing to mitochondrial degradation *via* ULK1 phosphorylation. In support of this, AMPK-dependent phosphorylation of ULK1 was recently shown to be critical for translocation of ULK1 to mitochondria and for mitophagy in response to hypoxic stress (266). In addition, AMPK may prepare mitochondria to undergo mitophagy by stimulating mitochondrial fission *via* phosphorylation of mitochondrial fission factor and facilitating isolation of damaged mitochondria structure (267, 268). Thus, AMPK emerges as a master regulator of mitochondrial homeostasis, coupling mitochondrial dynamics to mitophagy. Interestingly, skeletal muscle of AMPK β 1 β 2M-KO mice has fewer but larger mitochondria compared with WT mice (74). Furthermore, it has recently been shown that acute exercise-induced mitophagy is regulated through AMPK-dependent phosphorylation of Ulk1 in skeletal muscle (269). This provides evidence to support that AMPK is indeed involved in removal of dysfunctional mitochondria to preserve skeletal muscle function.

It is well known that in skeletal muscle, exercise and starvation induce physiologic levels of ROS acting as signaling molecules (270–272). In humans, the use of antioxidants can prevent some of the beneficial effects of exercise training, such as enhanced insulin sensitivity (273). Although the role of ROS in these contexts needs further investigation, induction of fast autophagy by ROS may be a mechanism by which starvation and exercise provide nutrients but also remove dysfunctional mitochondria in skeletal muscle. On one hand, ROS and reactive nitrogen species activate ataxia telangiectasia mutated kinase to recruit LKB1/AMPK and/or TSC1/2 to inhibit mTORC1 and activate autophagy/mitophagy (274–277). On the other hand, AMPK may also directly sense cytosolic redox homeostasis because AMPK has been proposed to be activated upon H₂O₂ exposure, particularly through S-glutathionylation of oxidized cysteines within the AMPK α and AMPK β subunits (278). Antioxidants that block basal mitochondrial ROS production, induced by acute nutrient starvation, also reduce AMPK phosphorylation and inhibit induction of autophagy in skeletal muscle (279). Starvation-induced AMPK activation and autophagy seem to be dependent on mitochondrial ROS based on findings in cell cultures showing that overexpression of SOD2 or stimulation with ROS scavenger N-acetylcysteine attenuates activation of AMPK and autophagy during starvation (280). In addition to mitophagy, recent studies have shown that accumulation of ubiquitinated proteins activates AMPK to induce autophagy in mouse embryonic fibroblasts and macrophages, and similar processes could also be important for skeletal muscle homeostasis (281). Altogether, these studies indicate that AMPK acts as a redox sensor in order to activate autophagy and mitophagy to promote protein degradation and remove dysfunctional mitochondria, thus avoiding

toxicity due to high production of ROS and reactive nitrogen species in mitochondria.

Taken together, these findings show that AMPK acts on anabolic and catabolic processes to regulate muscle protein metabolism. On one hand, it can decrease muscle mass by regulating protein synthesis; on the other hand, it can promote ubiquitin-proteasome and autophagy in response to nutrients and changes in the redox states (274–281).

AMPK IN THE CONTROL OF MUSCLE MITOCHONDRIAL FUNCTION AND CELLULAR ENERGY CHARGE

AMPK relays the signal of low energy charge through phosphorylation of proteins important for the ability of the cell to adapt to changes in energy availability (282). Thus, AMPK signaling to target proteins constitutes a cellular memory of energy stress. The ability of a cell to match substrate utilization with demand involves a coordinated regulation of genes encoding mitochondrial proteins, a process known as mitochondrial biogenesis (283). In response to repeated metabolic stress, AMPK orchestrates a cellular response to ensure sufficient mitochondrial adaptation, and various AMPK transgenic and KO mouse models have consistently shown altered capacity to regulate mitochondrial biogenesis in skeletal muscle compared with WT littermates (15, 72, 74, 128, 284–286). In skeletal muscle, AMPK controls mitochondrial biogenesis through PGC-1 α and nuclear respiratory factor 1 (15). AMPK has been shown to phosphorylate and activate PGC-1 α (287), which drives expression from the PGC-1 α promoter itself and from a multitude of metabolic and mitochondrial genes. PGC-1 α activity is also controlled by lysine acetylation mediated by general control non-repressible 5 (288) and deacetylation mediated by NAD-dependent sirtuin1 (SIRT 1) (289, 290).

Similarly to the AMP/ATP ratio, the ratio between the oxidized and reduced forms of NAD constitutes an index of cellular energy charge, and activation of AMPK with AICAR in C2C12 muscle cells has been found to increase SIRT1 activity by increasing NAD levels and thereby promote deacetylation of PGC-1 α (291). Although the mechanism by which cytosolic/nuclear NAD levels increase in response to AMPK activation is not completely clear, muscle contraction has been shown to increase cytosolic and mitochondrial NAD levels as well as the NAD/NADH ratio (292). Moreover, disruption of AMPK signaling has profound effects on food deprivation- and exercise-induced SIRT1-dependent activation of PGC-1 α signaling in skeletal muscle (293). This indicates a molecular circuit involving AMPK and NAD-dependent sirtuin signaling important for cellular metabolic function (Fig. 6).

Another level of control of PGC-1 α expression is exerted through AMPK-mediated phosphorylation and SIRT1-mediated deacetylation of FOXO3 required for full transcriptional control of the PGC-1 α gene (239, 291, 294). Although FOXO3 has been linked to initiation of autophagy in skeletal muscle (295), FOXO3 has also been shown to be important for muscle mitochondria to adapt to nutrient restriction (296). Collectively, regulation of mitochondrial biogenesis through PGC-1 α in response to fluctuations in muscle energy levels appears to require several phosphorylation and deacetylation events.

The NAD/NADH ratio is thought to increase in mitochondria in response to metabolic stress (*e.g.*, muscle contractions) when oxidation of NADH is not matched by substrate influx resulting in NAD generation (292, 297). As a consequence, SIRT3 is activated, resulting in a priming of mitochondria to increase ATP production. Overexpression of SIRT3 in mouse skeletal muscle has been shown to induce an oxidative fiber type switch (type II to type I), to enhance energy expenditure, and to decrease muscle mass (298). Although some of the observed phenotypes associated with SIRT3 overexpression may be

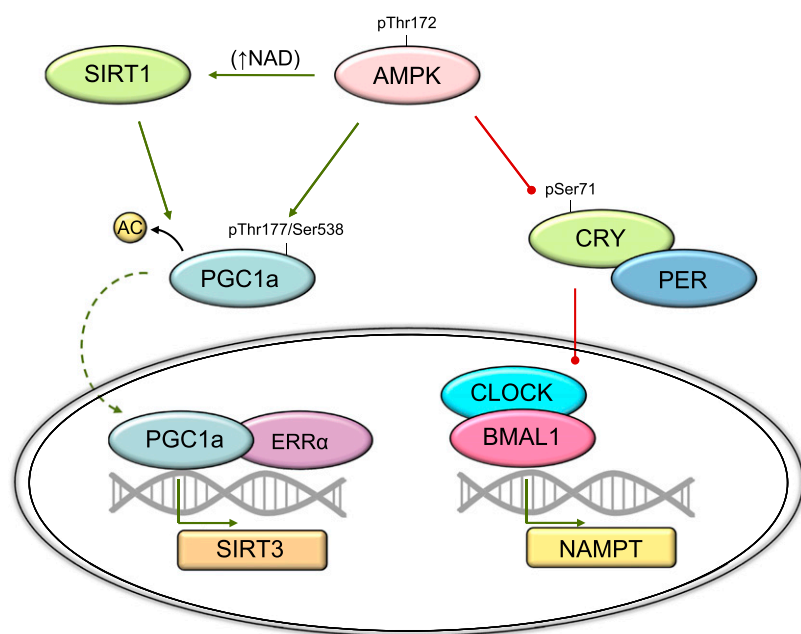


Figure 6. Role of AMPK in mitochondrial function and cellular energy charge. Activation of AMPK enhances phosphorylation of PGC-1 α . In addition, AMPK facilitates activation of SIRT1 by increasing intracellular NAD concentrations, leading to deacetylation and activation of PGC-1 α . After translocation to the nucleus, PGC-1 α interacts with ERR α to increase activity of the SIRT3 promoter, causing an increase in mitochondrial biogenesis. AMPK has been proposed to regulate mammalian NAMPT gene expression *via* phosphorylation of CRYPTOCHROME (CRY), which together with PERIOD (PER) forms an inhibitory complex of CLOCK/BMAL1. When phosphorylated, CRY is degraded, allowing NAMPT transcription by CLOCK/BMAL1. Green lines, promotive; red lines, inhibitory.

beneficial for metabolic health, these observations suggest that expression levels of SIRT3 should ideally be coordinated with other mitochondrial proteins. Indeed, in mouse muscle myotubes and brown adipocytes, PGC-1 α has been shown to control SIRT3 expression. PGC-1 α increases the activity of the SIRT3 promoter through an interaction with estrogen-related receptor (ERR) α and subsequent binding to an ERR binding element (299, 300) (Fig. 6). Moreover, mitochondrial biogenesis induced by overexpressing PGC-1 α in C2C12 myotubes has been found to be mediated through SIRT3 (299). Consistent with the idea that PGC-1 α regulates SIRT3 gene expression in skeletal muscle, SIRT3 mRNA and protein levels were found to be reduced by 50% in resting mouse muscle from PGC-1 α knockout mice (286). Furthermore, there are clear indications that the described signaling axis between AMPK and PGC-1 α includes SIRT3. Thus, overexpression of AMPK in primary mouse hepatocytes results in elevated levels of SIRT3 mRNA (301), and the exercise training-induced increase in SIRT3 protein abundance in mouse skeletal muscle is completely dependent on AMPK (286).

Knockout of the upstream AMPK kinase LKB1 in skeletal muscle affects the abundance of mitochondrial enzymes (99). In cultured liver cells, SIRT1 has been shown to deacetylate LKB1, resulting in its translocation to the cytosol where it activates AMPK (302). Moreover, the ability of AICAR to activate AMPK was blunted in primary myoblasts and hepatocytes from SIRT1 knockout mice (303). In contrast to these observations, AMPK activation measured by phosphorylation is not affected in response to an acute bout of exercise in skeletal muscle-specific SIRT1-KO mice (304). It is possible that secondary adaptations to muscle-specific deletion of SIRT1 could mask the true effects of SIRT1 to potentiate AMPK activation, but further studies are required to verify this potential feedback mechanism in skeletal muscle. In various human cultured cell lines, binding sites for FOXO3 have been found in the promoter region of LKB1, and AMPK activation induced by energy stress has been shown to increase LKB1 mRNA levels (305, 306). Whether a similar feedback mechanism is present in skeletal muscle has not been investigated.

Sirtuins break down NAD to nicotinamide (NAM) and 2-O-acetyl-ADPR. NAM negatively feeds back to inhibit these enzymes (307) and must be converted back to NAD to maintain sirtuin enzymatic activity. The rate-limiting enzyme in the main NAD salvage pathway in mammalian cells is nicotinamide phosphoribosyltransferase (NAMPT), accounting for most of the NAD generated in the cell, (308, 309). NAMPT converts NAM to nicotinamide mononucleotide, which is converted to NAD by nicotinamide mononucleotide adenylyltransferase, of which 3 isoforms exist (310). Thus, NAMPT may be important for the ability of cells to transmit signals associated with limited nutrient availability. In skeletal muscle, AMPK is important for maintaining NAMPT protein levels, and the AICAR-induced increase in NAMPT protein abundance is completely dependent on the AMPK α 2 subunit (311). Furthermore, circadian oscillation in NAMPT mRNA expression is markedly blunted

in skeletal muscle from whole-body AMPK α 1- and AMPK α 2-KO mice (312), and expression of core clock components is dysregulated in response to AICAR in skeletal muscle from AMPK γ 3-KO mice (313). These findings indicate that circadian regulation of metabolism integrates nucleotide sensing and regulation of NAD salvage capacity in skeletal muscle and implies that AMPK may directly regulate NAMPT expression in this tissue. Mammalian NAMPT gene expression has been found to be under the control of the clock machinery (CLOCK/BMAL1) (314, 315), and AMPK has been shown to directly phosphorylate CRYPTOCHROME at Ser71, which, together with PERIOD, forms an inhibitory complex of CLOCK/BMAL1 (316). Phosphorylation of CRYPTOCHROME by AMPK results in its ubiquitination and degradation, allowing clock-controlled genes (e.g., NAMPT) to be transcribed (Fig. 6) (316). It is assumed that the regulation of the circadian clock in muscle cells is regulated as described above. However, this needs further investigation.

AMPK appears to have profound effects on the ability of cells to adapt to conditions of low energy availability. Signaling through PGC-1 α , to mediate changes that enable proper mitochondrial adaptation, requires fine-tuned feedback loops involving consumption of NAD as a substrate in SIRT1- and SIRT3-dependent reactions. Activation of AMPK not only increases NAD levels but also maintains sufficient NAD regeneration capacity by controlling abundance of NAMPT. Although NAD regeneration capacity in young, healthy skeletal muscle may not be limiting for mitochondrial function and muscle health (317), levels of NAD in skeletal muscle decline with age (318, 319) and are suppressed in mitochondrial myopathies (320). Thus, targeting AMPK or NAD salvage systems may prove useful for counteracting age-related reductions in muscle function.

AMPK IN EXERCISE RECOVERY

After exercise, AMPK activity and downstream signaling decrease to levels observed in resting muscle typically within 3–7 h (131, 209, 222, 321, 322). The exact mechanism regulating AMPK activity in skeletal muscle several hours into recovery from a single bout of exercise is incompletely understood. However, it does not seem to involve changes in skeletal muscle adenosine nucleotide concentrations because these have been shown to return to pre-exercise levels 10 min after exhaustive exercise (323). Lower glycogen levels may account for some of the elevated AMPK activity observed in skeletal muscle after exercise but probably not all because AMPK activity reverses to resting levels before muscle glycogen levels have been fully resynthesized (131). Other factors that might explain elevated AMPK activity in exercise recovery include changes in heterotrimer localization and/or conformation that render the complex less sensitive for upstream phosphatases. In addition, activation of AMPK in nonmuscle cells situated or recruited to the muscle tissue could explain the slow reversal of AMPK activity observed in skeletal muscle after exercise.

AMPK-dependent regulation of muscle gene expression after exercise

In recovery from a single bout of exercise, the transcriptional activity of genes coding for key metabolic proteins increases, which is followed by an increase in corresponding mRNA (324). The repeated activation of these transient stimuli by regular exercise may result in the beneficial adaptations of exercise training (325). The effect of exercise-induced AMPK activation on the gene expression response appears to involve a number of cellular transcription factors and coactivators, including PGC-1 α (326). AMPK activation has been shown to increase gene expression of the 2 key glucose-handling intermediates, GLUT4 and HKII, by regulating histone deacetylase 5 (HDAC5) and cAMP response binding-element protein, respectively (327, 328). As such, a single injection of AICAR that elevates AMPK activity in skeletal muscle has been shown to increase gene expression of PGC-1 α , HKII, and GLUT4 in WT mice but not in skeletal muscle of AMPK α 2-deficient mice (329, 330).

Acute exercise induces a similar gene response at the level of GLUT4, PGC-1 α , and HKII in WT and AMPK α 2-deficient mice, suggesting that exercise-induced gene expression is orchestrated independently of AMPK (329, 330). These findings indicate that repetitive AMPK activation by pharmacological means is sufficient to increase gene expression, whereas signaling parameters other than AMPK may be essential for exercise-induced gene expression. An alternative scenario could be that redundant signaling, through the remaining AMPK α 1 complexes in the transgenic model used, may be sufficient to induce these metabolic changes. Recently, exercise- and training-induced gene expression was investigated in AMPK α mdKO mice (331). In this study, acute exercise-induced gene response at the level of COX-I, GLUT4, and VEGF increased in an AMPK-dependent manner. On the other hand, the acute exercise-induced gene response for HKII, CD36, and FATP1 increased similarly in skeletal muscle from AMPK α mdKO and WT littermates. Collectively, these observations suggest that AMPK is essential for the acute exercise-induced up-regulation of some but not all genes encoding key metabolic proteins. However, although AMPK seems to be essential for up-regulation of gene expression for some key metabolic proteins in response to acute exercise, this seems not to be the case for training-induced adaptations.

AMPK and postexercise muscle insulin sensitivity

Skeletal muscle glucose uptake increases markedly during exercise in an intensity-dependent manner and reverses to resting levels within a few hours of recovery (332). However, the ability of insulin to stimulate muscle glucose uptake is markedly enhanced in the period after exercise (333–336). The increase in insulin sensitivity of prior exercised skeletal muscle can persist for up to 48 h, facilitating muscle glycogen replenishment in recovery from exercise (336, 337). Insulin and exercise increase muscle glucose uptake by

increasing GLUT4 translocation to the plasma membrane (338–340), and it has therefore been proposed that prior exercise increases muscle insulin sensitivity by potentiating GLUT4 translocation in response to insulin (341). The underlying mechanism responsible for enhancing postexercise insulin sensitivity seems to be induced locally within the prior exercised muscle (342) and appears to be independent of elevated activity of the proximal insulin signaling cascade (343–347). Insulin and exercise signaling converge at the level of TBC1D4 (348–351), a Rab-GTPase involved in the regulation of GLUT4 translocation. It has been proposed that phosphorylation of TBC1D4 may provide a mechanism for exercise-induced improvements in muscle insulin sensitivity. Thus, after insulin stimulation, phosphorylation of TBC1D4 increases to higher levels in prior exercised muscle compared with rested muscle, concomitant with enhanced muscle insulin sensitivity (348, 352). A direct link between AMPK and TBC1D4 becomes evident by observations demonstrating that pharmacological activation of AMPK by AICAR in incubated muscle leads to increased phosphorylation of TBC1D4 (351), an effect that is blunted in AMPK-deficient mice (353). Furthermore, it has been demonstrated that recombinant AMPK increases TBC1D4 phosphorylation in cell-free assays (353) and that phosphorylation of Ser704 on TBC1D4 increases in response to AICAR and contraction in an AMPK α 2-dependent manner.

Supporting the role of AMPK in regulating muscle insulin sensitivity, prior pharmacological AMPK activation by AICAR enhances insulin sensitivity in rat skeletal muscle (354). This effect is absent in AMPK α mdKO mice but is seen in concert with increased site-specific phosphorylation of TBC1D4 in WT mice (129). The latter study furthermore demonstrated that prior AMPK activation by AICAR increased AMPK activity of the α 2 β 2 γ 3 complex and that the insulin-sensitizing effect of prior AICAR stimulation was dependent on the AMPK γ 3 subunit (129). In parallel, it was recently demonstrated that prior *in situ* contraction and treadmill exercise increase muscle and whole-body insulin sensitivity in WT mice but not in AMPK α mdKO mice (355). Furthermore, these findings were associated with enhanced insulin-stimulated phosphorylation of TBC1D4 at Thr642 and Ser704 in WT mice. Taken together, these findings suggest that AMPK activation is necessary to enhance muscle insulin sensitivity after acute exercise and muscle contraction (Fig. 7). However, exercise activates a broad range of signaling molecules that may play an important role in improving muscle insulin sensitivity after exercise. Yang and Holman (356) reported that insulin stimulation increases GLUT4 exocytosis, whereas AMPK activation induced by oxidative stress inhibits GLUT4 endocytosis in cardiomyocytes. In human skeletal muscle, heterotrimer-specific activation of AMPK in recovery from exercise has recently been demonstrated (209), implying that enhanced AMPK activation in recovery from exercise may improve insulin sensitivity by delaying GLUT4 endocytosis. More research is needed

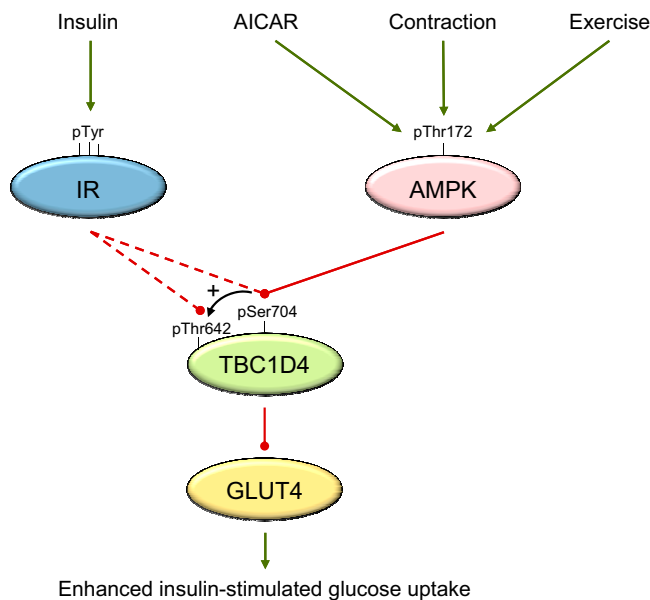


Figure 7. Involvement of AMPK in regulating improvements in muscle insulin sensitivity after exercise. AMPK phosphorylates TBC1D4 at Ser704 in response to AICAR, contraction, and exercise. This likely primes TBC1D4 for insulin signaling at Ser704 and Thr642, which enhances translocation of GLUT4 to the cell surface membrane, leading to improvements in insulin-stimulated muscle glucose uptake. Green lines, promotive; red lines, inhibitory; black line, priming; dashed lines, indirect.

to address the heterotrimeric regulation of AMPK in recovery from exercise and to transform these observations to physiologic endpoints, such as insulin sensitivity in human skeletal muscle.

AMPK in the regulation of the intracellular fate of glucose in skeletal muscle

The breakdown of muscle glycogen during exercise increases in an intensity-dependent manner (357). Upon recovery from exercise, glycogen synthesis is of high metabolic priority in expense of increased lipid oxidation (358, 359). Glycogen synthase (GS), the rate-limiting enzyme in glycogen synthesis, catalyzes the incorporation of UDP glucose into branching glycogen polymers (360). The activity of GS is regulated by the allosteric activator glucose-6-phosphate and covalently by phosphorylation at multiple inhibiting residues (361, 362). AMPK has been identified as a site 2 (Ser7) GS kinase *in vitro* (363), and in rodent skeletal muscle phosphorylation of GS increases in response to AMPK activation by AICAR in an AMPK α 2-dependent manner (10). This event decreases GS activity, however, because AICAR increases glucose uptake the subsequent intracellular rise of glucose-6-phosphate has been shown to overrule GS inhibition *via* covalent AMPK-induced phosphorylation (364).

The AMPK β subunit contains a GBD that allows the kinase to bind to glycogen (56, 106), suggesting that AMPK, in addition to its energy sensing role, acts as a glycogen sensor (365). According to this model, a pool of AMPK is responsive to changes in muscle glycogen

content. Thus, during high glycogen levels, activity of this AMPK pool increases, consequently decreasing GS activity by phosphorylation of site two. Conversely, when glycogen levels are low, activity of this AMPK pool decreases, which increases GS activity due to decreased phosphorylation of site two (365). Although this scenario is based on a subfraction of AMPK, it contradicts the general observations of glycogen as a negative regulator of AMPK activity (106, 107, 109). Furthermore, recent observations indicate that carbohydrate binding to the GBD reduces AMPK activation by increasing accessibility for α -Thr172 dephosphorylation (366). The subcellular localization and regulation of different AMPK complexes according to muscle glycogen content may explain some of these discrepancies. More recently, a novel phosphorylation site has been identified (β -Thr148) in the GBD (111). Thus, in HEK293T cells autophosphorylation of β -Thr148 prevents AMPK binding to glycogen. Low glycogen levels induced by glucose deprivation or AMPK activation by AICAR increase phosphorylation of β -Thr148, suggesting that the binding affinity of AMPK to glycogen is regulated in response to glycogen content and AMPK activation. The physiologic relevance of these findings needs to be tested in mature skeletal muscle where glycogen serves as an essential source of energy.

Mutation of the AMPK γ 3 subunit, leading to chronic activation, promotes excessive glycogen storage in mouse (125), pig (124), and human skeletal muscle (123). In line with this, genetic deletion of AMPK γ 3 or overexpression of an AMPK α 2 kinase dead construct in skeletal muscle markedly reduces glycogen resynthesis after exercise (125, 367). Based on these observations, it would be interesting to investigate whether a reduction in glycogen resynthesis is caused by the lack of an AMPK-dependent increase in muscle insulin sensitivity for glucose uptake after exercise.

Exercise has a major impact on substrate utilization in the postexercise period. Thus, fat oxidation is enhanced, whereas glucose oxidation is suppressed in favor of muscle glycogen resynthesis (359). Several lines of evidence suggest that AMPK is involved in the regulation of glucose oxidation in skeletal muscle. As such, AMPK activation by AICAR in rat skeletal muscle has been shown to enhance glucose oxidation in concert with elevated activity of pyruvate dehydrogenase (PDH), a protein regulating the entry of glucose into the tricarboxylic acid cycle for oxidation (368). In contrast, PDH activity is increased in glycolytic muscle from AMPK α 2-KO mice compared with WT littermates, albeit muscle PDH protein abundance is reduced by $\sim 20\%$ in AMPK α 2-KO mice (369). In line with this observation, glucose oxidation in recovery from *in vitro* muscle contraction is increased in AMPK γ 3-KO mice compared with WT littermates, indicating that the presence of AMPK is required for suppression of glucose oxidation, which may become highly relevant in exercise recovery (369). Pyruvate dehydrogenase kinase 4 inhibits PDH activity by phosphorylation, and a study in primary cardiomyocytes has shown that pharmacological activation of AMPK by AICAR increases pyruvate dehydrogenase kinase 4 mRNA expression. Collectively, these observations indicate that AMPK is involved in the regulation of glucose oxidation in skeletal muscle, but whether

AMPK-dependent regulation of glucose oxidation is important for fuel selection during exercise recovery has received surprisingly little attention. Supporting a role of AMPK in substrate selection, Fritzen *et al.* (370) observed enhanced glucose oxidation at a whole-body level in recovery from exercise in whole-body AMPK α 2-KO mice. However, enhanced glucose oxidation early in recovery from exercise was not observed in mice with a muscle-specific deletion of AMPK α 1/ α 2 (AMPK α mdKO) (73). Collectively, studies in a few AMPK-deficient mouse models indicate that skeletal muscle AMPK is involved in regulating the cellular fate of glucose in recovery from exercise at the level of insulin sensitivity (AMPK mdKO), glycogen synthesis (AMPK mdKO, AMPK kinase dead, and AMPK γ 3 KO), and glucose utilization (AMPK α 2 KO). Future studies should therefore aim to validate these findings in other AMPK-deficient models and integrate these aspects of AMPK regulation in recovery from exercise, ultimately transferring these observations to human settings.

AMPK IN THE REGULATION OF TRAINING-INDUCED ADAPTATIONS

Preceding the first studies linking AMPK to the regulation of exercise training-induced adaptations, it was shown that prolonged feeding of rats with β -guanidinopropionic acid (β -GPA; creatine analog) increased mitochondrial biogenesis as well as GLUT4 and HKII expression (371, 372). Bergeron *et al.* (13) extended these findings by showing that AMPK was activated in rats fed β -GPA, and the following year mitochondrial and metabolic adaptations to chronic β -GPA injections were shown to depend on AMPK α 2 because adaptations were abrogated in mice overexpressing a kinase-dead AMPK α 2 construct (15). Using AICAR, the role of AMPK in the adaptive response was strengthened. Only WT and not AMPK α 2-KO mice showed increased content of metabolic and mitochondrial proteins after 4 wk of subcutaneous AICAR injections (284). Collectively, these observations suggest that repeated artificial AMPK activation mimics some of the beneficial effects induced by regular exercise training (31). Similar to observations in the AMPK α -deficient mouse model, mice lacking PGC-1 α do not increase skeletal muscle content of metabolic and mitochondrial proteins in response to chronic AICAR injections (373), indicating that AMPK may act through PGC-1 α to induce exercise-like adaptations in skeletal muscle.

It is well established that endurance exercise training results in a more oxidative muscle phenotype, expressing increased content of metabolic and mitochondrial proteins (374, 375). Because AMPK is activated during exercise, it has been proposed that AMPK mediates the positive effects of exercise training on the skeletal muscle phenotype. However, solid evidence confirming a role of AMPK for *in vivo* exercise training-induced mitochondrial and metabolic adaptations is lacking. Indeed, in most studies (284, 285, 311, 331, 376, 377) but not all (286), exercise training-induced adaptations are not abrogated in mutant mice that lack functional AMPK α 2 or both AMPK α 1 and

AMPK α 2 in skeletal muscle. This was also demonstrated in a study comparing subgroups of AMPK α mdKO mice and WT littermates that had performed an equal exercise training volume during a 4-wk intervention period. Muscle content analyses revealed that only 1 out of 13 proteins measured increased more in WT than in AMPK α mdKO mice with exercise training (331).

A number of alternative mechanisms proposed to mediate exercise training-induced adaptations include the Ca²⁺/calmodulin-dependent protein kinase (CaMK) (378), p38 MAPK (379), and ROS (380), among others (381). These mechanisms could compensate for the loss of AMPK α activity in the AMPK α mdKO mouse model with respect to skeletal muscle adaptations induced by exercise and are discussed in the following sections.

The expression of GLUT4 is in part regulated by binding of the transcription factor myocyte enhancer factor 2 (MEF2) to the GLUT4 promoter (382, 383). In an unstimulated state, the activity of both MEF2 and the GLUT4 promoter is suppressed by binding of HDAC5 (327, 384, 385). However, phosphorylation of HDAC5 on Ser259 and Ser498 promotes the binding of 14-3-3 protein and the subsequent nuclear export of the complex (385, 386). Both of the HDAC5 serine residues are targeted by AMPK (327), and exercise enhances the binding affinity of MEF2 to the GLUT4 promoter (387). This delineates a pathway by which exercise stimulates the expression of GLUT4 through AMPK.

Nevertheless, acute exercise and exercise training have evoked similar adaptations in skeletal muscle GLUT4 gene expression and protein content, respectively, in WT and AMPK α 2-deficient mice, in addition to other metabolic and mitochondrial markers (284, 329, 330). This prompted McGee *et al.* (388) to investigate whether HDAC5 was involved in a compensatory mechanism to maintain the adaptive response to exercise in skeletal muscle from AMPK α -deficient mice (29). They found that exercise-induced HDAC5 phosphorylation of Ser259 and Ser498 was maintained in AMPK α 2 kinase dead mice. Another HDAC5 kinase, protein kinase D (PKD), displayed elevated phosphorylation of Ser498 in skeletal muscle from AMPK α 2 kinase dead mice compared with WT littermates immediately after exercise. Using myoblasts stably expressing constitutive active or inactive forms of PKD, they further showed that this was sufficient to regulate the expression of a range of metabolic and mitochondrial genes. It is therefore possible that exercise training adaptations in AMPK α 2-KO and AMPK α mdKO mice may be elicited by compensatory PKD regulation of HDAC5 phosphorylation. To verify this *in vivo*, a concomitant deletion of AMPK α 2 and PKD would be necessary. Since the results of McGee *et al.* (331) were published, it has been shown that the acute exercise-induced increase in gene expression (including GLUT4) was ablated in skeletal muscle from AMPK α mdKO mice, albeit exercise training-induced adaptations were similar to that of WT mice. In keeping with the intact adaptive responses in AMPK α 2-deficient mice (284, 329, 330), results from the AMPK α mdKO model indicate that compensatory elevated AMPK α 1 levels in skeletal muscle of AMPK α 2-deficient mice may compensate for the lack of AMPK α 2 in

acute exercise-induced but not exercise training-induced adaptations. Therefore, it may be necessary to delete AMPK α 1, AMPK α 2, and PKD concomitantly in mouse skeletal muscle to prevent exercise training adaptations of metabolic and mitochondrial proteins.

Another mechanism that may induce compensatory adaptations in AMPK α -deficient mice with exercise and exercise training includes the nuclear hormone receptor 4A (NR4A) subgroup. This subgroup is comprised of 3 orphan nuclear receptors: Nur77 (neuron-derived clone 77; NR4A1), Nurr1 (nuclear receptor related 1; NR4A2), and Nor-1 (neuron-derived orphan receptor 1; NR4A3), and findings have shown that at least 2 of the 3 receptors exert metabolic control in many tissues, including skeletal muscle (389). Interestingly, exercise potentially stimulates the gene expression of the NR4A subgroup in human skeletal muscle (325, 390). The rise in catecholamine concentration during exercise (391) may confer at least some of the mechanisms behind the induction of the NR4A genes because *in vivo* β -adrenergic stimulation increases their expression (392–394). Nuclear receptor signaling provides an alternative to the AMPK–PGC-1 α pathway in regulating the expression of genes related to metabolic control. Findings suggest that this alternative β -adrenergic–NR4A signaling pathway may be up-regulated in skeletal muscle from AMPK α mdKO mice during exercise. Thus, in response to an acute bout of treadmill exercise at the same relative intensity, blood epinephrine concentration increased 131% in AMPK α mdKO mice, compared with a 27% increase in the WT littermates (interaction $P = 0.07$) (73). Therefore, it seems likely that higher activation of β -adrenergic signaling in AMPK α mdKO mice compared with WT mice could induce compensatory mitochondrial and metabolic adaptations with exercise training in skeletal muscles of the AMPK α mdKO mice. Furthermore, in the nonexercised state, a microarray analysis of extensor digitorum longus muscle from AMPK α mdKO and WT mice shows that Nur77 signaling is up-regulated in the mdKO mouse model (73), perhaps in an attempt to maintain the basal expression of metabolic and mitochondrial proteins.

Currently, catecholamine concentrations have not been reported in studies of AMPK-deficient mice other than the aforementioned study. More studies are needed to determine the exact role of β -adrenergic signaling and NR4A nuclear receptors in adaptations to exercise training.

AMPK AND EXERCISE PERFORMANCE

Given that AMPK in skeletal muscle is implicated in many of the pathways otherwise involved in muscle contraction and muscle adaptation to exercise, the fact that muscle AMPK is physiologically activated by exercise is not surprising. One question that has been extensively studied over the years is how important AMPK is for muscle force development and exercise performance. Most mouse models with muscle deletions for various subunits of AMPK have therefore been studied in the setting of exercise. All models generated with skeletal muscle loss of function or KO of the AMPK α 2 catalytic subunit or its

preferred regulatory partner β 2 or the upstream AMPK kinase (LKB1) display a phenotype of decreased *in vivo* exercise performance (time to exhaustion and maximal running speed) when forced to run on a treadmill, whereas performance (accumulative distance) during voluntary exercise (wheel running) has been reported to be similar or reduced compared with WT littermates (**Table 2**). Often muscle from these models shows decreased glycogen content but also indices of lower oxidative capacity, mitochondria content/function, and fatty acid oxidation during exercise or contraction (72–74, 198, 331, 395, 396). These findings suggest that AMPK is important during development for obtaining a normal metabolic capacity and thus likely also a normal exercise capacity, at least in rodents.

Although most of these models exhibit decreased exercise capacity, the intracellular phenotype induced by lack of AMPK function varies considerably between different models. Although this may reflect potential compensatory effects in different mouse strains (72, 74), it may also hint at hitherto unknown functions of AMPK. Although induction of mRNA expression of key metabolic genes by a single exercise bout is compromised in AMPK α -deficient muscle, this does not translate to impaired adaptations at the protein level in response to exercise training (331). Thus, AMPK signaling seems dispensable for normal adaptation to exercise training for a range of metabolic parameters (284, 311, 331, 376, 377, 397, 398) as well as exercise training-induced improvements in running performance (331, 398). The interpretation of many of these observations with exercise training is impeded by the overall decreased exercise capacity in the transgenic models [as illustrated by Fentz *et al.* (331)], and thus no solid genetic evidence has been put forward supporting that AMPK in the adult muscle is a true mediator of adaptations to endurance exercise training.

Nonetheless, a growing and illegal market for substances activating AMPK and the discovery of AICAR in trash bins around hotels of the cycling teams at the 2009 Tour de France led the World Anti-Doping Agency (WADA) to add any substances activating AMPK to the prohibited list (399). Besides these behavioral observations, the arguments from WADA were exclusively based on studies performed on animal models indicating ergogenic effects of AICAR treatment (400). In clinical trials, AICAR has been tested for minimizing damage after cardiac reperfusion (401), and the metabolic consequences of AICAR infusion have been studied in a few human experiments, apparently without major effects on AMPK activity in skeletal muscle (144, 402, 403). Therefore, the action of WADA is based on weak scientific evidence from animal models and virtually none from human experimentation. However, because the pharmaceutical industry continues to search for novel specific AMPK agonists combating diseases like diabetes and cancer, the availability of such substances is likely going to increase, and the (mis)use of such in the hope for ergogenic purposes will likely follow.

From the abovementioned possible roles of AMPK in muscle function and metabolism, one might predict that AMPK activation promotes exercise endurance. It might

TABLE 2. Exercise capacity in various transgenic AMPK mouse models

Mouse model name	Target protein	Type of manipulation	Tissue specificity	Background strain	Running capacity ^a		Reference
					Forced	Voluntary	
α1KO	AMPKα1	Loss of function	Whole body	C57Bl/6J-Sv129	=	Unknown	370
α1 DN, also results in knockdown of AMPKα2	AMPKα1	Loss of function	Skeletal muscle (HSA)	Unknown	↓	Unknown	488, 489
α2KO	AMPKα2	Loss of function	Whole body	C57Bl/6J-Sv129	↓	=	198, 284, 370, 490
α2 kinase dead	AMPKα2	Loss of function	Heart + skeletal muscle (MCK)	C57Bl/6J	↓	↓; =	196, 311, 376, 377, 395–397, 491, 492
α2iTG (inactivating mutation in α2)	AMPKα2	Loss of function	Heart + skeletal muscle (MCK)	FVB	↓	=	285, 493
LKB1 DN	LKB1	Loss of function	Skeletal muscle (HSA)	Unknown	↓	Unknown	489
skmLKB1-KO	LKB1	Loss of function	Skeletal muscle (Myf6)	Unknown	↓	↓	398
LKB1-KO	LKB1	Loss of function	Heart + skeletal muscle (MCK)	FVB	↓	↓	99, 198
β1M-KO	AMPKβ1	Loss of function	Heart + skeletal muscle (MCK)	C57Bl/6J	=	Unknown	74
β2M-KO	AMPKβ2	Loss of function	Heart + skeletal muscle (MCK)	C57Bl/6J	↓	Unknown	74
β1β2M-KO	AMPKβ1 + AMPKβ2	Loss of function	Heart + skeletal muscle (MCK)	C57Bl/6J	↓	↓	74
α mdKO	AMPKα1 + AMPKα2	Loss of function	Skeletal muscle (HSA)	C57Bl/6J-Sv129	↓	↓	72, 73, 331
HA-R70Qy1 transgenic	AMPKγ1	Gain of function	Skeletal muscle (HSA)	FVB	↑	Unknown	126

DN, dominant negative; HSA, human skeletal actin; MCK, muscle creatine kinase; Myf6, myogenic factor 6. ^aThe symbols ↓, and ↑ denote whether the respective genetic mutation does not affect, decreases, or increases the running exercise capacity compared with WT littermates, respectively.

also promote fuel restoration in recovery from exercise, and it may be important in recovery from skeletal muscle injury. On the other hand, AMPK activation might impose a challenge for myofibrillar protein synthesis and hypertrophic responses to certain exercise training programs.

In the late 1960s, exercise physiologists began to illuminate the importance of pre-exercise muscle glycogen content and the importance of glucose/glycogen sparing (elevated fatty acid oxidation) during exercise as primary factors determining performance in endurance type of exercises. Thus, a muscle phenotype with high vascularization, high mitochondria capacity, potentiated/optimal oxidative handling of fatty acids, elevated glycogen content, and perhaps a fiber type distribution toward MyHC I and IIA were predicted to benefit exercise endurance. The question is then whether AMPK activation promotes such muscle phenotype.

Family members carrying a PRKAG3-activating mutation (AMPKγ3-R225W) provide the only human evidence so far that chronic AMPK activation leads to elevated skeletal muscle glycogen levels, decreased intramuscular triglyceride, and indices of increased oxidative capacity (primary muscle cells) (123, 404). It is not known whether this mutation or phenotype improves exercise endurance in humans. However, electromyography indices of rate of fatigue during a 60-s test of isometric contraction suggest that skeletal muscle of the carriers is

less prone to fatigue (404). Studies in pigs may further define the role of the PRKAG3-activating mutation given similar physiology to humans. The activating PRKAG3 mutation (R200Q) was identified in pigs (124) and leads to muscle glycogen accumulation as well as elevated muscle glycolytic and oxidative capacity (405). So far, only short-term treadmill exercise tests (~6–8 min) have been applied to the pig model, and no improvement in performance has been reported (406).

Although rodent models often display minor genetic variation compared with humans, the value of these models in relation to exercise performance and metabolism may be questioned. The endurance-like tests often applied are less physiologically relevant for the natural behavior of rodents, and indeed their metabolism is different in various aspects. Murine skeletal muscle has only 1/10 the amount of glycogen compared with human skeletal muscle, indicating a highly different whole-body glucose metabolism between and within organs. Furthermore, the role of muscle glycogen for exercise performance during treadmill exercise of mouse is still debated (407, 408). Nevertheless, introducing the activating PRKAG3 mutation in transgenic mice leads to a phenotype with elevated muscle glycogen levels, high oxidative capacity, and enhanced contractile performance in isolated glycolytic muscle *in vitro* (128, 409, 410). However, any performance improvement during swimming (410) or other

forms of *in vivo* exercise has not been reported so far. Activating mutations in PRKAG1 (R70Q or H151R) have also been introduced in skeletal (predominantly glycolytic) muscle of mice (126, 127). These models also display glycogen accumulation, and at least 1 model was reported to have improved treadmill exercise endurance by a factor of 2 (126). Thus, the phenotype obtained by expression of an activating AMPK mutation in skeletal muscle would be predicted to promote *in vivo* exercise endurance, but the evidence to support such consequence is scarce.

The question rises as to whether AMPK activation by genetic or pharmacologic means alone or in combination with exercise may lead to improved exercise performance. If mice carrying the activating PRKAG3 mutation perform an acute bout of swimming exercise, the gene response is greater in the carriers compared with the control animals (411). However, whether this translates to superior functional adaptations at the protein/organelle level and improved exercise performance remains unknown. In fact, in pigs, differences and similarities in enzyme activities (*e.g.*, HK, CS, LDH) observed at baseline between carriers and noncarriers of the activating PRKAG3 mutation are maintained after exercise training, suggesting that adaptations to exercise training are not potentiated by chronic elevated AMPK activity (406, 412). Long-term exercise training studies with detailed phenotypic evaluation might provide insights into any synergistic adaptive response to exercise training when performed in the context of genetically elevated basal AMPK activity. Several pharmacological studies using various indirect and direct AMPK agonists support the notion that AMPK activation is sufficient to drive muscle adaptation toward high glycogen content, glycolytic and oxidative capacity, and mitochondrial biogenesis. Three studies have addressed whether this also associates with improved exercise performance (400, 413, 414). The study by Narkar *et al.* (400) demonstrated that 4 wk of AICAR treatment induced an increased exercise endurance phenotype during treadmill running. However, this study did not apply AMPK-deficient animal models to address whether AMPK in skeletal muscle was indeed needed for this adaption. Marcinko *et al.* (413) treated insulin-resistant mice fed HFD with R419 (Fig. 4) for 6 wk (mitochondrial complex 1 inhibitor) and observed an improvement in treadmill exercise capacity (413). This effect was dependent on AMPK β 1 and β 2 expression in skeletal muscle. Similarly, Um *et al.* (414) found that resveratrol increased treadmill exercise capacity in HFD-fed WT mice but not in whole-body AMPK α 2- or AMPK α 1-KO mice. Thus, in a likely exercise-intolerant model (due to HFD), pharmacological activation of AMPK in skeletal muscle improves exercise performance. From the perspective of physical activity in health, this is a promising observation. However, the available data do not offer solid scientific proof that AMPK activation (genetic or pharmacological) associates with improved exercise performance in metabolic healthy organisms or in well-trained athletes. On the other hand, data to disprove this claim are also not strong. Although the role of muscle AMPK is unclear, exercise training in combination with metformin treatment of obese prediabetic subjects (415) and insulin-resistant subjects (416)

as well as exercise training in combination with resveratrol treatment of elderly persons (417) have been reported to induce similar or even less adaptive responses on factors such as Vo_{2peak} . Resveratrol treatment of animals in combination with exercise training has revealed improved adaptations compared with exercise alone (418, 419). This provides another example of the limited knowledge transfer between species. Based on the above-mentioned findings and the emergence of direct AMPK agonists that target β 2-containing complexes in skeletal muscle (PF-739 and MK-8722), future studies should aim at determining the potential for such activators to improve exercise capacity in both healthy and diseased models.

By various mechanisms, AMPK has been proposed to inhibit the regulation of mTORC1 signaling and thus to inhibit skeletal muscle strength and hypertrophy. In accordance with this hypothesis, mTORC1 activation in muscle by strength training is lost or compromised when high-intensity sprint cycle exercise is performed before or immediately after strength training (420, 421), whereas moderate-intensity endurance exercise does not produce this effect (422, 423). Interestingly, AICAR-induced AMPK activation dismisses muscle mTOR signaling in response to high frequent electrical muscle stimulation in rats (424). Furthermore, in a model of muscle overload, the AMPK α 1 complex is activated (425). In accordance with this finding, skeletal muscle growth after overload-induced muscle hypertrophy is enhanced in mice lacking the AMPK α 1 isoform, which relates to elevated mTOR signaling (426). In agreement with this, knockout of the primary AMPK α 2 upstream kinase LKB1 (425) and the AMPK γ 3 isoform (427) does not affect overload-induced muscle hypertrophy. The idea that AMPK α 1 is important in the interference between endurance and strength training is not straightforward, and studies of concurrent training should be performed with more detailed evaluation of AMPK heterotrimer complex activities to bring further insight to the AMPK–mTOR interaction. Although the rodent studies are in strong support for such interaction, it should not be forgotten that exercise regulates a range of other signals (428), some of which may act in such settings of interference (*e.g.*, PKA, MAPK, SIRT6).

AMPK: A KEY PLAYER IN SKELETAL MUSCLE PLASTICITY, REGENERATION, AND REPAIR

Metabolic and nonmetabolic functions of AMPK, including its role in plasticity of skeletal muscle, have been investigated in tissues of various species (49). Skeletal muscle displays a remarkable ability to adapt its phenotype in terms of size, composition, and metabolic properties in response to environmental stimuli. Skeletal muscle fibers rapidly adapt to drastic changes in energy demands during exercise through fine-tuning of the balance between catabolic and anabolic processes (49).

Chronic *in vivo* activation of AMPK by AICAR administration inhibits overload-induced skeletal muscle hypertrophy (429), suggesting a crucial role of AMPK for the control of myofiber size. The α 1 catalytic subunit of AMPK is activated by overload-induced hypertrophy (a

supraphysiological muscle loading) (425, 426, 430). Therefore, even if AMPK γ 3 seems to be dispensable for skeletal muscle hypertrophy induced by functional overload (427) and if the cross-section area of myofibers is reduced in skeletal muscle depleted of AMPK β 1 and β 2 (146, 431), it is not surprising that knocking out AMPK α 1 results in greater overload-induced hypertrophy (426). Of note, the degree of muscle cell hypertrophy in response to activation of the protein synthesis pathway is lower in AMPK α 2-deficient muscle cells (430). Furthermore, *in vitro* studies have demonstrated that AMPK negatively regulates hypertrophy of muscle cells (426, 430, 432, 433). Finally, AMPK α mdKO mice possess fibers with larger cross-sectional area as compared with WT mice, further supporting the notion that AMPK α negatively influences muscle growth (432). Because AMPK decreases mTORC1 activity *in vitro* through phosphorylation of TSC2 at Thr1345 (226) or raptor at Ser792 (229), these responses appear to be related to the deactivation of the signals involved in the protein synthesis pathway mTOR/p70S6K. Overload-induced hypertrophy results in the activation of molecular brakes that may feed back to attenuate the rate of skeletal muscle growth (434). Indeed, Hamilton *et al.* (434) have demonstrated that overload-induced stress results in the activation of AMPK α 1 and phosphorylation of TSC2 Thr1345, which limits activation of mTORC1. Finally, knockout of AMPK induces myotube hypertrophy and up-regulation of p70S6K phosphorylation in primary cultured myoblasts under nonstimulated conditions (432). Taken together, these findings show that AMPK may act as a negative regulator of skeletal muscle hypertrophy through down-regulation of protein synthesis pathways.

Exercise training results in a more oxidative muscle phenotype that is likely mediated by an oxidative fiber type switch. Whether AMPK triggers an oxidative slow twitch MyHC phenotype has not been demonstrated convincingly. Thus, one study suggests that skeletal muscle of the PRKAG1-activating mutation model has a higher proportion of IIA fibers (285). Some observations suggest a MyHC IIB toward a IIA conversion in skeletal muscle of pigs carrying the PRKAG3 activating mutation (412, 435). However, such differences have not been reported in the transgenic mouse model (128) or in human carriers (123) of the PRKAG3 activating mutation. Reports on fiber type composition in muscle of other AMPK transgenic mouse models are also somewhat limited. One study suggests that lack of functional AMPK α 2 leads to less MyHC type IIA/X-positive fibers (285), which is in part in line with observations of decreased MyHC IIX and increased MyHC IIB-positive fiber area in skeletal muscle of the LKB1-KO mouse model (436). However, another study found that the lack of both AMPK α 1 and AMPK α 2 leads to a muscle enriched in MyHC type I and perhaps also type IIA fibers (72). Studies of similar or other AMPK-deficient mouse models (AMPK kinase dead, β 2M-KO, β 1 β 2M-KO) report unchanged fiber type distribution (74, 146, 437). One study reports that lack of functional AMPK α 2 attenuates exercise-induced muscle fiber type

remodeling toward a distribution enriched in MyHC IIA/X fibers (285). Thus, genetic evidence for AMPK being an important mediator in promoting the expression of an oxidative slow-twitch MyHC phenotype is inconsistent. However, many of these studies are based on a limited number of observations and are often limited to a few muscle groups.

Besides its high adaptation capability, adult skeletal muscle possesses the capacity to regenerate after injury. Skeletal muscle regeneration requires the activation, migration, proliferation, and fusion of muscle stem cells to form new functional myofibers. The process of adult myogenesis requires the establishment of specific interactions of muscle cells with other cell types, including immune cells (438, 439). The inflammatory response during skeletal muscle regeneration is a spatially and temporally coordinated process. Whereas the first steps of the inflammatory response are associated with damage-associated macrophages that secrete proinflammatory cytokines, resolution of inflammation is associated with restorative macrophages that exhibit an anti-inflammatory phenotype and directly support myogenesis (440). The resolution of inflammation is a critical step, and its alteration impairs skeletal muscle regeneration (441, 442). Indeed, providing anti-inflammatory cues too early during the inflammatory phase or blocking anti-inflammatory cues after the first days of injury alters the regeneration process (443), showing that the temporal transition of proinflammatory to anti-inflammatory phases during skeletal muscle regeneration is crucial (444). In parallel to its role as an energy sensor, AMPK activation is associated with a decrease in the inflammatory response (445–447). Recent evidence has demonstrated that macrophage AMPK α 1 is crucial for the resolution of inflammation during skeletal muscle regeneration, linking metabolism and inflammation (448). Indeed, specific deletion of AMPK α 1 in macrophages induces a defect in macrophage skewing (*i.e.*, the transition from the damage-associated phenotype to the restorative macrophage phenotype) (449), triggering the impairment of skeletal muscle regeneration. During the regeneration process, AMPK α 1 activation, dependent on phosphorylation by CaMKK β but not by LKB1, is linked to the phagocytosis of apoptotic/necrotic muscle debris, triggering macrophage skewing (448).

AMPK is also involved in regulation of muscle stem cells (MuSCs) themselves, which, during regeneration, activate and recapitulate the myogenic program to repair damaged myofibers. MuSCs are capable of both differentiation to repair muscle tissue and self-renewal to replenish the stem cell pool, which is a crucial process to maintain the pool of MuSCs for further needs. Interestingly, LKB1 was shown to limit MuSC proliferation through the AMPK/mTORC1 pathway (450). The AMPK-regulated transcription factor FOXO3 enhances MuSC self-renewal *via* the activation of Notch signaling, an essential pathway for the maintenance of MuSC quiescence (451). In addition, regulation of stem cell fate seems to be dependent on the metabolic state of the cell (452, 453). In skeletal muscle, it has recently been shown that metabolic reprogramming promotes activation of MuSCs,

identifying a tight link between the metabolic state of MuSCs and epigenetic modifications that regulate their myogenic commitment (454). In this context, AMPK controls muscle stem cell fate. AMPK α 1-KO muscle stem cells displayed a high self-renewal rate and showed a Warburg-like switch of their metabolism to higher glycolysis through the regulation of the activity of LDH, a new functional target of AMPK in stem cells (455).

Contrary to healthy skeletal muscle, degenerating myopathies are characterized by permanent attempts of regeneration associated with persistent inflammation. Interestingly, animal models depleted for AMPK specifically in skeletal muscle fibers display a myopathic phenotype (72, 263, 431). Furthermore, chronic activation of AMPK by AICAR treatment of the dystrophic mdx mouse, a murine model for Duchenne myopathy, improves metabolic function and histologic pattern of the skeletal muscle. In this context, AMPK activation may not only target myofibers and mitochondrial metabolism but also inflammation in the myopathic muscle. Indeed, fibrosis, IL-6, and IL-10 levels are reduced in the muscle of AICAR-treated mdx mice (456). Deciphering the molecular pathways triggered by AMPK activation in myogenic (myofibers, MuSCs) and nonmyogenic cells (immune cells, fibroblastic cells, endothelial cells) will help to finely tune the potential benefits of activating this pathway to treat muscle disorders.

AMPK: A THERAPEUTIC TARGET?

AMPK and Duchenne muscular dystrophy

The inherited neuromuscular disorder Duchenne muscular dystrophy (DMD) causes life-limiting muscle wasting in boys and young men. The underlying etiology is a lack of functional dystrophin protein in muscle. Because no cure is available, ways to maintain muscular function are of significant interest to patients with DMD. Of particular interest for this review is the idea that AMPK activation might promote some aspects of the oxidative myogenic program and the observation that the slow oxidative myofiber phenotype is more resistant to dystrophin defects than the faster glycolytic myofiber phenotype (457).

By assembling with multiple partners, dystrophin provides a strong and elastic link between the intracellular cytoskeleton and the extracellular matrix and transmits mechanical and cellular signals (458). A lack of dystrophin leads to diminished sarcolemma integrity, mitochondria dysfunction, compromised calcium handling, myofibrillar protein degradation, apoptosis, and necrosis (459). Interestingly, a lack of both dystrophin and utrophin A (Dmd^{mdx}/Utrn KO) in the mouse mimics the human DMD disease better than a lack of dystrophin alone (460). Reintroducing utrophin A by overexpression in muscle of such models rescues the phenotype by decreasing fiber regeneration and by normalizing both dystrophin localization at the PM and expression of developmental proteins (461). Due to consequence similarities these proteins may have redundant roles within the cell, giving rise to this rescue effect of utrophin A. Utrophin A expression is

calcineurin/nuclear factor of activated T cell-dependent and thus is part of the oxidative fiber programming (462). As discussed previously, AMPK may play an important role in inducing the oxidative myogenic program, and observations from several animal studies in which mdx mice have been treated with AMPK activating compounds such as AICAR, metformin, and resveratrol suggest that targeting AMPK in skeletal muscle may ameliorate the pathology observed in patients with DMD (12, 463, 464). Furthermore, studies suggest that up-regulation of utrophin A is key to the functional adaptations in the mdx mouse model after chronic AICAR treatment (465), although genetic evidence linking these effects to AMPK action is lacking. In fact, the idea is challenged by the apparent lack of effect on utrophin A gene expression in muscle after both acute and chronic treatment with the direct pan-AMPK activator PF-739 despite other indices of an up-regulated oxidative muscle fiber program (161). Nevertheless, endurance-type exercise regimes in animal models of DMD (mdx mice) induce adaptations toward remodeling of the muscle phenotype (466). In line with this finding, it has been suggested that patients with DMD may benefit from exercise training programs attempting to avoid disuse-induced muscle atrophy and dysfunction (467, 468). Whether activation of AMPK in skeletal muscle during exercise is necessary for these effects of exercise needs to be clarified.

AMPK in obesity and diabetes

Obesity and related metabolic disorders such as type 2 diabetes and dyslipidemia are emerging as major global health challenges due to a complex interplay between obesity-favoring environmental factors, such as sedentary lifestyle, reduced physical activity levels, overconsumption of high-caloric food, and a permissive genetic prevalence (469, 470). Insulin resistance is one of the earliest hallmarks of the prediabetic state and arises from impaired glucose and fatty acid uptake and metabolism in a number of tissues, notably skeletal muscle, which is responsible for the majority of glucose disposal upon insulin stimulation and exercise. Given the importance of AMPK in the metabolic responses of skeletal muscle, a great deal of interest has focused on understanding AMPK signaling in skeletal muscle induced by either exercise or pharmacological activators as a treatment modality for type 2 diabetes. Thus, during the last decade there has been a major drive to develop small-molecule AMPK activators, and the identification of A769662 held promise in drug development for the treatment of T2D (156) until it was reported that A769662 predominantly activates AMPK β 1-containing complexes (70), which do not seem to exist in human skeletal muscle (69, 76). The importance of targeting the specific AMPK complexes in skeletal muscle has been highlighted by studies using compounds stimulating AMPK activity of both β 1- and β 2-containing complexes. It has been shown that the pan-AMPK activator 991 efficiently activates AMPK in isolated mouse skeletal muscle and elicits metabolic effects appropriate for treating T2D by stimulating glucose uptake and fatty acid oxidation (160, 471). The clinical relevance for the use of nonselective activators of all AMPK complexes to lower plasma glucose levels through

increasing skeletal muscle glucose uptake has recently been demonstrated in rodents and nonhuman primates (161, 162). Thus, acute dosing of the pan-AMPK agonist PF-739 activated AMPK in skeletal muscle and liver and was capable of lowering blood glucose levels of diet-induced obese mice (161). Treatment with PF-739 resulted in increased glucose disposal rate but had no impact on endogenous glucose production (161). In addition, the blood glucose-lowering effect of PF-739 was diminished in mice deficient of AMPK α 1 and AMPK α 2 in skeletal muscle but not in liver, indicating that activation of AMPK in skeletal muscle is the appropriate means for the treatment of patients with type 2 diabetes independently of hepatic AMPK activation (161). Similarly, the pan-AMPK activator MK-8722, a potent activator of β 2-containing complexes, enhanced glucose uptake in skeletal muscle *in vitro* and *in vivo* and improved glucose tolerance in diabetic rodents and nonhuman primates (162). However, although systemic pharmacological activation of AMPK by MK-8722 showed beneficial improvements of glucose homeostasis, potential cardiac safety concerns were reported. Long-term MK-8722 treatment was associated with reversible cardiac hypertrophy but without apparent adverse consequences (162). This effect could be reminiscent of the physiologic cardiac hypertrophy found in elite athletes due to cardiac AMPK activation during training (472).

Studies investigating AMPK activity in skeletal muscle of humans with obesity and type 2 diabetes have yielded mixed results. Some groups report decreased AMPK activity in skeletal muscle of obese people with or without type 2 diabetes compared with lean control subjects (321, 473), whereas others report intact regulation of AMPK activity and signaling both at rest and in response to exercise (120, 209, 474). Despite these observations, the contribution of reductions in muscle AMPK activity in the development of insulin resistance and obesity remains unclear. Indeed, knockout mouse models lacking AMPK specifically in skeletal muscle display comparable body mass, adiposity, glucose tolerance, and insulin sensitivity compared with control animals fed a control chow diet as well as a HFD (72, 74). These data indicate that loss of AMPK *per se* is not sufficient to induce HFD-induced obesity or insulin resistance. Nevertheless, there is substantial evidence suggesting that improvements in whole-body insulin sensitivity are associated with elevated AMPK activity. It is now well documented that physiologic (*e.g.*, exercise training) or pharmacological AMPK activation can combat insulin resistance and metabolic dysfunction caused by chronic nutrient excess (137, 475). Several studies have shown that pharmacological AMPK activation mimics the effects of regular exercise and calorie restriction and decreases the risk of progression of diabetes and the development of skeletal muscle insulin resistance (476–478). As such, a recent study using different genetically modified mouse models with altered AMPK activity has provided evidence to support that AMPK is essential for AICAR-induced elevation of muscle insulin sensitivity likely through phosphorylation of TBC1D4, a downstream target of insulin and AMPK signaling regulating GLUT4 translocation (129). In addition to its role in enhancing insulin-stimulated glucose uptake, repeated AICAR

treatment has been reported to induce improvements in insulin-stimulated fatty acid uptake and oxidation (479). Although AICAR treatment represents a promising strategy to improve muscle insulin sensitivity in type 2 diabetes, observations in humans have so far been inconclusive (402, 403, 480, 481), given its relatively high threshold for AMPK activation in skeletal muscle (144, 480, 481). On the other hand, the reported impact of *in vivo* AICAR infusion on suppressing hepatic glucose output in patients with type 2 diabetes (481) and animal models (478, 482, 483) may provide interesting new leads for future pharmacological interventions. Finally, findings suggest that AMPK may suppress inflammatory signaling in skeletal muscle (484), which may contribute to its insulin-sensitizing action because it is well established that obesity-linked type 2 diabetes is associated with a low-grade chronic inflammatory state (485). Indeed, both AICAR and metformin have been shown to exert anti-inflammatory effects in many cell types (*e.g.*, myocytes, adipocytes, and macrophages) as well as in mature skeletal muscle and adipose tissue of LPS-challenged rats, a model of metabolic endotoxemia (486). Based on these observations and on findings of intact resting and exercise-induced activation of AMPK in skeletal muscle of patients with type 2 diabetes (120, 209, 474), this signifies a potential role of AMPK in skeletal muscle as a pharmacological target for prevention and treatment of inflammation and insulin resistance in various dysmetabolic conditions.

CONCLUSIONS

AMPK has garnered considerable interest since its identification as an energy sensor and a critical regulator of cell metabolism in the 1980s. In skeletal muscle, numerous studies have highlighted the role of AMPK as a mediator of cell signaling pathways that are intrinsically linked to muscle function and metabolism (*i.e.*, glucose uptake, fatty acid oxidation, mitochondrial biogenesis, protein metabolism, and muscle plasticity). Although these observations are consistent with the idea that AMPK is essential for a healthy muscle phenotype, a lack of AMPK in skeletal muscle in general only leads to mild alterations in the muscle/whole-body phenotype. The simple explanation is that redundant mechanisms are put into play in the AMPK-deficient cell. However, another view could be that AMPK is dispensable under most physiologic situations, and only during severe energy perturbation (*e.g.*, exercise and starvation) does AMPK action become important for cell function and survival. In this view, skeletal muscle energy turnover at resting/normal conditions is very low, and the energy charge is only rarely challenged. Clearly, we need to dig deeper into the world of AMPK regulation and its role in physiology. For example, studies focusing on the regulation of the different heterotrimeric complexes, their specific roles, and the consequences of their subcellular localization would add a new dimension to the field. The observation of differential regulation of the heterotrimeric complexes opens up a largely unexplored avenue as to the downstream effectors of each of these and the mode of their regulation. The lack of proper and

heterotrimer selective AMPK inhibitors/activators has to a large extent limited the investigation of the physiologic regulatory role of AMPK in muscle metabolism. Newer transgenic models of inducible knockout/overexpression may offer a way for such exploration. Because muscle tissue contributes ~40% of the entire body mass, the general condition of this tissue is important for whole-body metabolism and overall health. Thus, being able to target AMPK specifically in skeletal muscle—perhaps through the unique expression of the AMPK γ 3 complex—may offer an attractive opportunity, and further investigation is warranted to reveal the full potential for such strategies. More work is also needed to delineate the exact role(s) of AMPK in mature skeletal muscle. Given that exercise increases overall health as well as AMPK activity in skeletal muscle, we believe that finding ways of activating AMPK in skeletal muscle (preferably by exercise) may constitute an effective medical treatment for a number of diseases and may oppose the detrimental effects of age on daily physical capacity. **[FJ]**

ACKNOWLEDGMENTS

This work was supported by grants from INSERM, Centre National de la Recherche Scientifique (CNRS), Université Paris Descartes, and Agence Nationale de la Recherche (to B.V. and M.F.); the Association Française contre les Myopathies, Ligue Nationale Contre le Cancer, and the Société Française de Myologie (to R.M.); the Canadian Institutes for Health Research (CIHR; FDN-143247) (to A.M.); the Danish Council for independent Research—Medical Sciences, the Lundbeck Foundation, and Novo Nordisk Foundation (to J.F.P.W.); and U.S. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases Grants DK054902 and DK059637 (to L.L.). J.T.T. was supported by the Novo Nordisk Foundation (Excellence Project Award NNF14OC0009315), and by the Danish Council for Independent Research (Research Project Grant DFF – 4004-00235). The authors declare no conflicts of interest.

AUTHOR CONTRIBUTIONS

All authors contributed to the concept, design, and writing of this article and experimental work described in the text.

REFERENCES

- Booth, F. W., and Baldwin, K. (1996) Muscle plasticity: energy demand and supply processes. In *Handbook of Physiology. Exercise: Regulation and Integration of Multiple Systems*, pp. 1075–1123, American Physiological Society, Bethesda, MD, USA
- Egan, B., and Zierath, J. R. (2013) Exercise metabolism and the molecular regulation of skeletal muscle adaptation. *Cell Metab.* **17**, 162–184
- Flück, M. (2006) Functional, structural and molecular plasticity of mammalian skeletal muscle in response to exercise stimuli. *J. Exp. Biol.* **209**, 2239–2248
- Bassel-Duby, R., and Olson, E. N. (2006) Signaling pathways in skeletal muscle remodeling. *Annu. Rev. Biochem.* **75**, 19–37
- Hood, D. A., Irrcher, I., Ljubicic, V., and Joseph, A.-M. (2006) Coordination of metabolic plasticity in skeletal muscle. *J. Exp. Biol.* **209**, 2265–2275
- Gaitanos, G. C., Williams, C., Boobis, L. H., and Brooks, S. (1993) Human muscle metabolism during intermittent maximal exercise. *J. Appl. Physiol.* **75**, 712–719
- Gowans, G. J., and Hardie, D. G. (2014) AMPK: a cellular energy sensor primarily regulated by AMP. *Biochem. Soc. Trans.* **42**, 71–75
- Hardie, D. G., Ross, F. A., and Hawley, S. A. (2012) AMPK: a nutrient and energy sensor that maintains energy homeostasis. *Nat. Rev. Mol. Cell Biol.* **13**, 251–262
- Merrill, G. F., Kurth, E. J., Hardie, D. G., and Winder, W. W. (1997) AICA riboside increases AMP-activated protein kinase, fatty acid oxidation, and glucose uptake in rat muscle. *Am. J. Physiol.* **273**, E1107–E1112
- Jørgensen, S. B., Nielsen, J. N., Birk, J. B., Olsen, G. S., Viollet, B., Andreelli, F., Schjerling, P., Vaulont, S., Hardie, D. G., Hansen, B. F., Richter, E. A., and Wojtaszewski, J. F. (2004) The α 2-5'AMP-activated protein kinase is a site 2 glycogen synthase kinase in skeletal muscle and is responsive to glucose loading. *Diabetes* **53**, 3074–3081
- Bolster, D. R., Crozier, S. J., Kimball, S. R., and Jefferson, L. S. (2002) AMP-activated protein kinase suppresses protein synthesis in rat skeletal muscle through down-regulated mammalian target of rapamycin (mTOR) signaling. *J. Biol. Chem.* **277**, 23977–23980
- Ljubicic, V., Miura, P., Burt, M., Boudreault, L., Khogali, S., Lunde, J. A., Renaud, J. M., and Jasmin, B. J. (2011) Chronic AMPK activation evokes the slow, oxidative myogenic program and triggers beneficial adaptations in mdx mouse skeletal muscle. *Hum. Mol. Genet.* **20**, 3478–3493
- Bergeron, R., Ren, J. M., Cadman, K. S., Moore, I. K., Perret, P., Pypaert, M., Young, L. H., Semenkovich, C. F., and Shulman, G. I. (2001) Chronic activation of AMP kinase results in NRF-1 activation and mitochondrial biogenesis. *Am. J. Physiol. Endocrinol. Metab.* **281**, E1340–E1346
- Winder, W. W., Holmes, B. F., Rubink, D. S., Jensen, E. B., Chen, M., and Holloszy, J. O. (2000) Activation of AMP-activated protein kinase increases mitochondrial enzymes in skeletal muscle. *J. Appl. Physiol.* **88**, 2219–2226
- Zong, H., Ren, J. M., Young, L. H., Pypaert, M., Mu, J., Birnbaum, M. J., and Shulman, G. I. (2002) AMP kinase is required for mitochondrial biogenesis in skeletal muscle in response to chronic energy deprivation. *Proc. Natl. Acad. Sci. USA* **99**, 15983–15987
- Winder, W. W., Hardie, D. G., Mustard, K. J., Greenwood, L. J., Paxton, B. E., Park, S. H., Rubink, D. S., and Taylor, E. B. (2003) Long-term regulation of AMP-activated protein kinase and acetyl-CoA carboxylase in skeletal muscle. *Biochem. Soc. Trans.* **31**, 182–185
- Nielsen, J. N., Jørgensen, S. B., Frøsig, C., Viollet, B., Andreelli, F., Vaulont, S., Kiens, B., Richter, E. A., and Wojtaszewski, J. F. (2003) A possible role for AMP-activated protein kinase in exercise-induced glucose utilization: insights from humans and transgenic animals. *Biochem. Soc. Trans.* **31**, 186–190
- Musi, N., Yu, H., and Goodyear, L. J. (2003) AMP-activated protein kinase regulation and action in skeletal muscle during exercise. *Biochem. Soc. Trans.* **31**, 191–195
- Musi, N., and Goodyear, L. J. (2003) AMP-activated protein kinase and muscle glucose uptake. *Acta Physiol. Scand.* **178**, 337–345
- Carling, D., Fryer, L. G., Woods, A., Daniel, T., Jarvie, S. L., and Whitrow, H. (2003) Bypassing the glucose/fatty acid cycle: AMP-activated protein kinase. *Biochem. Soc. Trans.* **31**, 1157–1160
- Wojtaszewski, J. F., Nielsen, J. N., Jørgensen, S. B., Frøsig, C., Birk, J. B., and Richter, E. A. (2003) Transgenic models: a scientific tool to understand exercise-induced metabolism: the regulatory role of AMPK (5'-AMP-activated protein kinase) in glucose transport and glycogen synthase activity in skeletal muscle. *Biochem. Soc. Trans.* **31**, 1290–1294
- Hardie, D. G. (2004) AMP-activated protein kinase: a key system mediating metabolic responses to exercise. *Med. Sci. Sports Exerc.* **36**, 28–34
- Aschenbach, W. G., Sakamoto, K., and Goodyear, L. J. (2004) 5' adenosine monophosphate-activated protein kinase, metabolism and exercise. *Sports Med.* **34**, 91–103
- Barnes, B. R., and Zierath, J. R. (2005) Role of AMP: activated protein kinase in the control of glucose homeostasis. *Curr. Mol. Med.* **5**, 341–348
- Hardie, D. G., and Sakamoto, K. (2006) AMPK: a key sensor of fuel and energy status in skeletal muscle. *Physiology (Bethesda)* **21**, 48–60
- Jørgensen, S. B., Richter, E. A., and Wojtaszewski, J. F. (2006) Role of AMPK in skeletal muscle metabolic regulation and adaptation in relation to exercise. *J. Physiol.* **574**, 17–31
- Fujii, N., Jessen, N., and Goodyear, L. J. (2006) AMP-activated protein kinase and the regulation of glucose transport. *Am. J. Physiol. Endocrinol. Metab.* **291**, E867–E877

28. Winder, W. W., Taylor, E. B., and Thomson, D. M. (2006) Role of AMP-activated protein kinase in the molecular adaptation to endurance exercise. *Med. Sci. Sports Exerc.* **38**, 1945–1949
29. Steinberg, G. R., and Jørgensen, S. B. (2007) The AMP-activated protein kinase: role in regulation of skeletal muscle metabolism and insulin sensitivity. *Mini Rev. Med. Chem.* **7**, 519–526
30. Winder, W. W., and Thomson, D. M. (2007) Cellular energy sensing and signaling by AMP-activated protein kinase. *Cell Biochem. Biophys.* **47**, 332–347
31. Jørgensen, S. B., Jensen, T. E., and Richter, E. A. (2007) Role of AMPK in skeletal muscle gene adaptation in relation to exercise. *Appl. Physiol. Nutr. Metab.* **32**, 904–911
32. Osler, M. E., and Zierath, J. R. (2008) Adenosine 5'-monophosphate-activated protein kinase regulation of fatty acid oxidation in skeletal muscle. *Endocrinology* **149**, 935–941
33. Koh, H.-J., Brandauer, J., and Goodyear, L. J. (2008) LKB1 and AMPK and the regulation of skeletal muscle metabolism. *Curr. Opin. Clin. Nutr. Metab. Care* **11**, 227–232
34. Jørgensen, S. B., and Rose, A. J. (2008) How is AMPK activity regulated in skeletal muscles during exercise? *Front. Biosci.* **13**, 5589–5604
35. Long, Y. C., and Zierath, J. R. (2008) Influence of AMP-activated protein kinase and calcineurin on metabolic networks in skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **295**, E545–E552
36. Treebak, J. T., and Wojtaszewski, J. F. (2008) Role of 5'AMP-activated protein kinase in skeletal muscle. *Int. J. Obes.* **32**(Suppl 4), S13–S17
37. Witczak, C. A., Sharoff, C. G., and Goodyear, L. J. (2008) AMP-activated protein kinase in skeletal muscle: from structure and localization to its role as a master regulator of cellular metabolism. *Cell. Mol. Life Sci.* **65**, 3737–3755
38. Jensen, T. E., Wojtaszewski, J. F., and Richter, E. A. (2009) AMP-activated protein kinase in contraction regulation of skeletal muscle metabolism: necessary and/or sufficient? *Acta Physiol. (Oxf.)* **196**, 155–174
39. Thomson, D. M., and Winder, W. W. (2009) AMP-activated protein kinase control of fat metabolism in skeletal muscle. *Acta Physiol. (Oxf.)* **196**, 147–154
40. McGee, S. L., and Hargreaves, M. (2010) AMPK-mediated regulation of transcription in skeletal muscle. *Clin. Sci.* **118**, 507–518
41. Hardie, D. G. (2011) Energy sensing by the AMP-activated protein kinase and its effects on muscle metabolism. *Proc. Nutr. Soc.* **70**, 92–99
42. Sanchez, A. M., Candau, R. B., Csibi, A., Pagano, A. F., Raibon, A., and Bernardi, H. (2012) The role of AMP-activated protein kinase in the coordination of skeletal muscle turnover and energy homeostasis. *Am. J. Physiol. Cell Physiol.* **303**, C475–C485
43. O'Neill, H. M., Holloway, G. P., and Steinberg, G. R. (2013) AMPK regulation of fatty acid metabolism and mitochondrial biogenesis: implications for obesity. *Mol. Cell. Endocrinol.* **366**, 135–151
44. Friedrichsen, M., Mortensen, B., Pehmøller, C., Birk, J. B., and Wojtaszewski, J. F. (2013) Exercise-induced AMPK activity in skeletal muscle: role in glucose uptake and insulin sensitivity. *Mol. Cell. Endocrinol.* **366**, 204–214
45. O'Neill, H. M. (2013) AMPK and exercise: glucose uptake and insulin sensitivity. *Diabetes Metab. J.* **37**, 1–21
46. Ijubicic, V., and Jasmin, B. J. (2013) AMP-activated protein kinase at the nexus of therapeutic skeletal muscle plasticity in Duchenne muscular dystrophy. *Trends Mol. Med.* **19**, 614–624
47. Jessen, N., Sundelin, E. I., and Möller, A. B. (2014) AMP kinase in exercise adaptation of skeletal muscle. *Drug Discov. Today* **19**, 999–1002
48. Marcinko, K., and Steinberg, G. R. (2014) The role of AMPK in controlling metabolism and mitochondrial biogenesis during exercise. *Exp. Physiol.* **99**, 1581–1585
49. Mounier, R., Théret, M., Lantier, L., Foretz, M., and Viollet, B. (2015) Expanding roles for AMPK in skeletal muscle plasticity. *Trends Endocrinol. Metab.* **26**, 275–286
50. Hardie, D. G. (2007) AMP-activated/SNF1 protein kinases: conserved guardians of cellular energy. *Nat. Rev. Mol. Cell Biol.* **8**, 774–785
51. Hawley, S. A., Davison, M., Woods, A., Davies, S. P., Beri, R. K., Carling, D., and Hardie, D. G. (1996) Characterization of the AMP-activated protein kinase from rat liver and identification of threonine 172 as the major site at which it phosphorylates AMP-activated protein kinase. *J. Biol. Chem.* **271**, 27879–27887
52. Stapleton, D., Mitchell, K. I., Gao, G., Widmer, J., Michell, B. J., Teh, T., House, C. M., Fernandez, C. S., Cox, T., Witters, L. A., and Kemp, B. E. (1996) Mammalian AMP-activated protein kinase subfamily. *J. Biol. Chem.* **271**, 611–614
53. Crute, B. E., Seefeld, K., Gamble, J., Kemp, B. E., and Witters, L. A. (1998) Functional domains of the alpha catalytic subunit of the AMP-activated protein kinase. *J. Biol. Chem.* **273**, 35347–35354
54. Iseli, T. J., Walter, M., van Denderen, B. J., Katsis, F., Witters, L. A., Kemp, B. E., Michell, B. J., and Stapleton, D. (2005) AMP-activated protein kinase beta subunit tethers alpha and gamma subunits via its C-terminal sequence (186–270). *J. Biol. Chem.* **280**, 13395–13400
55. Hudson, E. R., Pan, D. A., James, J., Lucocq, J. M., Hawley, S. A., Green, K. A., Baba, O., Terashima, T., and Hardie, D. G. (2003) A novel domain in AMP-activated protein kinase causes glycogen storage bodies similar to those seen in hereditary cardiac arrhythmias. *Curr. Biol.* **13**, 861–866
56. Polekhina, G., Gupta, A., Michell, B. J., van Denderen, B., Murthy, S., Feil, S. C., Jennings, I. G., Campbell, D. J., Witters, L. A., Parker, M. W., Kemp, B. E., and Stapleton, D. (2003) AMPK β subunit targets metabolic stress sensing to glycogen. *Curr. Biol.* **13**, 867–871
57. Scott, J. W., Hawley, S. A., Green, K. A., Anis, M., Stewart, G., Scullion, G. A., Norman, D. G., and Hardie, D. G. (2004) CBS domains form energy-sensing modules whose binding of adenosine ligands is disrupted by disease mutations. *J. Clin. Invest.* **113**, 274–284
58. Kazgan, N., Williams, T., Forsberg, L. J., and Brenman, J. E. (2010) Identification of a nuclear export signal in the catalytic subunit of AMP-activated protein kinase. *Mol. Biol. Cell* **21**, 3433–3442
59. Warden, S. M., Richardson, C., O'Donnell, J. Jr., Stapleton, D., Kemp, B. E., and Witters, L. A. (2001) Post-translational modifications of the beta-1 subunit of AMP-activated protein kinase affect enzyme activity and cellular localization. *Biochem. J.* **354**, 275–283
60. Oakhill, J. S., Chen, Z.-P., Scott, J. W., Steel, R., Castelli, L. A., Ling, N., Macaulay, S. L., and Kemp, B. E. (2010) β -Subunit myristoylation is the gatekeeper for initiating metabolic stress sensing by AMP-activated protein kinase (AMPK). *Proc. Natl. Acad. Sci. USA* **107**, 19237–19241
61. Liang, J., Xu, Z.-X., Ding, Z., Lu, Y., Yu, Q., Werle, K. D., Zhou, G., Park, Y.-Y., Peng, G., Gambello, M. J., and Mills, G. B. (2015) Myristoylation confers noncanonical AMPK functions in autophagy selectivity and mitochondrial surveillance. *Nat. Commun.* **6**, 7926
62. Pinter, K., Jefferson, A., Czibik, G., Watkins, H., and Redwood, C. (2012) Subunit composition of AMPK trimers present in the cytoskeletal apparatus: implications for drug target identification. *Cell Cycle* **11**, 917–921
63. Pinter, K., Grignani, R. T., Czibik, G., Farza, H., Watkins, H., and Redwood, C. (2012) Embryonic expression of AMPK γ subunits and the identification of a novel $\gamma 2$ transcript variant in adult heart. *J. Mol. Cell. Cardiol.* **53**, 342–349
64. Pinter, K., Grignani, R. T., Watkins, H., and Redwood, C. (2013) Localisation of AMPK γ subunits in cardiac and skeletal muscles. *J. Muscle Res. Cell Motil.* **34**, 369–378
65. Jensen, T. E., Ross, F. A., Kleinert, M., Sylow, L., Knudsen, J. R., Gowans, G. J., Hardie, D. G., and Richter, E. A. (2015) PT-1 selectively activates AMPK- $\gamma 1$ complexes in mouse skeletal muscle, but activates all three γ subunit complexes in cultured human cells by inhibiting the respiratory chain. *Biochem. J.* **467**, 461–472
66. Rajamohan, F., Reyes, A. R., Frisbie, R. K., Hoth, L. R., Sahasrabudhe, P., Magyar, R., Landro, J. A., Withka, J. M., Caspers, N. L., Calabrese, M. F., Ward, J., and Kurumbail, R. G. (2016) Probing the enzyme kinetics, allosteric modulation and activation of $\alpha 1$ - and $\alpha 2$ -subunit-containing AMP-activated protein kinase (AMPK) heterotrimeric complexes by pharmacological and physiological activators. *Biochem. J.* **473**, 581–592
67. Oakhill, J. S., Steel, R., Chen, Z.-P., Scott, J. W., Ling, N., Tam, S., and Kemp, B. E. (2011) AMPK is a direct adenylate charge-regulated protein kinase. *Science* **332**, 1433–1435
68. Cheung, P. C., Salt, I. P., Davies, S. P., Hardie, D. G., and Carling, D. (2000) Characterization of AMP-activated protein kinase gamma-subunit isoforms and their role in AMP binding. *Biochem. J.* **346**, 659–669
69. Wojtaszewski, J. F. P., Birk, J. B., Frøsig, C., Holten, M., Pilegaard, H., and Dela, F. (2005) 5'AMP activated protein kinase expression in human skeletal muscle: effects of strength training and type 2 diabetes. *J. Physiol.* **564**, 563–573
70. Treebak, J. T., Birk, J. B., Hansen, B. F., Olsen, G. S., and Wojtaszewski, J. F. (2009) A-769662 activates AMPK beta1-containing complexes but induces glucose uptake through a PI3-kinase-dependent pathway in mouse skeletal muscle. *Am. J. Physiol. Cell Physiol.* **297**, C1041–C1052
71. Kristensen, D. E., Albers, P. H., Prats, C., Baba, O., Birk, J. B., and Wojtaszewski, J. F. (2015) Human muscle fibre type-specific

- regulation of AMPK and downstream targets by exercise. *J. Physiol.* **593**, 2053–2069
72. Lantier, L., Fentz, J., Mounier, R., Leclerc, J., Treebak, J. T., Pehmøller, C., Sanz, N., Sakakibara, I., Saint-Amand, E., Rimbaud, S., Maire, P., Marette, A., Ventura-Clapier, R., Ferry, A., Wojtaszewski, J. F., Foretz, M., and Viollet, B. (2014) AMPK controls exercise endurance, mitochondrial oxidative capacity, and skeletal muscle integrity. *FASEB J.* **28**, 3211–3224
73. Fentz, J., Kjøbsted, R., Birk, J. B., Jordy, A. B., Jeppesen, J., Thorsen, K., Schjerling, P., Kiens, B., Jessen, N., Viollet, B., and Wojtaszewski, J. F. (2015) AMPK α is critical for enhancing skeletal muscle fatty acid utilization during in vivo exercise in mice. *FASEB J.* **29**, 1725–1738
74. O'Neill, H. M., Maarbjerg, S. J., Crane, J. D., Jeppesen, J., Jørgensen, S. B., Schertzer, J. D., Shyroka, O., Kiens, B., van Denderen, B. J., Tarnopolsky, M. A., Kemp, B. E., Richter, E. A., and Steinberg, G. R. (2011) AMP-activated protein kinase (AMPK) β 1/ β 2 muscle null mice reveal an essential role for AMPK in maintaining mitochondrial content and glucose uptake during exercise. *Proc. Natl. Acad. Sci. USA* **108**, 16092–16097
75. Mahlapuu, M., Johansson, C., Lindgren, K., Hjälm, G., Barnes, B. R., Krook, A., Zierath, J. R., Andersson, L., and Marklund, S. (2004) Expression profiling of the gamma-subunit isoforms of AMP-activated protein kinase suggests a major role for gamma3 in white skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **286**, E194–E200
76. Birk, J. B., and Wojtaszewski, J. F. (2006) Predominant α 2/ β 2/ γ 3 AMPK activation during exercise in human skeletal muscle. *J. Physiol.* **577**, 1021–1032
77. Iseli, T. J., Oakhill, J. S., Bailey, M. F., Wee, S., Walter, M., van Denderen, B. J., Castelli, L. A., Katsis, F., Witters, L. A., Stapleton, D., Macaulay, S. L., Michell, B. J., and Kemp, B. E. (2008) AMP-activated protein kinase subunit interactions: β 1: γ 3 association requires β 1 Thr-263 and Tyr-267. *J. Biol. Chem.* **283**, 4799–4807
78. Manning, G., Whyte, D. B., Martinez, R., Hunter, T., and Sudarsanam, S. (2002) The protein kinase complement of the human genome. *Science* **298**, 1912–1934
79. Lizcano, J. M., Göransson, O., Toth, R., Deak, M., Morrice, N. A., Boudeau, J., Hawley, S. A., Udd, L., Mäkelä, T. P., Hardie, D. G., and Alessi, D. R. (2004) LKB1 is a master kinase that activates 13 kinases of the AMPK subfamily, including MARK/PAR-1. *EMBO J.* **23**, 833–843
80. Al-Hakim, A. K., Göransson, O., Deak, M., Toth, R., Campbell, D. G., Morrice, N. A., Prescott, A. R., and Alessi, D. R. (2005) 14-3-3 cooperates with LKB1 to regulate the activity and localization of QSK and SIK. *J. Cell Sci.* **118**, 5661–5673
81. Koh, H.-J., Toyoda, T., Fujii, N., Jung, M. M., Rathod, A., Middelbeek, R. J.-W., Lessard, S. J., Treebak, J. T., Tsuchihara, K., Esumi, H., Richter, E. A., Wojtaszewski, J. F., Hirshman, M. F., and Goodyear, L. J. (2010) Sucrose nonfermenting AMPK-related kinase (SNARK) mediates contraction-stimulated glucose transport in mouse skeletal muscle. *Proc. Natl. Acad. Sci. USA* **107**, 15541–15546
82. Lefebvre, D. L., Bai, Y., Shahmoly, N., Sharma, M., Poon, R., Drucker, D. J., and Rosen, C. F. (2001) Identification and characterization of a novel sucrose-non-fermenting protein kinase/AMP-activated protein kinase-related protein kinase, SNARK. *Biochem. J.* **355**, 297–305
83. Lessard, S. J., Rivas, D. A., So, K., Koh, H., Queiroz, A. L., Hirshman, M. F., Fielding, R. A., and Goodyear, L. J. (2016) The AMPK-related kinase SNARK regulates muscle mass and myocyte survival. *J. Clin. Invest.* **126**, 560–570
84. Broberg, S., and Sahlin, K. (1989) Adenine nucleotide degradation in human skeletal muscle during prolonged exercise. *J. Appl. Physiol.* **67**, 116–122
85. Norman, B., Sollevi, A., Kaijser, L., and Jansson, E. (1987) ATP breakdown products in human skeletal muscle during prolonged exercise to exhaustion. *Clin. Physiol.* **7**, 503–510
86. Bateman, A. (1997) The structure of a domain common to archaeobacteria and the homocystinuria disease protein. *Trends Biochem. Sci.* **22**, 12–13
87. Carling, D., Clarke, P. R., Zammit, V. A., and Hardie, D. G. (1989) Purification and characterization of the AMP-activated protein kinase. Copurification of acetyl-CoA carboxylase kinase and 3-hydroxy-3-methylglutaryl-CoA reductase kinase activities. *Eur. J. Biochem.* **186**, 129–136
88. Ferrer, A., Caelles, C., Massot, N., and Hegardt, F. G. (1985) Activation of rat liver cytosolic 3-hydroxy-3-methylglutaryl coenzyme A reductase kinase by adenosine 5'-monophosphate. *Biochem. Biophys. Res. Commun.* **132**, 497–504
89. Kemp, B. E. (2004) Bateman domains and adenosine derivatives form a binding contract. *J. Clin. Invest.* **113**, 182–184
90. Xiao, B., Sanders, M. J., Underwood, E., Heath, R., Mayer, F. V., Carmena, D., Jing, C., Walker, P. A., Eccleston, J. F., Haire, L. F., Saiu, P., Howell, S. A., Aasland, R., Martin, S. R., Carling, D., and Gambin, S. J. (2011) Structure of mammalian AMPK and its regulation by ADP. *Nature* **472**, 230–233
91. Suter, M., Riek, U., Tuerk, R., Schlattner, U., Wallimann, T., and Neumann, D. (2006) Dissecting the role of 5'-AMP for allosteric stimulation, activation, and deactivation of AMP-activated protein kinase. *J. Biol. Chem.* **281**, 32207–32216
92. Davies, S. P., Helps, N. R., Cohen, P. T., and Hardie, D. G. (1995) 5'-AMP inhibits dephosphorylation, as well as promoting phosphorylation, of the AMP-activated protein kinase: studies using bacterially expressed human protein phosphatase-2C α and native bovine protein phosphatase-2AC. *FEBS Lett.* **377**, 421–425
93. Garcia-Haro, L., Garcia-Gimeno, M. A., Neumann, D., Beullens, M., Bollen, M., and Sanz, P. (2010) The PP1-R6 protein phosphatase holoenzyme is involved in the glucose-induced dephosphorylation and inactivation of AMP-activated protein kinase, a key regulator of insulin secretion, in MIN6 beta cells. *FASEB J.* **24**, 5080–5091
94. Sanders, M. J., Grondin, P. O., Hegarty, B. D., Snowden, M. A., and Carling, D. (2007) Investigating the mechanism for AMP activation of the AMP-activated protein kinase cascade. *Biochem. J.* **403**, 139–148
95. Wu, Y., Song, P., Xu, J., Zhang, M., and Zou, M. H. (2007) Activation of protein phosphatase 2A by palmitate inhibits AMP-activated protein kinase. *J. Biol. Chem.* **282**, 9777–9788
96. Koh, H.-J., Arnolds, D. E., Fujii, N., Tran, T. T., Rogers, M. J., Jessen, N., Li, Y., Liew, C. W., Ho, R. C., Hirshman, M. F., Kulkarni, R. N., Kahn, C. R., and Goodyear, L. J. (2006) Skeletal muscle-selective knockout of LKB1 increases insulin sensitivity, improves glucose homeostasis, and decreases TRB3. *Mol. Cell. Biol.* **26**, 8217–8227
97. Sakamoto, K., McCarthy, A., Smith, D., Green, K. A., Grahame Hardie, D., Ashworth, A., and Alessi, D. R. (2005) Deficiency of LKB1 in skeletal muscle prevents AMPK activation and glucose uptake during contraction. *EMBO J.* **24**, 1810–1820
98. Smith, C. D., Compton, R. A., Bowler, J. S., Kemp, J. T., Sudweeks, S. N., Thomson, D. M., and Winder, W. W. (2011) Characterization of the liver kinase B1-mouse protein-25-Ste-20-related adaptor protein complex in adult mouse skeletal muscle. *J. Appl. Physiol.* **111**, 1622–1628
99. Thomson, D. M., Porter, B. B., Tall, J. H., Kim, H.-J., Barrow, J. R., and Winder, W. W. (2007) Skeletal muscle and heart LKB1 deficiency causes decreased voluntary running and reduced muscle mitochondrial marker enzyme expression in mice. *Am. J. Physiol. Endocrinol. Metab.* **292**, E196–E202
100. Jensen, T. E., Rose, A. J., Jørgensen, S. B., Brandt, N., Schjerling, P., Wojtaszewski, J. F. P., and Richter, E. A. (2007) Possible CaMKK-dependent regulation of AMPK phosphorylation and glucose uptake at the onset of mild tetanic skeletal muscle contraction. *Am. J. Physiol. Endocrinol. Metab.* **292**, E1308–E1317
101. Treebak, J. T., Birk, J. B., Rose, A. J., Kiens, B., Richter, E. A., and Wojtaszewski, J. F. (2007) AS160 phosphorylation is associated with activation of α 2 β 2 γ 1- but not α 2 β 2 γ 3-AMPK trimeric complex in skeletal muscle during exercise in humans. *Am. J. Physiol. Endocrinol. Metab.* **292**, E715–E722
102. Hardie, D. G. (2005) New roles for the LKB1 $\rightarrow$ AMPK pathway. *Curr. Opin. Cell Biol.* **17**, 167–173
103. Sakamoto, K., Göransson, O., Hardie, D. G., and Alessi, D. R. (2004) Activity of LKB1 and AMPK-related kinases in skeletal muscle: effects of contraction, phenformin, and AICAR. *Am. J. Physiol. Endocrinol. Metab.* **287**, E310–E317
104. Hawley, S. A., Pan, D. A., Mustard, K. J., Ross, L., Bain, J., Edelman, A. M., Frenguelli, B. G., and Hardie, D. G. (2005) Calmodulin-dependent protein kinase kinase-beta is an alternative upstream kinase for AMP-activated protein kinase. *Cell Metab.* **2**, 9–19
105. Woods, A., Dickerson, K., Heath, R., Hong, S.-P., Momcilovic, M., Johnstone, S. R., Carlson, M., and Carling, D. (2005) Ca²⁺/calmodulin-dependent protein kinase kinase-beta acts upstream of AMP-activated protein kinase in mammalian cells. *Cell Metab.* **2**, 21–33
106. McBride, A., Ghilagaber, S., Nikolaev, A., and Hardie, D. G. (2009) The glycogen-binding domain on the AMPK β subunit allows the kinase to act as a glycogen sensor. *Cell Metab.* **9**, 23–34
107. Wojtaszewski, J. F. P., MacDonald, C., Nielsen, J. N., Hellsten, Y., Hardie, D. G., Kemp, B. E., Kiens, B., and Richter, E. A. (2003)

- Regulation of 5'AMP-activated protein kinase activity and substrate utilization in exercising human skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **284**, E813–E822
108. Derave, W., Ai, H., Ihlemann, J., Witters, L. A., Kristiansen, S., Richter, E. A., and Ploug, T. (2000) Dissociation of AMP-activated protein kinase activation and glucose transport in contracting slow-twitch muscle. *Diabetes* **49**, 1281–1287
109. Wojtaszewski, J. F., Jørgensen, S. B., Hellsten, Y., Hardie, D. G., and Richter, E. A. (2002) Glycogen-dependent effects of 5-aminoimidazole-4-carboxamide (AICA)-riboside on AMP-activated protein kinase and glycogen synthase activities in rat skeletal muscle. *Diabetes* **51**, 284–292
110. McConell, G. K., Lee-Young, R. S., Chen, Z.-P., Stepto, N. K., Huynh, N. N., Stephens, T. J., Canny, B. J., and Kemp, B. E. (2005) Short-term exercise training in humans reduces AMPK signalling during prolonged exercise independent of muscle glycogen. *J. Physiol.* **568**, 665–676
111. Oligschläger, Y., Miglianico, M., Chanda, D., Scholz, R., Thali, R. F., Tuerk, R., Stapleton, D. I., Gooley, P. R., and Neumann, D. (2015) The recruitment of AMP-activated protein kinase to glycogen is regulated by autophosphorylation. *J. Biol. Chem.* **290**, 11715–11728
112. Valentine, R. J., Coughlan, K. A., Ruderman, N. B., and Saha, A. K. (2014) Insulin inhibits AMPK activity and phosphorylates AMPK Ser^{485/491} through Akt in hepatocytes, myotubes and incubated rat skeletal muscle. *Arch. Biochem. Biophys.* **562**, 62–69
113. Winder, W. W., and Hardie, D. G. (1996) Inactivation of acetyl-CoA carboxylase and activation of AMP-activated protein kinase in muscle during exercise. *Am. J. Physiol.* **270**, E299–E304
114. Chen, Z. P., McConell, G. K., Michell, B. J., Snow, R. J., Canny, B. J., and Kemp, B. E. (2000) AMPK signaling in contracting human skeletal muscle: acetyl-CoA carboxylase and NO synthase phosphorylation. *Am. J. Physiol. Endocrinol. Metab.* **279**, E1202–E1206
115. Fujii, N., Hayashi, T., Hirshman, M. F., Smith, J. T., Habinowski, S. A., Kaijser, L., Mu, J., Ljungqvist, O., Birnbaum, M. J., Witters, L. A., Thorell, A., and Goodyear, L. J. (2000) Exercise induces isoform-specific increase in 5'AMP-activated protein kinase activity in human skeletal muscle. *Biochem. Biophys. Res. Commun.* **273**, 1150–1155
116. Wojtaszewski, J. F., Nielsen, P., Hansen, B. F., Richter, E. A., and Kiens, B. (2000) Isoform-specific and exercise intensity-dependent activation of 5'AMP-activated protein kinase in human skeletal muscle. *J. Physiol.* **528**, 221–226
117. Chen, Z. P., Stephens, T. J., Murthy, S., Canny, B. J., Hargreaves, M., Witters, L. A., Kemp, B. E., and McConell, G. K. (2003) Effect of exercise intensity on skeletal muscle AMPK signaling in humans. *Diabetes* **52**, 2205–2212
118. Stephens, T. J., Chen, Z.-P., Canny, B. J., Michell, B. J., Kemp, B. E., and McConell, G. K. (2002) Progressive increase in human skeletal muscle AMPKalpha2 activity and ACC phosphorylation during exercise. *Am. J. Physiol. Endocrinol. Metab.* **282**, E688–E694
119. Wojtaszewski, J. F. P., Mourtzakis, M., Hillig, T., Saltin, B., and Pilegaard, H. (2002) Dissociation of AMPK activity and ACCbeta phosphorylation in human muscle during prolonged exercise. *Biochem. Biophys. Res. Commun.* **298**, 309–316
120. Musi, N., Fujii, N., Hirshman, M. F., Ekberg, I., Frøberg, S., Ljungqvist, O., Thorell, A., and Goodyear, L. J. (2001) AMP-activated protein kinase (AMPK) is activated in muscle of subjects with type 2 diabetes during exercise. *Diabetes* **50**, 921–927
121. Salt, I., Celler, J. W., Hawley, S. A., Prescott, A., Woods, A., Carling, D., and Hardie, D. G. (1998) AMP-activated protein kinase: greater AMP dependence, and preferential nuclear localization, of complexes containing the alpha2 isoform. *Biochem. J.* **334** (Pt 1), 177–187
122. Treebak, J. T., Pehmøller, C., Kristensen, J. M., Kjøbsted, R., Birk, J. B., Schjerling, P., Richter, E. A., Goodyear, L. J., and Wojtaszewski, J. F. (2014) Acute exercise and physiological insulin induce distinct phosphorylation signatures on TBC1D1 and TBC1D4 proteins in human skeletal muscle. *J. Physiol.* **592**, 351–375
123. Costford, S. R., Kavaslar, N., Ahituv, N., Chaudhry, S. N., Schackwitz, W. S., Dent, R., Pennacchio, L. A., McPherson, R., and Harper, M. E. (2007) Gain-of-function R225W mutation in human AMPKgamma(3) causing increased glycogen and decreased triglyceride in skeletal muscle. *PLoS One* **2**, e903
124. Milan, D., Jeon, J. T., Looft, C., Amarger, V., Robic, A., Thelander, M., Rogel-Gaillard, C., Paul, S., Iannuccelli, N., Rask, L., Ronne, H., Lundström, K., Reinsch, N., Gellin, J., Kalm, E., Roy, P. L., Chardon, P., and Andersson, L. (2000) A mutation in PRKAG3 associated with excess glycogen content in pig skeletal muscle. *Science* **288**, 1248–1251
125. Barnes, B. R., Marklund, S., Steiler, T. L., Walter, M., Hjälm, G., Amarger, V., Mahlapuu, M., Leng, Y., Johansson, C., Galuska, D., Lindgren, K., Abbrink, M., Stapleton, D., Zierath, J. R., and Andersson, L. (2004) The 5'-AMP-activated protein kinase gamma3 isoform has a key role in carbohydrate and lipid metabolism in glycolytic skeletal muscle. *J. Biol. Chem.* **279**, 38441–38447
126. Barré, L., Richardson, C., Hirshman, M. F., Brozinick, J., Fiering, S., Kemp, B. E., Goodyear, L. J., and Witters, L. A. (2007) Genetic model for the chronic activation of skeletal muscle AMP-activated protein kinase leads to glycogen accumulation. *Am. J. Physiol. Endocrinol. Metab.* **292**, E802–E811
127. Schönke, M., Myers, M. G., Jr., Zierath, J. R., and Björnholm, M. (2015) Skeletal muscle AMP-activated protein kinase γ 1 (H151R) overexpression enhances whole body energy homeostasis and insulin sensitivity. *Am. J. Physiol. Endocrinol. Metab.* **309**, E679–E690
128. Garcia-Roves, P. M., Osler, M. E., Holmström, M. H., and Zierath, J. R. (2008) Gain-of-function R225Q mutation in AMP-activated protein kinase γ 3 subunit increases mitochondrial biogenesis in glycolytic skeletal muscle. *J. Biol. Chem.* **283**, 35724–35734
129. Kjøbsted, R., Treebak, J. T., Fentz, J., Lantier, L., Viollet, B., Birk, J. B., Schjerling, P., Björnholm, M., Zierath, J. R., and Wojtaszewski, J. F. (2015) Prior AICAR stimulation increases insulin sensitivity in mouse skeletal muscle in an AMPK-dependent manner. *Diabetes* **64**, 2042–2055
130. Nielsen, J. N., Mustard, K. J., Graham, D. A., Yu, H., MacDonald, C. S., Pilegaard, H., Goodyear, L. J., Hardie, D. G., Richter, E. A., and Wojtaszewski, J. F. (2003) 5'-AMP-activated protein kinase activity and subunit expression in exercise-trained human skeletal muscle. *J. Appl. Physiol.* (1985). **94**, 631–641
131. Mortensen, B., Hingst, J. R., Frederiksen, N., Hansen, R. W. W., Christiansen, C. S., Iversen, N., Friedrichsen, M., Birk, J. B., Pilegaard, H., Hellsten, Y., Vaag, A., and Wojtaszewski, J. F. P. (2013) Effect of birth weight and 12 weeks of exercise training on exercise-induced AMPK signaling in human skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **304**, E1379–E1390
132. Lee-Young, R. S., Canny, B. J., Myers, D. E., and McConell, G. K. (2009) AMPK activation is fiber type specific in human skeletal muscle: effects of exercise and short-term exercise training. *J. Appl. Physiol.* **107**, 283–289
133. Coffey, V. G., Zhong, Z., Shield, A., Canny, B. J., Chibalin, A. V., Zierath, J. R., and Hawley, J. A. (2006) Early signaling responses to divergent exercise stimuli in skeletal muscle from well-trained humans. *FASEB J.* **20**, 190–192
134. Burgomaster, K. A., Howarth, K. R., Phillips, S. M., Rakobowchuk, M., Macdonald, M. J., McGee, S. L., and Gibala, M. J. (2008) Similar metabolic adaptations during exercise after low volume sprint interval and traditional endurance training in humans. *J. Physiol.* **586**, 151–160
135. Frøsig, C., Jørgensen, S. B., Hardie, D. G., Richter, E. A., and Wojtaszewski, J. F. (2004) 5'-AMP-activated protein kinase activity and protein expression are regulated by endurance training in human skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **286**, E411–E417
136. Lendoye, E., Sibille, B., Rousseau, A. S., Murdaca, J., Grimaldi, P. A., and Lopez, P. (2011) PPARbeta activation induces rapid changes of both AMPK subunit expression and AMPK activation in mouse skeletal muscle. *Mol. Endocrinol.* **25**, 1487–1498
137. Winder, W. W., and Hardie, D. G. (1999) AMP-activated protein kinase, a metabolic master switch: possible roles in type 2 diabetes. *Am. J. Physiol.* **277**, E1–E10
138. Grahame Hardie, D. (2016) Regulation of AMP-activated protein kinase by natural and synthetic activators. *Acta Pharm. Sin.* **B6**, 1–19
139. Sullivan, J. E., Carey, F., Carling, D., and Beri, R. K. (1994) Characterisation of 5'-AMP-activated protein kinase in human liver using specific peptide substrates and the effects of 5'-AMP analogues on enzyme activity. *Biochem. Biophys. Res. Commun.* **200**, 1551–1556
140. Gadalla, A. E., Pearson, T., Currie, A. J., Dale, N., Hawley, S. A., Sheehan, M., Hirst, W., Michel, A. D., Randall, A., Hardie, D. G., and Frenguelli, B. G. (2004) AICA riboside both activates AMP-activated protein kinase and competes with adenosine for the nucleoside transporter in the CA1 region of the rat hippocampus. *J. Neurochem.* **88**, 1272–1282
141. Day, P., Sharff, A., Parra, L., Cleasby, A., Williams, M., Hörer, S., Nar, H., Redemann, N., Tickle, I., and Yon, J. (2007) Structure of a CBS-domain pair from the regulatory γ 1 subunit of human AMPK in complex with AMP and ZMP. *Acta Crystallogr. D Biol. Crystallogr.* **63**, 587–596

142. Adams, J., Chen, Z.-P., Van Denderen, B. J. W., Morton, C. J., Parker, M. W., Witters, L. A., Stapleton, D., and Kemp, B. E. (2004) Intrasteric control of AMPK via the gamma1 subunit AMP allosteric regulatory site. *Protein Sci.* **13**, 155–165
143. Daiguan-Fornier, B., and Pinson, B. (2012) 5-Aminoimidazole-4-carboxamide-1-beta-D-ribofuranosyl 5'-monophosphate (AICAR), a highly conserved purine intermediate with multiple effects. *Metabolites* **2**, 292–302
144. Dixon, R., Gourzis, J., McDermott, D., Fujitaki, J., Dewland, P., and Gruber, H. (1991) AICA-riboside: safety, tolerance, and pharmacokinetics of a novel adenosine-regulating agent. *J. Clin. Pharmacol.* **31**, 342–347
145. Jørgensen, S. B., Viollet, B., Andreelli, F., Frøsig, C., Birk, J. B., Schjerling, P., Vaulont, S., Richter, E. A., and Wojtaszewski, J. F. (2004) Knockout of the alpha2 but not alpha1 5'-AMP-activated protein kinase isoform abolishes 5-aminoimidazole-4-carboxamide-1-beta-4-ribofuranosidebut not contraction-induced glucose uptake in skeletal muscle. *J. Biol. Chem.* **279**, 1070–1079
146. Steinberg, G. R., O'Neill, H. M., Dzamko, N. L., Galic, S., Naim, T., Koopman, R., Jørgensen, S. B., Honeyman, J., Hewitt, K., Chen, Z. P., Schertzer, J. D., Scott, J. W., Koentgen, F., Lynch, G. S., Watt, M. J., van Denderen, B. J., Campbell, D. J., and Kemp, B. E. (2010) Whole body deletion of AMP-activated protein kinase $\beta 2$ reduces muscle AMPK activity and exercise capacity. *J. Biol. Chem.* **285**, 37198–37209
147. Gómez-Galeno, J. E., Dang, Q., Nguyen, T. H., Boyer, S. H., Grote, M. P., Sun, Z., Chen, M., Craig, W. A., van Poelje, P. D., MacKenna, D. A., Cable, E. E., Rolzin, P. A., Finn, P. D., Chi, B., Linemeyer, D. L., Hecker, S. J., and Erion, M. D. (2010) A potent and selective AMPK activator that inhibits de novo lipogenesis. *ACS Med. Chem. Lett.* **1**, 478–482
148. Hunter, R. W., Foretz, M., Bultot, L., Fullerton, M. D., Deak, M., Ross, F. A., Hawley, S. A., Shpiro, N., Viollet, B., Barron, D., Kemp, B. E., Steinberg, G. R., Hardie, D. G., and Sakamoto, K. (2014) Mechanism of action of compound-13: an $\alpha 1$ -selective small molecule activator of AMPK. *Chem. Biol.* **21**, 866–879
149. Langendorf, C. G., Ngoei, K. R. W., Scott, J. W., Ling, N. X. Y., Issa, S. M. A., Gorman, M. A., Parker, M. W., Sakamoto, K., Oakhill, J. S., and Kemp, B. E. (2016) Structural basis of allosteric and synergistic activation of AMPK by furan-2-phosphonic derivative C2 binding. *Nat. Commun.* **7**, 10912
150. Owen, M. R., Doran, E., and Halestrap, A. P. (2000) Evidence that metformin exerts its anti-diabetic effects through inhibition of complex I of the mitochondrial respiratory chain. *Biochem. J.* **348**, 607–614
151. Lee, Y. S., Kim, W. S., Kim, K. H., Yoon, M. J., Cho, H. J., Shen, Y., Ye, J. M., Lee, C. H., Oh, W. K., Kim, C. T., Hohnen-Behrens, C., Gosby, A., Kraegen, E. W., James, D. E., and Kim, J. B. (2006) Berberine, a natural plant product, activates AMP-activated protein kinase with beneficial metabolic effects in diabetic and insulin-resistant states. *Diabetes* **55**, 2256–2264
152. Zheng, J., and Ramirez, V. D. (2000) Inhibition of mitochondrial proton F₀F₁-ATPase/ATP synthase by polyphenolic phytochemicals. *Br. J. Pharmacol.* **130**, 1115–1123
153. Hawley, S. A., Ross, F. A., Chevtzoff, C., Green, K. A., Evans, A., Fogarty, S., Towler, M. C., Brown, L. J., Ogunbayo, O. A., Evans, A. M., and Hardie, D. G. (2010) Use of cells expressing γ subunit variants to identify diverse mechanisms of AMPK activation. *Cell Metab.* **11**, 554–565
154. Sarabia, V., Lam, L., Burdett, E., Leiter, L. A., and Klip, A. (1992) Glucose transport in human skeletal muscle cells in culture. Stimulation by insulin and metformin. *J. Clin. Invest.* **90**, 1386–1395
155. Kristensen, J. M., Treebak, J. T., Schjerling, P., Goodyear, L., and Wojtaszewski, J. F. (2014) Two weeks of metformin treatment induces AMPK-dependent enhancement of insulin-stimulated glucose uptake in mouse soleus muscle. *Am. J. Physiol. Endocrinol. Metab.* **306**, E1099–E1109
156. Cool, B., Zinker, B., Chiou, W., Kifle, L., Cao, N., Perham, M., Dickinson, R., Adler, A., Gagne, G., Iyengar, R., Zhao, G., Marsh, K., Kym, P., Jung, P., Camp, H. S., and Frevert, E. (2006) Identification and characterization of a small molecule AMPK activator that treats key components of type 2 diabetes and the metabolic syndrome. *Cell Metab.* **3**, 403–416
157. Calabrese, M. F., Rajamohan, F., Harris, M. S., Caspers, N. L., Magyar, R., Withka, J. M., Wang, H., Borzilleri, K. A., Sahasrabudhe, P. V., Hoth, L. R., Geoghegan, K. F., Han, S., Brown, J., Subashi, T. A., Reyes, A. R., Frisbie, R. K., Ward, J., Miller, R. A., Landro, J. A., Londregan, A. T., Carpino, P. A., Cabral, S., Smith, A. C., Conn, E. L., Cameron, K. O., Qiu, X., and Kurumbail, R. G. (2014) Structural basis for AMPK activation: natural and synthetic ligands regulate kinase activity from opposite poles by different molecular mechanisms. *Structure* **22**, 1161–1172
158. Sanders, M. J., Ali, Z. S., Hegarty, B. D., Heath, R., Snowden, M. A., and Carling, D. (2007) Defining the mechanism of activation of AMP-activated protein kinase by the small molecule A-769662, a member of the thienopyridone family. *J. Biol. Chem.* **282**, 32539–32548
159. Xiao, B., Sanders, M. J., Carmena, D., Bright, N. J., Haire, L. F., Underwood, E., Patel, B. R., Heath, R. B., Walker, P. A., Hallen, S., Giordanetto, F., Martin, S. R., Carling, D., and Gamblin, S. J. (2013) Structural basis of AMPK regulation by small molecule activators. *Nat. Commun.* **4**, 3017
160. Lai, Y.-C., Kviklyte, S., Vertommen, D., Lantier, L., Foretz, M., Viollet, B., Hallén, S., and Rider, M. H. (2014) A small-molecule benzimidazole derivative that potently activates AMPK to increase glucose transport in skeletal muscle: comparison with effects of contraction and other AMPK activators. *Biochem. J.* **460**, 363–375
161. Cokorinos, E. C., Delmore, J., Reyes, A. R., Albuquerque, B., Kjøbsted, R., Jørgensen, N. O., Tran, J.-L., Jatkar, A., Cialdea, K., Esquejo, R. M., Meissen, J., Calabrese, M. F., Cordes, J., Moccia, R., Tess, D., Salatto, C. T., Coskran, T. M., Opsahl, A. C., Flynn, D., Blatnik, M., Li, W., Kindt, E., Foretz, M., Viollet, B., Ward, J., Kurumbail, R. G., Kalgutkar, A. S., Wojtaszewski, J. F. P., Cameron, K. O., and Miller, R. A. (2017) Activation of skeletal muscle AMPK promotes glucose disposal and glucose lowering in non-human primates and mice. *Cell Metab.* **25**, 1147–1159.e10
162. Myers, R. W., Guan, H.-P., Ehrhart, J., Petrov, A., Prahalada, S., Tozzo, E., Yang, X., Kurtz, M. M., Trujillo, M., Gonzalez Trotter, D., Feng, D., Xu, S., Eiermann, G., Holahan, M. A., Rubins, D., Conarello, S., Niu, X., Souza, S. C., Miller, C., Liu, J., Lu, K., Feng, W., Li, Y., Painter, R. E., Milligan, J. A., He, H., Liu, F., Ogawa, A., Wisniewski, D., Rohm, R. J., Wang, L., Wang, L., Qian, Y., Zhu, W., Wang, H., Bennet, B., LaFranco Scheuch, L., Fernandez, G. E., Li, C., Klimas, M., Zhou, G., van Heek, M., Biftu, T., Weber, A., Kelley, D. E., Thornberry, N., Erion, M. D., Kemp, D. M., and Sebbat, I. K. (2017) Systemic pan-AMPK activator MK-8722 improves glucose homeostasis but induces cardiac hypertrophy. *Science* **357**, 507–511
163. Hawley, S. A., Fullerton, M. D., Ross, F. A., Schertzer, J. D., Chevtzoff, C., Walker, K. J., Pegg, M. W., Zibrova, D., Green, K. A., Mustard, J. J., Kemp, B. E., Sakamoto, K., Steinberg, G. R., and Hardie, D. G. (2012) The ancient drug salicylate directly activates AMP-activated protein kinase. *Science* **336**, 918–922
164. Serizawa, Y., Oshima, R., Yoshida, M., Sakon, I., Kitani, K., Goto, A., Tsuda, S., and Hayashi, T. (2014) Salicylate acutely stimulates 5'-AMP-activated protein kinase and insulin-independent glucose transport in rat skeletal muscles. *Biochem. Biophys. Res. Commun.* **453**, 81–85
165. Xiao, B., Heath, R., Saiu, P., Leiper, F. C., Leone, P., Jing, C., Walker, P. A., Haire, L., Eccleston, J. F., Davis, C. T., Martin, S. R., Carling, D., and Gamblin, S. J. (2007) Structural basis for AMP binding to mammalian AMP-activated protein kinase. *Nature* **449**, 496–500
166. Pedersen, B. K., and Saltin, B. (2015) Exercise as medicine: evidence for prescribing exercise as therapy in 26 different chronic diseases. *Scand. J. Med. Sci. Sports* **25**(Suppl 3), 1–72
167. Brandt, C., and Pedersen, B. K. (2010) The role of exercise-induced myokines in muscle homeostasis and the defense against chronic diseases. *J. Biomed. Biotechnol.* **2010**, 520258
168. Pedersen, B. K., and Febbraio, M. A. (2012) Muscles, exercise and obesity: skeletal muscle as a secretory organ. *Nat. Rev. Endocrinol.* **8**, 457–465
169. Whitham, M., and Febbraio, M. A. (2016) The ever-expanding myokineome: discovery challenges and therapeutic implications. *Nat. Rev. Drug Discov.* **15**, 719–729
170. Kern, P. A., Ranganathan, S., Li, C., Wood, L., and Ranganathan, G. (2001) Adipose tissue tumor necrosis factor and interleukin-6 expression in human obesity and insulin resistance. *Am. J. Physiol. Endocrinol. Metab.* **280**, E745–E751
171. Drenth, J. P., Van Uum, S. H., Van Deuren, M., Pesman, G. J., Van der Ven-Jongekrijg, J., and Van der Meer, J. W. (1995) Endurance run increases circulating IL-6 and IL-1ra but down-regulates ex vivo TNF-alpha and IL-1 beta production. *J. Appl. Physiol.* **79**, 1497–1503
172. Steensberg, A., van Hall, G., Osada, T., Sacchetti, M., Saltin, B., and Klarlund Pedersen, B. (2000) Production of interleukin-6 in contracting human skeletal muscles can account for the exercise-induced increase in plasma interleukin-6. *J. Physiol.* **529**, 237–242

173. Weigert, C., Düfer, M., Simon, P., Debre, E., Runge, H., Brodbeck, K., Häring, H. U., and Schleicher, E. D. (2007) Upregulation of IL-6 mRNA by IL-6 in skeletal muscle cells: role of IL-6 mRNA stabilization and Ca²⁺-dependent mechanisms. *Am. J. Physiol. Cell Physiol.* **293**, C1139–C1147
174. MacDonald, C., Wojtaszewski, J. F., Pedersen, B. K., Kiens, B., and Richter, E. A. (2003) Interleukin-6 release from human skeletal muscle during exercise: relation to AMPK activity. *J. Appl. Physiol.* **95**, 2273–2277
175. Lauritzen, H. P., Brandauer, J., Schjerling, P., Koh, H. J., Treebak, J. T., Hirshman, M. F., Galbo, H., and Goodyear, L. J. (2013) Contraction and AICAR stimulate IL-6 vesicle depletion from skeletal muscle fibers in vivo. *Diabetes* **62**, 3081–3092
176. Lihn, A. S., Pedersen, S. B., Lund, S., and Richelsen, B. (2008) The anti-diabetic AMPK activator AICAR reduces IL-6 and IL-8 in human adipose tissue and skeletal muscle cells. *Mol. Cell. Endocrinol.* **292**, 36–41
177. Glund, S., Treebak, J. T., Long, Y. C., Barres, R., Viollet, B., Wojtaszewski, J. F., and Zierath, J. R. (2009) Role of adenosine 5'-monophosphate-activated protein kinase in interleukin-6 release from isolated mouse skeletal muscle. *Endocrinology* **150**, 600–606
178. Carey, A. L., Steinberg, G. R., Macaulay, S. L., Thomas, W. G., Holmes, A. G., Ramm, G., Prelovsek, O., Hohnen-Behrens, C., Watt, M. J., James, D. E., Kemp, B. E., Pedersen, B. K., and Febbraio, M. A. (2006) Interleukin-6 increases insulin-stimulated glucose disposal in humans and glucose uptake and fatty acid oxidation in vitro via AMP-activated protein kinase. *Diabetes* **55**, 2688–2697
179. Ruderman, N. B., Keller, C., Richard, A. M., Saha, A. K., Luo, Z., Xiang, X., Giralt, M., Ritov, V. B., Menshikova, E. V., Kelley, D. E., Hidalgo, J., Pedersen, B. K., and Kelly, M. (2006) Interleukin-6 regulation of AMP-activated protein kinase: potential role in the systemic response to exercise and prevention of the metabolic syndrome. *Diabetes* **55**(Suppl 2), S48–S54
180. Kelly, M., Keller, C., Avilucea, P. R., Keller, P., Luo, Z., Xiang, X., Giralt, M., Hidalgo, J., Saha, A. K., Pedersen, B. K., and Ruderman, N. B. (2004) AMPK activity is diminished in tissues of IL-6 knockout mice: the effect of exercise. *Biochem. Biophys. Res. Commun.* **320**, 449–454
181. Zhang, C., Li, Y., Wu, Y., Wang, L., Wang, X., and Du, J. (2013) Interleukin-6/signal transducer and activator of transcription 3 (STAT3) pathway is essential for macrophage infiltration and myoblast proliferation during muscle regeneration. *J. Biol. Chem.* **288**, 1489–1499
182. Tsujinaka, T., Ebisui, C., Fujita, J., Kishibuchi, M., Morimoto, T., Ogawa, A., Katsume, A., Ohsugi, Y., Kominami, E., and Monden, M. (1995) Muscle undergoes atrophy in association with increase of lysosomal cathepsin activity in interleukin-6 transgenic mouse. *Biochem. Biophys. Res. Commun.* **207**, 168–174
183. Scheller, J., Chalaris, A., Schmidt-Arras, D., and Rose-John, S. (2011) The pro- and anti-inflammatory properties of the cytokine interleukin-6. *Biochim. Biophys. Acta* **1813**, 878–888
184. Chan, M. H., Carey, A. L., Watt, M. J., and Febbraio, M. A. (2004) Cytokine gene expression in human skeletal muscle during concentric contraction: evidence that IL-8, like IL-6, is influenced by glycogen availability. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **287**, R322–R327
185. Tamura, Y., Watanabe, K., Kantani, T., Hayashi, J., Ishida, N., and Kaneki, M. (2011) Upregulation of circulating IL-15 by treadmill running in healthy individuals: is IL-15 an endocrine mediator of the beneficial effects of endurance exercise? *Endocr. J.* **58**, 211–215
186. Brenmoehl, J., Albrecht, E., Komolka, K., Schering, L., Langhammer, M., Hoeflich, A., and Maak, S. (2014) Irisin is elevated in skeletal muscle and serum of mice immediately after acute exercise. *Int. J. Biol. Sci.* **10**, 338–349
187. Matthews, V. B., Åström, M. B., Chan, M. H. S., Bruce, C. R., Krabbe, K. S., Prelovsek, O., Åkerström, T., Yfanti, C., Broholm, C., Mortensen, O. H., Penkowa, M., Hojman, P., Zankari, A., Watt, M. J., Bruunsgaard, H., Pedersen, B. K., and Febbraio, M. A. (2009) Brain-derived neurotrophic factor is produced by skeletal muscle cells in response to contraction and enhances fat oxidation via activation of AMP-activated protein kinase [Erratum]. *Diabetologia* **52**, 1409–1418
188. Broholm, C., Mortensen, O. H., Nielsen, S., Åkerström, T., Zankari, A., Dahl, B., and Pedersen, B. K. (2008) Exercise induces expression of leukaemia inhibitory factor in human skeletal muscle. *J. Physiol.* **586**, 2195–2201
189. Abbott, M. J., Constantinescu, S., and Turcotte, L. P. (2012) AMP-activated protein kinase $\alpha 2$ is an essential signal in the regulation of insulin-stimulated fatty acid uptake in control-fed and high-fat-fed mice. *Exp. Physiol.* **97**, 603–617
190. Crane, J. D., MacNeil, L. G., Lally, J. S., Ford, R. J., Bujak, A. L., Brar, I. K., Kemp, B. E., Raha, S., Steinberg, G. R., and Tarnopolsky, M. A. (2015) Exercise-stimulated interleukin-15 is controlled by AMPK and regulates skin metabolism and aging. *Aging Cell* **14**, 625–634
191. Lally, J. S., Ford, R. J., Johar, J., Crane, J. D., Kemp, B. E., and Steinberg, G. R. (2015) Skeletal muscle AMPK is essential for the maintenance of FNDC5 expression. *Physiol. Rep.* **3**, 1–9
192. Huh, J. Y., Mougios, V., Kabasakalis, A., Fatouros, I., Siopi, A., Douroudos, I. I., Filippaios, A., Panagiotou, G., Park, K. H., and Mantzoros, C. S. (2014) Exercise-induced irisin secretion is independent of age or fitness level and increased irisin may directly modulate muscle metabolism through AMPK activation. *J. Clin. Endocrinol. Metab.* **99**, E2154–E2161
193. Brandt, N., O'Neill, H. M., Kleinert, M., Schjerling, P., Vernet, E., Steinberg, G. R., Richter, E. A., and Jørgensen, S. B. (2015) Leukemia inhibitory factor increases glucose uptake in mouse skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **309**, E142–E153
194. Ajuwon, K. M., and Spurlock, M. E. (2004) Direct regulation of lipolysis by interleukin-15 in primary pig adipocytes. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **287**, R608–R611
195. Krolopp, J. E., Thornton, S. M., and Abbott, M. J. (2016) IL-15 activates the Jak3/STAT3 signaling pathway to mediate glucose uptake in skeletal muscle cells. *Front. Physiol.* **7**, 626
196. Mu, J., Brozinick, J. T., Jr., Valladares, O., Bucan, M., and Birnbaum, M. J. (2001) A role for AMP-activated protein kinase in contraction- and hypoxia-regulated glucose transport in skeletal muscle. *Mol. Cell* **7**, 1085–1094
197. Dzamko, N., Schertzer, J. D., Ryall, J. G., Steel, R., Macaulay, S. L., Wee, S., Chen, Z.-P., Michell, B. J., Oakhill, J. S., Watt, M. J., Jørgensen, S. B., Lynch, G. S., Kemp, B. E., and Steinberg, G. R. (2008) AMPK-independent pathways regulate skeletal muscle fatty acid oxidation. *J. Physiol.* **586**, 5819–5831
198. Jeppesen, J., Maarbjerg, S. J., Jordy, A. B., Fritzen, A. M., Pehmøller, C., Sylow, L., Serup, A. K., Jessen, N., Thorsen, K., Prats, C., Qvortrup, K., Dyck, J. R. B., Hunter, R. W., Sakamoto, K., Thomson, D. M., Schjerling, P., Wojtaszewski, J. F., Richter, E. A., and Kiens, B. (2013) LKB1 regulates lipid oxidation during exercise independently of AMPK. *Diabetes* **62**, 1490–1499
199. Fujii, N., Hirshman, M. F., Kane, E. M., Ho, R. C., Peter, L. E., Seifert, M. M., and Goodyear, L. J. (2005) AMP-activated protein kinase $\alpha 2$ activity is not essential for contraction- and hyperosmolarity-induced glucose transport in skeletal muscle. *J. Biol. Chem.* **280**, 39033–39041
200. Merry, T. L., Steinberg, G. R., Lynch, G. S., and McConell, G. K. (2010) Skeletal muscle glucose uptake during contraction is regulated by nitric oxide and ROS independently of AMPK. *Am. J. Physiol. Endocrinol. Metab.* **298**, E577–E585
201. Jensen, T. E., Schjerling, P., Viollet, B., Wojtaszewski, J. F. P., and Richter, E. A. (2008) AMPK $\alpha 1$ activation is required for stimulation of glucose uptake by twitch contraction, but not by H₂O₂, in mouse skeletal muscle. *PLoS One* **3**, e2102
202. Lefort, N., St-Amand, E., Morasse, S., Côté, C. H., and Marette, A. (2008) The α -subunit of AMPK is essential for submaximal contraction-mediated glucose transport in skeletal muscle in vitro. *Am. J. Physiol. Endocrinol. Metab.* **295**, E1447–E1454
203. Glatz, J. F., Luiken, J. J., and Bonen, A. (2010) Membrane fatty acid transporters as regulators of lipid metabolism: implications for metabolic disease. *Physiol. Rev.* **90**, 367–417
204. Kurth-Kraczek, E. J., Hirshman, M. F., Goodyear, L. J., and Winder, W. W. (1999) 5' AMP-activated protein kinase activation causes GLUT4 translocation in skeletal muscle. *Diabetes* **48**, 1667–1671
205. Chavez, J. A., Roach, W. G., Keller, S. R., Lane, W. S., and Lienhard, G. E. (2008) Inhibition of GLUT4 translocation by Tbc1d1, a Rab GTPase-activating protein abundant in skeletal muscle, is partially relieved by AMP-activated protein kinase activation. *J. Biol. Chem.* **283**, 9187–9195
206. Pehmøller, C., Treebak, J. T., Birk, J. B., Chen, S., Mackintosh, C., Hardie, D. G., Richter, E. A., and Wojtaszewski, J. F. (2009) Genetic disruption of AMPK signaling abolishes both contraction- and insulin-stimulated TBC1D1 phosphorylation and 14-3-3 binding in mouse skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **297**, E665–E675
207. Frøsig, C., Pehmøller, C., Birk, J. B., Richter, E. A., and Wojtaszewski, J. F. (2010) Exercise-induced TBC1D1 Ser237 phosphorylation and 14-3-3 protein binding capacity in human skeletal muscle. *J. Physiol.* **588**, 4539–4548

208. Vichaiwong, K., Purohit, S., An, D., Toyoda, T., Jessen, N., Hirshman, M. F., and Goodyear, L. J. (2010) Contraction regulates site-specific phosphorylation of TBC1D1 in skeletal muscle. *Biochem. J.* **431**, 311–320
209. Kjøbsted, R., Pedersen, A. J., Hingst, J. R., Sabaratnam, R., Birk, J. B., Kristensen, J. M., Højlund, K., and Wojtaszewski, J. F. (2016) Intact regulation of the AMPK signaling network in response to exercise and insulin in skeletal muscle of male patients with type 2 diabetes: illumination of AMPK activation in recovery from exercise. *Diabetes* **65**, 1219–1230
210. Szekeres, F., Chadt, A., Tom, R. Z., Deshmukh, A. S., Chibalin, A. V., Bjornholm, M., Al-Hasani, H., and Zierath, J. R. (2012) The Rab-GTPase-activating protein TBC1D1 regulates skeletal muscle glucose metabolism. *Am. J. P. Endocrinol. Metab.* **303**, E524–E533
211. Dokas, J., Chadt, A., Nolden, T., Himmelbauer, H., Zierath, J. R., Joost, H. G., and Al-Hasani, H. (2013) Conventional knockout of Tbc1d1 in mice impairs insulin- and AICAR-stimulated glucose uptake in skeletal muscle. *Endocrinology* **154**, 3502–3514
212. Stöckli, J., Meoli, C. C., Hoffman, N. J., Fazakerley, D. J., Pant, H., Cslasby, M. E., Ma, X., Kleinert, M., Brandon, A. E., Lopez, J. A., Cooney, G. J., and James, D. E. (2015) The RabGAP TBC1D1 plays a central role in exercise-regulated glucose metabolism in skeletal muscle. *Diabetes* **64**, 1914–1922
213. Chadt, A., Immisch, A., de Wendt, C., Springer, C., Zhou, Z., Stermann, T., Holman, G. D., Löffing-Cueni, D., Löffing, J., Joost, H.-G., and Al-Hasani, H. (2015) Deletion of both Rab-GTPase-activating proteins TBC1D1 and TBC1D4 in mice eliminates insulin- and AICAR-stimulated glucose transport [corrected]. *Diabetes* **64**, 746–759
214. McMahon, J. M., Signori, E., Wells, K. E., Fazio, V. M., and Wells, D. J. (2001) Optimisation of electrotransfer of plasmid into skeletal muscle by pretreatment with hyaluronidase: increased expression with reduced muscle damage. *Gene Ther.* **8**, 1264–1270
215. Chen, L., Chen, Q., Xie, B., Quan, C., Sheng, Y., Zhu, S., Rong, P., Zhou, S., Sakamoto, K., MacKintosh, C., Wang, H. Y., and Chen, S. (2016) Disruption of the AMPK-TBC1D1 nexus increases lipogenic gene expression and causes obesity in mice via promoting IGF1 secretion. *Proc. Natl. Acad. Sci. USA* **113**, 7219–7224
216. Chen, Q., Xie, B., Zhu, S., Rong, P., Sheng, Y., Ducommun, S., Chen, L., Quan, C., Li, M., Sakamoto, K., MacKintosh, C., Chen, S., and Wang, H. Y. (2017) A Tbc1d1 (Ser231Ala)-knockin mutation partially impairs AICAR- but not exercise-induced muscle glucose uptake in mice. *Diabetologia* **60**, 336–345
217. O'Neill, H. M., Lally, J. S., Galic, S., Thomas, M., Azizi, P. D., Fullerton, M. D., Smith, B. K., Pulinilkunnil, T., Chen, Z., Samaan, M. C., Jorgensen, S. B., Dyck, J. R. B., Holloway, G. P., Hawke, T. J., van Denderen, B. J., Kemp, B. E., and Steinberg, G. R. (2014) AMPK phosphorylation of ACC2 is required for skeletal muscle fatty acid oxidation and insulin sensitivity in mice. *Diabetologia* **57**, 1693–1702
218. O'Neill, H. M., Lally, J. S., Galic, S., Pulinilkunnil, T., Ford, R. J., Dyck, J. R., van Denderen, B. J., Kemp, B. E., and Steinberg, G. R. (2015) Skeletal muscle ACC2 S212 phosphorylation is not required for the control of fatty acid oxidation during exercise. *Physiol. Rep.* **3**, 1–10
219. Jeppesen, J., Albers, P. H., Rose, A. J., Birk, J. B., Schjerling, P., Dzamko, N., Steinberg, G. R., and Kiens, B. (2011) Contraction-induced skeletal muscle FAT/CD36 trafficking and FA uptake is AMPK independent. *J. Lipid Res.* **52**, 699–711
220. Raney, M. A., and Turcotte, L. P. (2006) Regulation of contraction-induced FA uptake and oxidation by AMPK and ERK1/2 is intensity dependent in rodent muscle. *Am. J. Physiol. Endocrinol. Metab.* **291**, E1220–E1227
221. Rasmussen, B. B., Hancock, C. R., and Winder, W. W. (1998) Postexercise recovery of skeletal muscle malonyl-CoA, acetyl-CoA carboxylase, and AMP-activated protein kinase. *J. Appl. Physiol.* **85**, 1629–1634
222. Frøsig, C., Roepstorff, C., Brandt, N., Maarbjerg, S. J., Birk, J. B., Wojtaszewski, J. F. P., Richter, E. A., and Kiens, B. (2009) Reduced malonyl-CoA content in recovery from exercise correlates with improved insulin-stimulated glucose uptake in human skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **296**, E787–E795
223. Bar-Peled, L., and Sabatini, D. M. (2014) Regulation of mTORC1 by amino acids. *Trends Cell Biol.* **24**, 400–406
224. Yu, X., and Long, Y. C. (2015) Autophagy modulates amino acid signaling network in myotubes: differential effects on mTORC1 pathway and the integrated stress response. *FASEB J.* **29**, 394–407
225. Dibble, C. C., and Manning, B. D. (2013) Signal integration by mTORC1 coordinates nutrient input with biosynthetic output. *Nat. Cell Biol.* **15**, 555–564
226. Inoki, K., Zhu, T., and Guan, K.-L. (2003) TSC2 mediates cellular energy response to control cell growth and survival. *Cell* **115**, 577–590
227. Cheng, S. W., Fryer, L. G., Carling, D., and Shepherd, P. R. (2004) Thr2446 is a novel mammalian target of rapamycin (mTOR) phosphorylation site regulated by nutrient status. *J. Biol. Chem.* **279**, 15719–15722
228. Dreyer, H. C., Fujita, S., Cadenas, J. G., Chinkes, D. L., Volpi, E., and Rasmussen, B. B. (2006) Resistance exercise increases AMPK activity and reduces 4E-BP1 phosphorylation and protein synthesis in human skeletal muscle. *J. Physiol.* **576**, 613–624
229. Gwinn, D. M., Shackelford, D. B., Egan, D. F., Mihaylova, M. M., Mery, A., Vasquez, D. S., Turk, B. E., and Shaw, R. J. (2008) AMPK phosphorylation of raptor mediates a metabolic checkpoint. *Mol. Cell* **30**, 214–226
230. Horman, S., Browne, G., Krause, U., Patel, J., Vertommen, D., Bertrand, L., Lavoie, A., Hue, L., Proud, C., and Rider, M. (2002) Activation of AMP-activated protein kinase leads to the phosphorylation of elongation factor 2 and an inhibition of protein synthesis. *Curr. Biol.* **12**, 1419–1423
231. Aguilar, V., Alliouachene, S., Sotiropoulos, A., Sobering, A., Athea, Y., Djouadi, F., Miraux, S., Thiaudière, E., Foretz, M., Viollet, B., Dirolez, P., Bastin, J., Benit, P., Rustin, P., Carling, D., Sandri, M., Ventura-Clapier, R., and Pende, M. (2007) S6 kinase deletion suppresses muscle growth adaptations to nutrient availability by activating AMP kinase. *Cell Metab.* **5**, 476–487
232. Carlson, C. J., White, M. F., and Rondinone, C. M. (2004) Mammalian target of rapamycin regulates IRS-1 serine 307 phosphorylation. *Biochem. Biophys. Res. Commun.* **316**, 533–539
233. Harrington, L. S., Findlay, G. M., Gray, A., Tolkacheva, T., Wigfield, S., Rebholz, H., Barnett, J., Leslie, N. R., Cheng, S., Shepherd, P. R., Gout, I., Downes, C. P., and Lamb, R. F. (2004) The TSC1-2 tumor suppressor controls insulin-PI3K signaling via regulation of IRS proteins. *J. Cell Biol.* **166**, 213–223
234. Tremblay, F., Brûlé, S., Hee Um, S., Li, Y., Masuda, K., Roden, M., Sun, X.-J., Krebs, M., Polakiewicz, R. D., Thomas, G., and Marette, A. (2007) Identification of IRS-1 Ser-1101 as a target of S6K1 in nutrient- and obesity-induced insulin resistance. *Proc. Natl. Acad. Sci. USA* **104**, 14056–14061
235. Um, S. H., Frigerio, F., Watanabe, M., Picard, F., Joaquin, M., Sticker, M., Fumagalli, S., Allegrini, P. R., Kozma, S. C., Auwerx, J., and Thomas, G. (2004) Absence of S6K1 protects against age- and diet-induced obesity while enhancing insulin sensitivity. *Nature* **431**, 200–205
236. Zhang, Y., and Manning, B. D. (2015) mTORC1 signaling activates NRF1 to increase cellular proteasome levels. *Cell Cycle* **14**, 2011–2017
237. Bodine, S. C., Latres, E., Baumhueter, S., Lai, V. K., Nunez, L., Clarke, B. A., Poueymirou, W. T., Panaro, F. J., Na, E., Dharmarajan, K., Pan, Z. Q., Valenzuela, D. M., DeChiara, T. M., Stitt, T. N., Yancopoulos, G. D., and Glass, D. J. (2001) Identification of ubiquitin ligases required for skeletal muscle atrophy. *Science* **294**, 1704–1708
238. Gomes, M. D., Lecker, S. H., Jagoe, R. T., Navon, A., and Goldberg, A. L. (2001) Atrogin-1, a muscle-specific F-box protein highly expressed during muscle atrophy. *Proc. Natl. Acad. Sci. USA* **98**, 14440–14445
239. Greer, E. L., Oskoui, P. R., Banko, M. R., Maniar, J. M., Gygi, M. P., Gygi, S. P., and Brunet, A. (2007) The energy sensor AMP-activated protein kinase directly regulates the mammalian FOXO3 transcription factor. *J. Biol. Chem.* **282**, 30107–30119
240. Kamei, Y., Miura, S., Suzuki, M., Kai, Y., Mizukami, J., Taniguchi, T., Mochida, K., Hata, T., Matsuda, J., Aburatani, H., Nishino, I., and Ezaki, O. (2004) Skeletal muscle FOXO1 (FKHR) transgenic mice have less skeletal muscle mass, down-regulated Type I (slow twitch/red muscle) fiber genes, and impaired glycemic control. *J. Biol. Chem.* **279**, 41114–41123
241. Sandri, M., Sandri, C., Gilbert, A., Skurk, C., Calabria, E., Picard, A., Walsh, K., Schiaffino, S., Lecker, S. H., and Goldberg, A. L. (2004) Foxo transcription factors induce the atrophy-related ubiquitin ligase atrogin-1 and cause skeletal muscle atrophy. *Cell* **117**, 399–412
242. Mammucari, C., Milan, G., Romanello, V., Masiero, E., Rudolf, R., Del Piccolo, P., Burden, S. J., Di Lisi, R., Sandri, C., Zhao, J., Goldberg, A. L., Schiaffino, S., and Sandri, M. (2007) FoxO3 controls autophagy in skeletal muscle in vivo. *Cell Metab.* **6**, 458–471
243. Zhao, J., Brault, J. J., Schild, A., Cao, P., Sandri, M., Schiaffino, S., Lecker, S. H., and Goldberg, A. L. (2007) FoxO3 coordinately activates protein degradation by the autophagic/lysosomal and proteasomal pathways in atrophying muscle cells. *Cell Metab.* **6**, 472–483
244. Singh, R., and Cuervo, A. M. (2011) Autophagy in the cellular energetic balance. *Cell Metab.* **13**, 495–504

245. Chan, E. Y. (2009) mTORC1 phosphorylates the ULK1-mAtg13-FIP200 autophagy regulatory complex. *Sci. Signal.* **2**, pe51
246. Hosokawa, N., Hara, T., Kaizuka, T., Kishi, C., Takamura, A., Miura, Y., Iemura, S., Natsume, T., Takehana, K., Yamada, N., Guan, J. L., Oshiro, N., and Mizushima, N. (2009) Nutrient-dependent mTORC1 association with the ULK1-Atg13-FIP200 complex required for autophagy. *Mol. Biol. Cell* **20**, 1981–1991
247. Jung, C. H., Jun, C. B., Ro, S.-H., Kim, Y.-M., Otto, N. M., Cao, J., Kundu, M., and Kim, D.-H. (2009) ULK-Atg13-FIP200 complexes mediate mTOR signaling to the autophagy machinery. *Mol. Biol. Cell* **20**, 1992–2003
248. Mizushima, N., Yamamoto, A., Matsui, M., Yoshimori, T., and Ohsumi, Y. (2004) In vivo analysis of autophagy in response to nutrient starvation using transgenic mice expressing a fluorescent autophagosome marker. *Mol. Biol. Cell* **15**, 1101–1111
249. Egan, D. F., Shackelford, D. B., Mihaylova, M. M., Gelino, S., Kohnz, R. A., Mair, W., Vasquez, D. S., Joshi, A., Gwinn, D. M., Taylor, R., Asara, J. M., Fitzpatrick, J., Dillin, A., Viollet, B., Kundu, M., Hansen, M., and Shaw, R. J. (2011) Phosphorylation of ULK1 (hATG1) by AMP-activated protein kinase connects energy sensing to mitophagy. *Science* **331**, 456–461
250. Kim, J., Kundu, M., Viollet, B., and Guan, K.-L. (2011) AMPK and mTOR regulate autophagy through direct phosphorylation of Ulk1. *Nat. Cell Biol.* **13**, 132–141
251. Lee, J. W., Park, S., Takahashi, Y., and Wang, H. G. (2010) The association of AMPK with ULK1 regulates autophagy. *PLoS One* **5**, e15394
252. Shang, L., Chen, S., Du, F., Li, S., Zhao, L., and Wang, X. (2011) Nutrient starvation elicits an acute autophagic response mediated by Ulk1 dephosphorylation and its subsequent dissociation from AMPK. *Proc. Natl. Acad. Sci. USA* **108**, 4788–4793
253. Fritzen, A. M., Madsen, A. B., Kleinert, M., Treebak, J. T., Lundsgaard, A.-M., Jensen, T. E., Richter, E. A., Wojtaszewski, J., Kiens, B., and Frøsig, C. (2016) Regulation of autophagy in human skeletal muscle: effects of exercise, exercise training and insulin stimulation. *J. Physiol.* **594**, 745–761
254. Fritzen, A. M., Frøsig, C., Jeppesen, J., Jensen, T. E., Lundsgaard, A. M., Serup, A. K., Schjerling, P., Proud, C. G., Richter, E. A., and Kiens, B. (2016) Role of AMPK in regulation of LC3 lipidation as a marker of autophagy in skeletal muscle. *Cell. Signal.* **28**, 663–674
255. Chang, N. C., Nguyen, M., Bourdon, J., Risse, P. A., Martin, J., Daniale, G., Rizzuto, R., Petrof, B. J., and Shore, G. C. (2012) Bcl-2-associated autophagy regulator Naf-1 required for maintenance of skeletal muscle. *Hum. Mol. Genet.* **21**, 2277–2287
256. Masiero, E., Agatea, L., Mammucari, C., Blaauw, B., Loro, E., Komatsu, M., Metzger, D., Reggiani, C., Schiaffino, S., and Sandri, M. (2009) Autophagy is required to maintain muscle mass. *Cell Metab.* **10**, 507–515
257. Masiero, E., and Sandri, M. (2010) Autophagy inhibition induces atrophy and myopathy in adult skeletal muscles. *Autophagy* **6**, 307–309
258. Nemazany, I., Blaauw, B., Paolini, C., Caillaud, C., Protasi, F., Mueller, A., Proikas-Cezanne, T., Russell, R. C., Guan, K. L., Nishino, I., Sandri, M., Pende, M., and Panasyuk, G. (2013) Defects of Vps15 in skeletal muscles lead to autophagic vacuolar myopathy and lysosomal disease. *EMBO Mol. Med.* **5**, 870–890
259. Raben, N., Hill, V., Shea, L., Takikita, S., Baum, R., Mizushima, N., Ralston, E., and Plotz, P. (2008) Suppression of autophagy in skeletal muscle uncovers the accumulation of ubiquitinated proteins and their potential role in muscle damage in Pompe disease. *Hum. Mol. Genet.* **17**, 3897–3908
260. Sandri, M. (2010) Autophagy in health and disease: 3. Involvement of autophagy in muscle atrophy. *Am. J. Physiol. Cell Physiol.* **298**, C1291–C1297
261. Kaushik, S., and Cuervo, A. M. (2016) AMPK-dependent phosphorylation of lipid droplet protein PLIN2 triggers its degradation by CMA. *Autophagy* **12**, 432–438
262. Felig, P., Pozefsky, T., Marliss, E., and Cahill, G. F., Jr. (1970) Alanine: key role in gluconeogenesis. *Science* **167**, 1003–1004
263. Bujak, A. L., Crane, J. D., Lally, J. S., Ford, R. J., Kang, S. J., Rebalka, I. A., Green, A. E., Kemp, B. E., Hawke, T. J., Schertzer, J. D., and Steinberg, G. R. (2015) AMPK activation of muscle autophagy prevents fasting-induced hypoglycemia and myopathy during aging. *Cell Metab.* **21**, 883–890
264. Sica, V., Galluzzi, L., Bravo-San Pedro, J. M., Izzo, V., Maiuri, M. C., and Kroemer, G. (2015) Organelle-specific initiation of autophagy. *Mol. Cell* **59**, 522–539
265. Romanello, V., Guadagnin, E., Gomes, L., Roder, I., Sandri, C., Petersen, Y., Milan, G., Masiero, E., Del Piccolo, P., Foretz, M., Scorrano, L., Rudolf, R., and Sandri, M. (2010) Mitochondrial fission and remodelling contributes to muscle atrophy. *EMBO J.* **29**, 1774–1785
266. Tian, W., Li, W., Chen, Y., Yan, Z., Huang, X., Zhuang, H., Zhong, W., Chen, Y., Wu, W., Lin, C., Chen, H., Hou, X., Zhang, L., Sui, S., Zhao, B., Hu, Z., Li, L., and Feng, D. (2015) Phosphorylation of ULK1 by AMPK regulates translocation of ULK1 to mitochondria and mitophagy. *FEBS Lett.* **589**, 1847–1854
267. Ducommun, S., Deak, M., Sumpton, D., Ford, R. J., Núñez Galindo, A., Kussmann, M., Viollet, B., Steinberg, G. R., Foretz, M., Dayon, L., Morrice, N. A., and Sakamoto, K. (2015) Motif affinity and mass spectrometry proteomic approach for the discovery of cellular AMPK targets: identification of mitochondrial fission factor as a new AMPK substrate. *Cell. Signal.* **27**, 978–988
268. Toyama, E. Q., Herzig, S., Courchet, J., Lewis, T. L., Loson, O. C., Hellberg, K., Young, N. P., Chen, H., Polleux, F., Chan, D. C., and Shaw, R. J. (2016) AMP-activated protein kinase mediates mitochondrial fission in response to energy stress. *Science* **351**, 275–281
269. Laker, R. C., Drake, J. C., Wilson, R. J., Lira, V. A., Lewellen, B. M., Ryall, K. A., Fisher, C. C., Zhang, M., Saucerman, J. J., Goodyear, L. J., Kundu, M., and Yan, Z. (2017) Ampk phosphorylation of Ulk1 is required for targeting of mitochondria to lysosomes in exercise-induced mitophagy. *Nat. Commun.* **8**, 548
270. Chen, Y., Azad, M. B., and Gibson, S. B. (2009) Superoxide is the major reactive oxygen species regulating autophagy. *Cell Death Differ.* **16**, 1040–1052
271. Davies, K. J., Quintanilha, A. T., Brooks, G. A., and Packer, L. (1982) Free radicals and tissue damage produced by exercise. *Biochem. Biophys. Res. Commun.* **107**, 1198–1205
272. Powers, S. K., Duarte, J., Kavazis, A. N., and Talbert, E. E. (2010) Reactive oxygen species are signalling molecules for skeletal muscle adaptation. *Exp. Physiol.* **95**, 1–9
273. Ristow, M., Zarse, K., Oberbach, A., Klötting, N., Birringer, M., Kiehntopf, M., Stumvoll, M., Kahn, C. R., and Blüher, M. (2009) Antioxidants prevent health-promoting effects of physical exercise in humans. *Proc. Natl. Acad. Sci. USA* **106**, 8665–8670
274. Alexander, A., Cai, S. L., Kim, J., Nanez, A., Sahin, M., MacLean, K. H., Inoki, K., Guan, K. L., Shen, J., Person, M. D., Kuşewitt, D., Mills, G. B., Kastan, M. B., and Walker, C. L. (2010) ATM signals to TSC2 in the cytoplasm to regulate mTORC1 in response to ROS [Erratum]. *Proc. Natl. Acad. Sci. USA* **107**, 4153–4158
275. Tripathi, D. N., Chowdhury, R., Trudel, L. J., Tee, A. R., Slack, R. S., Walker, C. L., and Wogan, G. N. (2013) Reactive nitrogen species regulate autophagy through ATM-AMPK-TSC2-mediated suppression of mTORC1. *Proc. Natl. Acad. Sci. USA* **110**, E2950–E2957
276. Zhang, J., Kim, J., Alexander, A., Cai, S., Tripathi, D. N., Dere, R., Tee, A. R., Tait-Mulder, J., Di Nardo, A., Han, J. M., Kwiatkowski, E., Dunlop, E. A., Dodd, K. M., Folkert, R. D., Faust, P. L., Kastan, M. B., Sahin, M., and Walker, C. L. (2013) A tuberous sclerosis complex signalling node at the peroxisome regulates mTORC1 and autophagy in response to ROS. *Nat. Cell Biol.* **15**, 1186–1196
277. Zhang, J., Tripathi, D. N., Jing, J., Alexander, A., Kim, J., Powell, R. T., Dere, R., Tait-Mulder, J., Lee, J.-H., Paull, T. T., Pandita, R. K., Charaka, V. K., Pandita, T. K., Kastan, M. B., and Walker, C. L. (2015) ATM functions at the peroxisome to induce pexophagy in response to ROS. *Nat. Cell Biol.* **17**, 1259–1269
278. Zmijewski, J. W., Banerjee, S., Bae, H., Friggeri, A., Lazarowski, E. R., and Abraham, E. (2010) Exposure to hydrogen peroxide induces oxidation and activation of AMP-activated protein kinase. *J. Biol. Chem.* **285**, 33154–33164
279. Rahman, M., Mofarrahi, M., Kristof, A. S., Nkengfack, B., Harel, S., and Hussain, S. N. (2014) Reactive oxygen species regulation of autophagy in skeletal muscles. *Antioxid. Redox Signal.* **20**, 443–459
280. Li, L., Chen, Y., and Gibson, S. B. (2013) Starvation-induced autophagy is regulated by mitochondrial reactive oxygen species leading to AMPK activation. *Cell. Signal.* **25**, 50–65
281. Jiang, S., Park, D. W., Gao, Y., Ravi, S., Darley-Usmar, V., Abraham, E., and Zmijewski, J. W. (2015) Participation of proteasome-ubiquitin protein degradation in autophagy and the activation of AMP-activated protein kinase. *Cell. Signal.* **27**, 1186–1197
282. Hardie, D. G., and Ashford, M. L. (2014) AMPK: regulating energy balance at the cellular and whole body levels. *Physiology (Bethesda)* **29**, 99–107

283. Hood, D. A., Balaban, A., Connor, M. K., Craig, E. E., Nishio, M. L., Rezvani, M., and Takahashi, M. (1994) Mitochondrial biogenesis in striated muscle. *Can. J. Appl. Physiol.* **19**, 12–48
284. Jørgensen, S. B., Treebak, J. T., Viollet, B., Schjerling, P., Vaulont, S., Wojtaszewski, J. F., and Richter, E. A. (2007) Role of AMPK α 2 in basal, training-, and AICAR-induced GLUT4, hexokinase II, and mitochondrial protein expression in mouse muscle. *Am. J. Physiol. Endocrinol. Metab.* **292**, E331–E339
285. Röckl, K. S., Hirshman, M. F., Brandauer, J., Fujii, N., Witters, L. A., and Goodyear, L. J. (2007) Skeletal muscle adaptation to exercise training: AMP-activated protein kinase mediates muscle fiber type shift. *Diabetes* **56**, 2062–2069
286. Brandauer, J., Andersen, M. A., Kellezi, H., Risis, S., Frøsig, C., Vienberg, S. G., and Treebak, J. T. (2015) AMP-activated protein kinase controls exercise training- and AICAR-induced increases in SIRT3 and MnSOD. *Front. Physiol.* **6**, 85
287. Jäger, S., Handschin, C., St-Pierre, J., and Spiegelman, B. M. (2007) AMP-activated protein kinase (AMPK) action in skeletal muscle via direct phosphorylation of PGC-1 α . *Proc. Natl. Acad. Sci. USA* **104**, 12017–12022
288. Lerin, C., Rodgers, J. T., Kalume, D. E., Kim, S. H., Pandey, A., and Puigserver, P. (2006) GCN5 acetyltransferase complex controls glucose metabolism through transcriptional repression of PGC-1 α . *Cell Metab.* **3**, 429–438
289. Nemoto, S., Fergusson, M. M., and Finkel, T. (2005) SIRT1 functionally interacts with the metabolic regulator and transcriptional coactivator PGC-1 α . *J. Biol. Chem.* **280**, 16456–16460
290. Gerhart-Hines, Z., Rodgers, J. T., Bare, O., Lerin, C., Kim, S. H., Mostoslavsky, R., Alt, F. W., Wu, Z., and Puigserver, P. (2007) Metabolic control of muscle mitochondrial function and fatty acid oxidation through SIRT1/PGC-1 α . *EMBO J.* **26**, 1913–1923
291. Cantó, C., Gerhart-Hines, Z., Feige, J. N., Lagouge, M., Noriega, L., Milne, J. C., Elliott, P. J., Puigserver, P., and Auwerx, J. (2009) AMPK regulates energy expenditure by modulating NAD⁺ metabolism and SIRT1 activity. *Nature* **458**, 1056–1060
292. Edington, D. W., and McCafferty, W. B. (1973) Mitochondrial size distribution analysis in the soleus muscle of trained and aged rats. *Experientia* **29**, 692–693
293. Cantó, C., Jiang, L. Q., Deshmukh, A. S., Matak, C., Coste, A., Lagouge, M., Zierath, J. R., and Auwerx, J. (2010) Interdependence of AMPK and SIRT1 for metabolic adaptation to fasting and exercise in skeletal muscle. *Cell Metab.* **11**, 213–219
294. Olmos, Y., Sánchez-Gómez, F. J., Wild, B., García-Quintana, N., Cabezas, S., Lamas, S., and Monsalve, M. (2013) Sirt1 regulation of antioxidant genes is dependent on the formation of a FoxO3a/PGC-1 α complex. *Antioxid. Redox Signal.* **19**, 1507–1521
295. Sanchez, A. M., Csibi, A., Raibon, A., Cornille, K., Gay, S., Bernardi, H., and Candau, R. (2012) AMPK promotes skeletal muscle autophagy through activation of forkhead FoxO3a and interaction with Ulk1. *J. Cell. Biochem.* **113**, 695–710
296. Peserico, A., Chiacchiera, F., Grossi, V., Matrone, A., Latorre, D., Simonatto, M., Fusella, A., Ryall, J. G., Finley, L. W. S., Haigis, M. C., Villani, G., Puri, P. L., Sartorelli, V., and Simone, C. (2013) A novel AMPK-dependent FoxO3a-SIRT3 intramitochondrial complex sensing glucose levels. *Cell. Mol. Life Sci.* **70**, 2015–2029
297. Jöbsis, F. F., and Stainsby, W. N. (1968) Oxidation of NADH during contractions of circled mammalian skeletal muscle. *Respir. Physiol.* **4**, 292–300
298. Lin, L., Chen, K., Abdel Khalek, W., Ward III, J. L., Yang, H., Chabi, B., Wrutniak-Cabello, C., and Tong, Q. (2014) Regulation of skeletal muscle oxidative capacity and muscle mass by SIRT3. *PLoS One* **9**, e85636
299. Kong, X., Wang, R., Xue, Y., Liu, X., Zhang, H., Chen, Y., Fang, F., and Chang, Y. (2010) Sirtuin 3, a new target of PGC-1 α , plays an important role in the suppression of ROS and mitochondrial biogenesis. *PLoS One* **5**, e11707
300. Giralt, A., Hondares, E., Villena, J. A., Ribas, F., Díaz-Delfin, J., Giralt, M., Iglesias, R., and Villarroja, F. (2011) Peroxisome receptor- γ coactivator-1 α controls transcription of the Sirt3 gene, an essential component of the thermogenic brown adipocyte phenotype. *J. Biol. Chem.* **286**, 16958–16966
301. Bulter, M., Aatsinki, S. M., Izzi, V., and Hakkola, J. (2012) Metformin reduces hepatic expression of SIRT3, the mitochondrial deacetylase controlling energy metabolism. *PLoS One* **7**, e49863
302. Lan, F., Cacicedo, J. M., Ruderman, N., and Ido, Y. (2008) SIRT1 modulation of the acetylation status, cytosolic localization, and activity of LKB1: possible role in AMP-activated protein kinase activation. *J. Biol. Chem.* **283**, 27628–27635
303. Price, N. L., Gomes, A. P., Ling, A. J., Duarte, F. V., Martin-Montalvo, A., North, B. J., Agarwal, B., Ye, L., Ramadori, G., Teodoro, J. S., Hubbard, B. P., Varela, A. T., Davis, J. G., Varamini, B., Hafner, A., Moaddel, R., Rolo, A. P., Coppari, R., Palmeira, C. M., de Cabo, R., Baur, J. A., and Sinclair, D. A. (2012) SIRT1 is required for AMPK activation and the beneficial effects of resveratrol on mitochondrial function. *Cell Metab.* **15**, 675–690
304. Philp, A., Chen, A., Lan, D., Meyer, G. A., Murphy, A. N., Knapp, A. E., Olfert, I. M., McCurdy, C. E., Marcotte, G. R., Hogan, M. C., Baar, K., and Schenk, S. (2011) Sirtuin 1 (SIRT1) deacetylase activity is not required for mitochondrial biogenesis or peroxisome proliferator-activated receptor- γ coactivator-1 α (PGC-1 α) deacetylation following endurance exercise. *J. Biol. Chem.* **286**, 30561–30570
305. Lützner, N., Kalbacher, H., Krones-Herzig, A., and Rösl, F. (2012) FOXO3 is a glucocorticoid receptor target and regulates LKB1 and its own expression based on cellular AMP levels via a positive autoregulatory loop. *PLoS One* **7**, e42166
306. Lützner, N., De-Castro Arce, J., and Rösl, F. (2012) Gene expression of the tumour suppressor LKB1 is mediated by Sp1, NF-Y and FOXO transcription factors. *PLoS One* **7**, e32590
307. Bitterman, K. J., Anderson, R. M., Cohen, H. Y., Latorre-Esteves, M., and Sinclair, D. A. (2002) Inhibition of silencing and accelerated aging by nicotinamide, a putative negative regulator of yeast sir2 and human SIRT1. *J. Biol. Chem.* **277**, 45099–45107
308. Revollo, J. R., Grimm, A. A., and Imai, S. (2004) The NAD biosynthesis pathway mediated by nicotinamide phosphoribosyltransferase regulates Sir2 activity in mammalian cells. *J. Biol. Chem.* **279**, 50754–50763
309. Garten, A., Petzold, S., Körner, A., Imai, S., and Kiess, W. (2009) Namp1: linking NAD biology, metabolism and cancer. *Trends Endocrinol. Metab.* **20**, 130–138
310. Lau, C., Niere, M., and Ziegler, M. (2009) The NMN/NaMN adenylyltransferase (NMNAT) protein family. *Front. Biosci. (Landmark Ed.)* **14**, 410–431
311. Brandauer, J., Vienberg, S. G., Andersen, M. A., Ringholm, S., Risis, S., Larsen, P. S., Kristensen, J. M., Frøsig, C., Leick, L., Fentz, J., Jørgensen, S., Kiens, B., Wojtaszewski, J. F., Richter, E. A., Zierath, J. R., Goodyear, L. J., Pilegaard, H., and Treebak, J. T. (2013) AMP-activated protein kinase regulates nicotinamide phosphoribosyl transferase expression in skeletal muscle. *J. Physiol.* **591**, 5207–5220
312. Um, J. H., Pendergast, J. S., Springer, D. A., Foretz, M., Viollet, B., Brown, A., Kim, M. K., Yamazaki, S., and Chung, J. H. (2011) AMPK regulates circadian rhythms in a tissue- and isoform-specific manner. *PLoS One* **6**, e18450
313. Vieira, E., Nilsson, E. C., Nerstedt, A., Ormestad, M., Long, Y. C., Garcia-Roves, P. M., Zierath, J. R., and Mahlapuu, M. (2008) Relationship between AMPK and the transcriptional balance of clock-related genes in skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **295**, E1032–E1037
314. Ramsey, K. M., Yoshino, J., Brace, C. S., Abrassart, D., Kobayashi, Y., Marcheva, B., Hong, H.-K., Chong, J. L., Buhr, E. D., Lee, C., Takahashi, J. S., Imai, S., and Bass, J. (2009) Circadian clock feedback cycle through NAMPT-mediated NAD(+) biosynthesis. *Science* **324**, 651–654
315. Nakahata, Y., Sahar, S., Astarita, G., Kaluzova, M., and Sassone-Corsi, P. (2009) Circadian control of the NAD⁺ salvage pathway by CLOCK-SIRT1. *Science* **324**, 654–657
316. Lamia, K. A., Sachdeva, U. M., DiTacchio, L., Williams, E. C., Alvarez, J. G., Egan, D. F., Vasquez, D. S., Juguilon, H., Panda, S., Shaw, R. J., Thompson, C. B., and Evans, R. M. (2009) AMPK regulates the circadian clock by cryptochrome phosphorylation and degradation. *Science* **326**, 437–440
317. Frederick, D. W., Davis, J. G., Dávila, A., Jr., Agarwal, B., Michan, S., Puchowicz, M. A., Nakamaru-Ogiso, E., and Baur, J. A. (2015) Increasing NAD synthesis in muscle via nicotinamide phosphoribosyltransferase is not sufficient to promote oxidative metabolism. *J. Biol. Chem.* **290**, 1546–1558
318. Gomes, A. P., Price, N. L., Ling, A. J., Moslehi, J. J., Montgomery, M. K., Rajman, L., White, J. P., Teodoro, J. S., Wrann, C. D., Hubbard, B. P., Mercken, E. M., Palmeira, C. M., de Cabo, R., Rolo, A. P., Turner, N., Bell, E. L., and Sinclair, D. A. (2013) Declining NAD(+) induces a pseudohypoxic state disrupting nuclear-mitochondrial communication during aging. *Cell* **155**, 1624–1638

319. Johnson, M. L., Irving, B. A., Lanza, I. R., Vendelbo, M. H., Konopka, A. R., Robinson, M. M., Henderson, G. C., Klaus, K. A., Morse, D. M., Heppelmann, C., Bergen III, H. R., Dasari, S., Schimke, J. M., Jakaitis, D. R., and Nair, K. S. (2015) Differential effect of endurance training on mitochondrial protein damage, degradation, and acetylation in the context of aging. *J. Gerontol. A Biol. Sci. Med. Sci.* **70**, 1386–1393
320. Khan, N. A., Auranen, M., Paetau, I., Pirinen, E., Euro, L., Forsström, S., Pasila, L., Velagapudi, V., Carroll, C. J., Auwerx, J., and Suomalainen, A. (2014) Effective treatment of mitochondrial myopathy by nicotinamide riboside, a vitamin B3. *EMBO Mol. Med.* **6**, 721–731
321. Sriwijitkamol, A., Coletta, D. K., Wajsborg, E., Balbontin, G. B., Reyna, S. M., Barrientes, J., Eagan, P. A., Jenkinson, C. P., Cersosimo, E., DeFronzo, R. A., Sakamoto, K., and Musi, N. (2007) Effect of acute exercise on AMPK signaling in skeletal muscle of subjects with type 2 diabetes: a time-course and dose-response study. *Diabetes* **56**, 836–848
322. Lee-Young, R. S., Koufogiannis, G., Canny, B. J., and McConell, G. K. (2008) Acute exercise does not cause sustained elevations in AMPK signaling or expression. *Med. Sci. Sports Exerc.* **40**, 1490–1494
323. Hellsten, Y., Richter, E. A., Kiens, B., and Bangsbo, J. (1999) AMP deamination and purine exchange in human skeletal muscle during and after intense exercise. *J. Physiol.* **520**, 909–920
324. Pilegaard, H., Ordway, G. A., Saltin, B., and Neufer, P. D. (2000) Transcriptional regulation of gene expression in human skeletal muscle during recovery from exercise. *Am. J. Physiol. Endocrinol. Metab.* **279**, E806–E814
325. Mahoney, D. J., Parise, G., Melov, S., Safdar, A., and Tarnopolsky, M. A. (2005) Analysis of global mRNA expression in human skeletal muscle during recovery from endurance exercise. *FASEB J.* **19**, 1498–1500
326. Scarpulla, R. C. (2002) Nuclear activators and coactivators in mammalian mitochondrial biogenesis. *Biochim. Biophys. Acta* **1576**, 1–14
327. McGee, S. L., van Denderen, B. J., Howlett, K. F., Mollica, J., Schertzer, J. D., Kemp, B. E., and Hargreaves, M. (2008) AMP-activated protein kinase regulates GLUT4 transcription by phosphorylating histone deacetylase 5. *Diabetes* **57**, 860–867
328. Thomson, D. M., Herway, S. T., Fillmore, N., Kim, H., Brown, J. D., Barrow, J. R., and Winder, W. W. (2008) AMP-activated protein kinase phosphorylates transcription factors of the CREB family. *J. Appl. Physiol.* (1985). **104**, 429–438
329. Jørgensen, S. B., Wojtaszewski, J. F. P., Viollet, B., Andreelli, F., Birk, J. B., Hellsten, Y., Schjerling, P., Vaulont, S., Neufer, P. D., Richter, E. A., and Pilegaard, H. (2005) Effects of α -AMPK knockout on exercise-induced gene activation in mouse skeletal muscle. *FASEB J.* **19**, 1146–1148
330. Holmes, B. F., Lang, D. B., Birnbaum, M. J., Mu, J., and Dohm, G. L. (2004) AMP kinase is not required for the GLUT4 response to exercise and denervation in skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **287**, E739–E743
331. Fentz, J., Kjøbsted, R., Kristensen, C. M., Hingst, J. R., Birk, J. B., Gudiksen, A., Foretz, M., Schjerling, P., Viollet, B., Pilegaard, H., and Wojtaszewski, J. F. (2015) AMPK α is essential for acute exercise-induced gene responses but not for exercise training-induced adaptations in mouse skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **309**, E900–E914
332. Wojtaszewski, J. F., and Richter, E. A. (2006) Effects of acute exercise and training on insulin action and sensitivity: focus on molecular mechanisms in muscle. *Essays Biochem.* **42**, 31–46
333. Richter, E. A., Garetto, L. P., Goodman, M. N., and Ruderman, N. B. (1982) Muscle glucose metabolism following exercise in the rat: increased sensitivity to insulin. *J. Clin. Invest.* **69**, 785–793
334. Richter, E. A., Mikines, K. J., Galbo, H., and Kiens, B. (1989) Effect of exercise on insulin action in human skeletal muscle. *J. Appl. Physiol.* **66**, 876–885
335. Cartee, G. D., and Holloszy, J. O. (1990) Exercise increases susceptibility of muscle glucose transport to activation by various stimuli. *Am. J. Physiol.* **258**, E390–E393
336. Cartee, G. D., Young, D. A., Sleeper, M. D., Zierath, J., Wallberg-Henriksson, H., and Holloszy, J. O. (1989) Prolonged increase in insulin-stimulated glucose transport in muscle after exercise. *Am. J. Physiol.* **256**, E494–E499
337. Mikines, K. J., Sonne, B., Farrell, P. A., Tronier, B., and Galbo, H. (1988) Effect of physical exercise on sensitivity and responsiveness to insulin in humans. *Am. J. Physiol.* **254**, E248–E259
338. Douen, A. G., Ramlal, T., Rastogi, S., Bilan, P. J., Cartee, G. D., Vranic, M., Holloszy, J. O., and Klip, A. (1990) Exercise induces recruitment of the “insulin-responsive glucose transporter”: evidence for distinct intracellular insulin- and exercise-recruitable transporter pools in skeletal muscle. *J. Biol. Chem.* **265**, 13427–13430
339. Klip, A., Ramlal, T., Young, D. A., and Holloszy, J. O. (1987) Insulin-induced translocation of glucose transporters in rat hindlimb muscles. *FEBS Lett.* **224**, 224–230
340. Zisman, A., Peroni, O. D., Abel, E. D., Michael, M. D., Mauvais-Jarvis, F., Lowell, B. B., Wojtaszewski, J. F., Hirshman, M. F., Virkamaki, A., Goodyear, L. J., Kahn, C. R., and Kahn, B. B. (2000) Targeted disruption of the glucose transporter 4 selectively in muscle causes insulin resistance and glucose intolerance. *Nat. Med.* **6**, 924–928
341. Hansen, P. A., Nolte, L. A., Chen, M. M., and Holloszy, J. O. (1998) Increased GLUT-4 translocation mediates enhanced insulin sensitivity of muscle glucose transport after exercise. *J. Appl. Physiol.* (1985). **85**, 1218–1222
342. Richter, E. A., Garetto, L. P., Goodman, M. N., and Ruderman, N. B. (1984) Enhanced muscle glucose metabolism after exercise: modulation by local factors. *Am. J. Physiol.* **246**, E476–E482
343. Funai, K., Schweitzer, G. G., Sharma, N., Kanzaki, M., and Cartee, G. D. (2009) Increased AS160 phosphorylation, but not TBC1D1 phosphorylation, with increased postexercise insulin sensitivity in rat skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **297**, E242–E251
344. Funai, K., Schweitzer, G. G., Castorena, C. M., Kanzaki, M., and Cartee, G. D. (2010) In vivo exercise followed by in vitro contraction additively elevates subsequent insulin-stimulated glucose transport by rat skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **298**, E999–E1010
345. Hamada, T., Arias, E. B., and Cartee, G. D. (2006) Increased submaximal insulin-stimulated glucose uptake in mouse skeletal muscle after treadmill exercise. *J. Appl. Physiol.* **101**, 1368–1376
346. Wojtaszewski, J. F. P., Hansen, B. F., Kiens, B., and Richter, E. A. (1997) Insulin signaling in human skeletal muscle: time course and effect of exercise. *Diabetes* **46**, 1775–1781
347. Wojtaszewski, J. F., Hansen, B. F., Gade, Kiens, B., Markuns, J. F., Goodyear, L. J., and Richter, E. A. (2000) Insulin signaling and insulin sensitivity after exercise in human skeletal muscle. *Diabetes* **49**, 325–331
348. Arias, E. B., Kim, J., Funai, K., and Cartee, G. D. (2007) Prior exercise increases phosphorylation of Akt substrate of 160 kDa (AS160) in rat skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **292**, E1191–E1200
349. Kane, S., Sano, H., Liu, S. C. H., Asara, J. M., Lane, W. S., Garner, C. C., and Lienhard, G. E. (2002) A method to identify serine kinase substrates: Akt phosphorylates a novel adipocyte protein with a Rab GTPase-activating protein (GAP) domain. *J. Biol. Chem.* **277**, 22115–22118
350. Sano, H., Kane, S., Sano, E., Miinea, C. P., Asara, J. M., Lane, W. S., Garner, C. W., and Lienhard, G. E. (2003) Insulin-stimulated phosphorylation of a Rab GTPase-activating protein regulates GLUT4 translocation. *J. Biol. Chem.* **278**, 14599–14602
351. Bruss, M. D., Arias, E. B., Lienhard, G. E., and Cartee, G. D. (2005) Increased phosphorylation of Akt substrate of 160 kDa (AS160) in rat skeletal muscle in response to insulin or contractile activity. *Diabetes* **54**, 41–50
352. Treebak, J. T., Frøsig, C., Pehmøller, C., Chen, S., Maarbjerg, S. J., Brandt, N., MacKintosh, C., Zierath, J. R., Hardie, D. G., Kiens, B., Richter, E. A., Pilegaard, H., and Wojtaszewski, J. F. (2009) Potential role of TBC1D4 in enhanced post-exercise insulin action in human skeletal muscle. *Diabetologia* **52**, 891–900
353. Treebak, J. T., Glund, S., Deshmukh, A., Klein, D. K., Long, Y. C., Jensen, T. E., Jørgensen, S. B., Viollet, B., Andersson, L., Neumann, D., Wallimann, T., Richter, E. A., Chibalin, A. V., Zierath, J. R., and Wojtaszewski, J. F. (2006) AMPK-mediated AS160 phosphorylation in skeletal muscle is dependent on AMPK catalytic and regulatory subunits. *Diabetes* **55**, 2051–2058
354. Fisher, J. S., Gao, J., Han, D.-H., Holloszy, J. O., and Nolte, L. A. (2002) Activation of AMP kinase enhances sensitivity of muscle glucose transport to insulin. *Am. J. Physiol. Endocrinol. Metab.* **282**, E18–E23
355. Kjøbsted, R., Munk-Hansen, N., Birk, J. B., Foretz, M., Viollet, B., Bjørnholm, M., Zierath, J. R., Treebak, J. T., and Wojtaszewski, J. F. (2017) Enhanced muscle insulin sensitivity after contraction/exercise is mediated by AMPK. *Diabetes* **66**, 598–612
356. Yang, J., and Holman, G. D. (2005) Insulin and contraction stimulate exocytosis, but increased AMP-activated protein kinase activity resulting from oxidative metabolism stress slows endocytosis of GLUT4 in cardiomyocytes. *J. Biol. Chem.* **280**, 4070–4078

357. Romijn, J. A., Coyle, E. F., Sidossis, L. S., Gastaldelli, A., Horowitz, J. F., Endert, E., and Wolfe, R. R. (1993) Regulation of endogenous fat and carbohydrate metabolism in relation to exercise intensity and duration. *Am. J. Physiol.* **265**, E380–E391
358. Kiens, B., and Richter, E. A. (1998) Utilization of skeletal muscle triacylglycerol during postexercise recovery in humans. *Am. J. Physiol.* **275**, E332–E337
359. Kimber, N. E., Heigenhauser, G. J., Spriet, L. L., and Dyck, D. J. (2003) Skeletal muscle fat and carbohydrate metabolism during recovery from glycogen-depleting exercise in humans. *J. Physiol.* **548**, 919–927
360. Roach, P. J., Depaoli-Roach, A. A., Hurley, T. D., and Tagliabracchi, V. S. (2012) Glycogen and its metabolism: some new developments and old themes. *Biochem. J.* **441**, 763–787
361. Friedman, D. L., and Lerner, J. (1963) Studies on UDPG-alpha-glucan transglucosylase. III. Interconversion of two forms of muscle UDPG-alpha-glucan transglucosylase by a phosphorylation-dephosphorylation reaction sequence. *Biochemistry* **2**, 669–675
362. Leloir, L. F., Olavarria, J. M., Goldemberg, S. H., and Carminatti, H. (1959) Biosynthesis of glycogen from uridine diphosphate glucose. *Arch. Biochem. Biophys.* **81**, 508–520
363. Carling, D., and Hardie, D. G. (1989) The substrate and sequence specificity of the AMP-activated protein kinase: phosphorylation of glycogen synthase and phosphorylase kinase. *Biochim. Biophys. Acta* **1012**, 81–86
364. Hunter, R. W., Treebak, J. T., Wojtaszewski, J. F., and Sakamoto, K. (2011) Molecular mechanism by which AMP-activated protein kinase activation promotes glycogen accumulation in muscle. *Diabetes* **60**, 766–774
365. McBride, A., and Hardie, D. G. (2009) AMP-activated protein kinase: a sensor of glycogen as well as AMP and ATP? *Acta Physiol. (Oxf.)* **196**, 99–113
366. Li, X., Wang, L., Zhou, X. E., Ke, J., de Waal, P. W., Gu, X., Tan, M. H., Wang, D., Wu, D., Xu, H. E., and Melcher, K. (2015) Structural basis of AMPK regulation by adenine nucleotides and glycogen. *Cell Res.* **25**, 50–66
367. Mu, J., Barton, E. R., and Birnbaum, M. J. (2003) Selective suppression of AMP-activated protein kinase in skeletal muscle: update on 'lazy mice'. *Biochem. Soc. Trans.* **31**, 236–241
368. Smith, A. C., Bruce, C. R., and Dyck, D. J. (2005) AMP kinase activation with AICAR simultaneously increases fatty acid and glucose oxidation in resting rat soleus muscle. *J. Physiol.* **565**, 537–546
369. Klein, D. K., Pilegaard, H., Treebak, J. T., Jensen, T. E., Viollet, B., Schjerling, P., and Wojtaszewski, J. F. (2007) Lack of AMPKalpha2 enhances pyruvate dehydrogenase activity during exercise. *Am. J. Physiol. Endocrinol. Metab.* **293**, E1242–E1249
370. Fritzen, A. M., Lundsgaard, A. M., Jeppesen, J., Christiansen, M. L., Biersø, R., Dyck, J. R., Pilegaard, H., and Kiens, B. (2015) 5'-AMP activated protein kinase α_2 controls substrate metabolism during post-exercise recovery via regulation of pyruvate dehydrogenase kinase 4. *J. Physiol.* **593**, 4765–4780
371. Shoubridge, E. A., Challiss, R. A., Hayes, D. J., and Radda, G. K. (1985) Biochemical adaptation in the skeletal muscle of rats depleted of creatine with the substrate analogue beta-guanidinopropionic acid. *Biochem. J.* **232**, 125–131
372. Ren, J. M., Semenkovich, C. F., and Holloszy, J. O. (1993) Adaptation of muscle to creatine depletion: effect on GLUT-4 glucose transporter expression. *Am. J. Physiol.* **264**, C146–C150
373. Leick, L., Fentz, J., Biersø, R. S., Knudsen, J. G., Jeppesen, J., Kiens, B., Wojtaszewski, J. F., and Pilegaard, H. (2010) PGC-1alpha is required for AICAR-induced expression of GLUT4 and mitochondrial proteins in mouse skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **299**, E456–E465
374. Holloszy, J. O. (1967) Biochemical adaptations in muscle: effects of exercise on mitochondrial oxygen uptake and respiratory enzyme activity in skeletal muscle. *J. Biol. Chem.* **242**, 2278–2282
375. Holloszy, J. O. (2011) Regulation of mitochondrial biogenesis and GLUT4 expression by exercise. *Compr. Physiol.* **1**, 921–940
376. Zwetsloot, K. A., Westerkamp, L. M., Holmes, B. F., and Gavin, T. P. (2008) AMPK regulates basal skeletal muscle capillarization and VEGF expression, but is not necessary for the angiogenic response to exercise. *J. Physiol.* **586**, 6021–6035
377. Morissette, M. P., Susser, S. E., Stammers, A. N., O'Hara, K. A., Gardiner, P. F., Sheppard, P., Moffatt, T. L., and Duhamel, T. A. (2014) Differential regulation of the fiber type-specific gene expression of the sarcoplasmic reticulum calcium-ATPase isoforms induced by exercise training. *J. Appl. Physiol.* **117**, 544–555
378. Wright, D. C., Geiger, P. C., Han, D.-H., Jones, T. E., and Holloszy, J. O. (2007) Calcium induces increases in peroxisome proliferator-activated receptor gamma coactivator-lalpha and mitochondrial biogenesis by a pathway leading to p38 mitogen-activated protein kinase activation. *J. Biol. Chem.* **282**, 18793–18799
379. Pogozelski, A. R., Geng, T., Li, P., Yin, X., Lira, V. A., Zhang, M., Chi, J.-T., and Yan, Z. (2009) p38gamma mitogen-activated protein kinase is a key regulator in skeletal muscle metabolic adaptation in mice. *PLoS One* **4**, e7934
380. Irrcher, I., Ljubicic, V., and Hood, D. A. (2009) Interactions between ROS and AMP kinase activity in the regulation of PGC-1 α transcription in skeletal muscle cells. *Am. J. Physiol. Cell Physiol.* **296**, C116–C123
381. Baar, K. (2014) Nutrition and the adaptation to endurance training. *Sports Med.* **44** (Suppl 1) S5–S12
382. Ojuka, E. O., Jones, T. E., Nolte, L. A., Chen, M., Wamhoff, B. R., Sturek, M., and Holloszy, J. O. (2002) Regulation of GLUT4 biogenesis in muscle: evidence for involvement of AMPK and Ca(2+). *Am. J. Physiol. Endocrinol. Metab.* **282**, E1008–E1013
383. Mora, S., and Pessin, J. E. (2000) The MEF2A isoform is required for striated muscle-specific expression of the insulin-responsive GLUT4 glucose transporter. *J. Biol. Chem.* **275**, 16323–16328
384. Lemerrier, C., Verdel, A., Galloo, B., Curtet, S., Brocard, M. P., and Khochbin, S. (2000) mHDA1/HDAC5 histone deacetylase interacts with and represses MEF2A transcriptional activity. *J. Biol. Chem.* **275**, 15594–15599
385. McKinsey, T. A., Zhang, C. L., Lu, J., and Olson, E. N. (2000) Signal-dependent nuclear export of a histone deacetylase regulates muscle differentiation. *Nature* **408**, 106–111
386. McGee, S. L., and Hargreaves, M. (2004) Exercise and myocyte enhancer factor 2 regulation in human skeletal muscle. *Diabetes* **53**, 1208–1214
387. McGee, S. L., Sparling, D., Olson, A. L., and Hargreaves, M. (2006) Exercise increases MEF2- and GEF DNA-binding activity in human skeletal muscle. *FASEB J.* **20**, 348–349
388. McGee, S. L., Swinton, C., Morrison, S., Gaur, V., Campbell, D. E., Jorgensen, S. B., Kemp, B. E., Baar, K., Steinberg, G. R., and Hargreaves, M. (2014) Compensatory regulation of HDAC5 in muscle maintains metabolic adaptive responses and metabolism in response to energetic stress. *FASEB J.* **28**, 3384–3395
389. Pearen, M. A., and Muscat, G. E. (2010) Mini-review: nuclear hormone receptor 4A signaling: implications for metabolic disease. *Mol. Endocrinol.* **24**, 1891–1903
390. Catoire, M., Mensink, M., Boekschoten, M. V., Hangelbroek, R., Müller, M., Schrauwen, P., and Kersten, S. (2012) Pronounced effects of acute endurance exercise on gene expression in resting and exercising human skeletal muscle. *PLoS One* **7**, e51066
391. Christensen, N. J., Galbo, H., Hansen, J. F., Hesse, B., Richter, E. A., and Trap-Jensen, J. (1979) Catecholamines and exercise. *Diabetes* **28**(Suppl 1), 58–62
392. Pearen, M. A., Myers, S. A., Raichur, S., Ryall, J. G., Lynch, G. S., and Muscat, G. E. (2008) The orphan nuclear receptor, NOR-1, a target of beta-adrenergic signaling, regulates gene expression that controls oxidative metabolism in skeletal muscle. *Endocrinology* **149**, 2853–2865
393. Pearen, M. A., Ryall, J. G., Maxwell, M. A., Ohkura, N., Lynch, G. S., and Muscat, G. E. (2006) The orphan nuclear receptor, NOR-1, is a target of beta-adrenergic signaling in skeletal muscle. *Endocrinology* **147**, 5217–5227
394. Myers, S. A., Eriksson, N., Burow, R., Wang, S.-C., and Muscat, G. E. (2009) Beta-adrenergic signaling regulates NR4A nuclear receptor and metabolic gene expression in multiple tissues. *Mol. Cell. Endocrinol.* **309**, 101–108
395. Maarbjerg, S. J., Jørgensen, S. B., Rose, A. J., Jeppesen, J., Jensen, T. E., Treebak, J. T., Birk, J. B., Schjerling, P., Wojtaszewski, J. F., and Richter, E. A. (2009) Genetic impairment of AMPKalpha2 signaling does not reduce muscle glucose uptake during treadmill exercise in mice. *Am. J. Physiol. Endocrinol. Metab.* **297**, E924–E934
396. Lee-Young, R. S., Griffie, S. R., Lynes, S. E., Bracy, D. P., Ayala, J. E., McGuinness, O. P., and Wasserman, D. H. (2009) Skeletal muscle AMP-activated protein kinase is essential for the metabolic response to exercise in vivo. *J. Biol. Chem.* **284**, 23925–23934
397. Abbott, M. J., and Turcotte, L. P. (2014) AMPK- α_2 is involved in exercise training-induced adaptations in insulin-stimulated metabolism in skeletal muscle following high-fat diet. *J. Appl. Physiol.* **117**, 869–879

398. Tanner, C. B., Madsen, S. R., Hallowell, D. M., Goring, D. M. J., Moore, T. M., Hardman, S. E., Heninger, M. R., Atwood, D. R., and Thomson, D. M. (2013) Mitochondrial and performance adaptations to exercise training in mice lacking skeletal muscle LKB1. *Am. J. Physiol. Endocrinol. Metab.* **305**, E1018–E1029
399. Piper, T., Thomas, A., Baume, N., Sobolevsky, T., Saugy, M., Rodchenkov, G., Schänzer, W., and Thevis, M. (2014) Determination of $^{13}\text{C}/^{12}\text{C}$ ratios of endogenous urinary 5-amino-imidazole-4-carboxamide β -D-ribofuranoside (AICAR). *Rapid Commun. Mass Spectrom.* **28**, 1194–1202
400. Narkar, V. A., Downes, M., Yu, R. T., Embler, E., Wang, Y. X., Banayo, E., Mihaylova, M. M., Nelson, M. C., Zou, Y., Jugulion, H., Kang, H., Shaw, R. J., and Evans, R. M. (2008) AMPK and PPAR δ agonists are exercise mimetics. *Cell* **134**, 405–415
401. Mangano, D. T., Miao, Y., Tudor, I. C., and Dietzel, C.; Investigators of the Multicenter Study of Perioperative Ischemia (McSPI) Research Group; Ischemia Research and Education Foundation (IREF). (2006) Post-reperfusion myocardial infarction: long-term survival improvement using adenosine regulation with acadesine. *J. Am. Coll. Cardiol.* **48**, 206–214
402. Cuthbertson, D. J., Babraj, J. A., Mustard, K. J. W., Towler, M. C., Green, K. A., Wackerhage, H., Leese, G. P., Baar, K., Thomason-Hughes, M., Sutherland, C., Hardie, D. G., and Rennie, M. J. (2007) 5-aminoimidazole-4-carboxamide 1- β -D-ribofuranoside acutely stimulates skeletal muscle 2-deoxyglucose uptake in healthy men. *Diabetes* **56**, 2078–2084
403. Babraj, J. A., Mustard, K., Sutherland, C., Towler, M. C., Chen, S., Smith, K., Green, K., Leese, G., Hardie, D. G., Rennie, M. J., and Cuthbertson, D. J. (2009) Blunting of AICAR-induced human skeletal muscle glucose uptake in type 2 diabetes is dependent on age rather than diabetic status. *Am. J. Physiol. Endocrinol. Metab.* **296**, E1042–E1048
404. Crawford, S. A., Costford, S. R., Aguer, C., Thomas, S. C., deKemp, R. A., DaSilva, J. N., Lafontaine, D., Kendall, M., Dent, R., Beanlands, R. S., McPherson, R., and Harper, M. E. (2010) Naturally occurring R225W mutation of the gene encoding AMP-activated protein kinase (AMPK) γ 3 results in increased oxidative capacity and glucose uptake in human primary myotubes. *Diabetologia* **53**, 1986–1997
405. Granlund, A., Kotova, O., Benziene, B., Galuska, D., Jensen-Waern, M., Chibalin, A. V., and Essén-Gustavsson, B. (2010) Effects of exercise on muscle glycogen synthesis signalling and enzyme activities in pigs carrying the PRKAG3 mutation. *Exp. Physiol.* **95**, 541–549
406. Essén-Gustavsson, B., Granlund, A., Benziene, B., Jensen-Waern, M., and Chibalin, A. V. (2011) Muscle glycogen resynthesis, signalling and metabolic responses following acute exercise in exercise-trained pigs carrying the PRKAG3 mutation. *Exp. Physiol.* **96**, 927–937
407. Pederson, B. A., Cope, C. R., Schroeder, J. M., Smith, M. W., Irimia, J. M., Thurberg, B. L., DePaoli-Roach, A. A., and Roach, P. J. (2005) Exercise capacity of mice genetically lacking muscle glycogen synthase: in mice, muscle glycogen is not essential for exercise. *J. Biol. Chem.* **280**, 17260–17265
408. Xirouchaki, C. E., Mangiafico, S. P., Bate, K., Ruan, Z., Huang, A. M., Tedjoswito, B. W., Lamont, B., Pong, W., Favaloro, J., Blair, A. R., Zajac, J. D., Proietto, J., and Andrikopoulos, S. (2016) Impaired glucose metabolism and exercise capacity with muscle-specific glycogen synthase 1 (gyl1) deletion in adult mice. *Mol. Metab.* **5**, 221–232
409. Yu, H., Hirshman, M. F., Fujii, N., Pomerleau, J. M., Peter, L. E., and Goodyear, L. J. (2006) Muscle-specific overexpression of wild type and R225Q mutant AMP-activated protein kinase γ 3-subunit differentially regulates glycogen accumulation. *Am. J. Physiol. Endocrinol. Metab.* **291**, E557–E565
410. Barnes, B. R., Glund, S., Long, Y. C., Hjälml, G., Andersson, L., and Zierath, J. R. (2005) 5'-AMP-activated protein kinase regulates skeletal muscle glycogen content and ergogenics. *FASEB J.* **19**, 773–779
411. Barnes, B. R., Long, Y. C., Steiler, T. L., Leng, Y., Galuska, D., Wojtaszewski, J. F., Andersson, L., and Zierath, J. R. (2005) Changes in exercise-induced gene expression in 5'-AMP-activated protein kinase γ 3-null and γ 3 R225Q transgenic mice. *Diabetes* **54**, 3484–3489
412. Granlund, A., Jensen-Waern, M., and Essén-Gustavsson, B. (2011) The influence of the PRKAG3 mutation on glycogen, enzyme activities and fibre types in different skeletal muscles of exercise trained pigs. *Acta Vet. Scand.* **53**, 20
413. Marcinko, K., Bujak, A. L., Lally, J. S. V., Ford, R. J., Wong, T. H., Smith, B. K., Kemp, B. E., Jenkins, Y., Li, W., Kinsella, T. M., Hitoshi, Y., and Steinberg, G. R. (2015) The AMPK activator R419 improves exercise capacity and skeletal muscle insulin sensitivity in obese mice. *Mol. Metab.* **4**, 643–651
414. Um, J.-H., Park, S.-J., Kang, H., Yang, S., Foretz, M., McBurney, M. W., Kim, M. K., Viollet, B., and Chung, J. H. (2010) AMP-activated protein kinase-deficient mice are resistant to the metabolic effects of resveratrol. *Diabetes* **59**, 554–563
415. Malin, S. K., and Braun, B. (2013) Effect of metformin on substrate utilization after exercise training in adults with impaired glucose tolerance. *Appl. Physiol. Nutr. Metab.* **38**, 427–430
416. Cadeddu, C., Nocco, S., Lucia, C., Deidda, M., Bina, A., Fabio, O., Bandinu, S., Cossu, E., Baroni, M. G., and Mercuro, G. (2014) Effects of metformin and exercise training, alone or in association, on cardio-pulmonary performance and quality of life in insulin resistance patients. *Cardiovasc. Diabetol.* **13**, 93
417. Gliemann, L., Schmidt, J. F., Olesen, J., Biensø, R. S., Peronard, S. L., Grandjean, S. U., Mortensen, S. P., Nyberg, M., Bangsbo, J., Pilegaard, H., and Hellsten, Y. (2013) Resveratrol blunts the positive effects of exercise training on cardiovascular health in aged men. *J. Physiol.* **591**, 5047–5059
418. Hart, N., Sarga, L., Csenge, Z., Koltai, E., Koch, L. G., Britton, S. L., Davies, K. J., Kouretas, D., Wessner, B., and Radak, Z. (2013) Resveratrol enhances exercise training responses in rats selectively bred for high running performance. *Food Chem. Toxicol.* **61**, 53–59
419. Dolinsky, V. W., Jones, K. E., Sidhu, R. S., Haykowsky, M., Czubyrt, M. P., Gordon, T., and Dyck, J. R. (2012) Improvements in skeletal muscle strength and cardiac function induced by resveratrol during exercise training contribute to enhanced exercise performance in rats. *J. Physiol.* **590**, 2783–2799
420. Coffey, V. G., Jemiolo, B., Edge, J., Garnham, A. P., Trappe, S. W., and Hawley, J. A. (2009) Effect of consecutive repeated sprint and resistance exercise bouts on acute adaptive responses in human skeletal muscle. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **297**, R1441–R1451
421. Coffey, V. G., Pilegaard, H., Garnham, A. P., O'Brien, B. J., and Hawley, J. A. (2009) Consecutive bouts of diverse contractile activity alter acute responses in human skeletal muscle. *J. Appl. Physiol.* **106**, 1187–1197
422. Lundberg, T. R., Fernandez-Gonzalo, R., Gustafsson, T., and Tesch, P. A. (2012) Aerobic exercise alters skeletal muscle molecular responses to resistance exercise. *Med. Sci. Sports Exerc.* **44**, 1680–1688
423. Apró, W., Wang, L., Pontén, M., Blomstrand, E., and Sahlin, K. (2013) Resistance exercise induced mTORC1 signaling is not impaired by subsequent endurance exercise in human skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **305**, E22–E32
424. Thomson, D. M., Fick, C. A., and Gordon, S. E. (2008) AMPK activation attenuates S6K1, 4E-BP1, and eEF2 signaling responses to high-frequency electrically stimulated skeletal muscle contractions. *J. Appl. Physiol.* **104**, 625–632
425. McGee, S. L., Mustard, K. J., Hardie, D. G., and Baar, K. (2008) Normal hypertrophy accompanied by phosphorylation and activation of AMP-activated protein kinase α 1 following overload in LKB1 knockout mice. *J. Physiol.* **586**, 1731–1741
426. Mounier, R., Lantier, L., Leclerc, J., Sotiropoulos, A., Pende, M., Daegelen, D., Sakamoto, K., Foretz, M., and Viollet, B. (2009) Important role for AMPK α 1 in limiting skeletal muscle cell hypertrophy. *FASEB J.* **23**, 2264–2273
427. Riedl, I., Osler, M. E., Björnholm, M., Egan, B., Nader, G. A., Chibalin, A. V., and Zierath, J. R. (2016) AMPK γ 3 is dispensable for skeletal muscle hypertrophy induced by functional overload. *Am. J. Physiol. Endocrinol. Metab.* **310**, E461–E472
428. Hoffman, N. J., Parker, B. L., Chaudhuri, R., Fisher-Wellman, K. H., Kleinert, M., Humphrey, S. J., Yang, P., Holliday, M., Trefely, S., Frazierley, D. J., Stöckli, J., Burchfield, J. G., Jensen, T. E., Jothi, R., Kiens, B., Wojtaszewski, J. F. P., Richter, E. A., and James, D. E. (2015) Global phosphoproteomic analysis of human skeletal muscle reveals a network of exercise-regulated kinases and AMPK substrates [Erratum]. *Cell Metab.* **22**, 922–935
429. Gordon, S. E., Lake, J. A., Westerkamp, C. M., and Thomson, D. M. (2008) Does AMP-activated protein kinase negatively mediate aged fast-twitch skeletal muscle mass? *Exerc. Sport Sci. Rev.* **36**, 179–186
430. Mounier, R., Lantier, L., Leclerc, J., Sotiropoulos, A., Foretz, M., and Viollet, B. (2011) Antagonistic control of muscle cell size by AMPK and mTORC1. *Cell Cycle* **10**, 2640–2646

431. Thomas, M. M., Wang, D. C., D'Souza, D. M., Krause, M. P., Layne, A. S., Criswell, D. S., O'Neill, H. M., Connor, M. K., Anderson, J. E., Kemp, B. E., Steinberg, G. R., and Hawke, T. J. (2014) Muscle-specific AMPK $\beta 1\beta 2$ -null mice display a myopathy due to loss of capillary density in nonpostural muscles. *FASEB J.* **28**, 2098–2107
432. Lantier, L., Mounier, R., Leclerc, J., Pende, M., Foretz, M., and Viollet, B. (2010) Coordinated maintenance of muscle cell size control by AMP-activated protein kinase. *FASEB J.* **24**, 3555–3561
433. Egawa, T., Ohno, Y., Goto, A., Ikuta, A., Suzuki, M., Ohira, T., Yokoyama, S., Sugiura, T., Ohira, Y., Yoshioka, T., and Goto, K. (2014) AICAR-induced activation of AMPK negatively regulates myotube hypertrophy through the HSP72-mediated pathway in C2C12 skeletal muscle cells. *Am. J. Physiol. Endocrinol. Metab.* **306**, E344–E354
434. Hamilton, D. L., Philp, A., MacKenzie, M. G., Patton, A., Towler, M. C., Gallagher, I. J., Bodine, S. C., and Baar, K. (2014) Molecular brakes regulating mTORC1 activation in skeletal muscle following synergist ablation. *Am. J. Physiol. Endocrinol. Metab.* **307**, E365–E373
435. Lebre, B., Le Roy, P., Monin, G., Lefaucheur, L., Caritez, J. C., Talmant, A., Elsen, J. M., and Sellier, P. (1999) Influence of the three RN genotypes on chemical composition, enzyme activities, and myofiber characteristics of porcine skeletal muscle. *J. Anim. Sci.* **77**, 1482–1489
436. Thomson, D. M., Hancock, C. R., Evanson, B. G., Kenney, S. G., Malan, B. B., Mongillo, A. D., Brown, J. D., Hepworth, S., Fillmore, N., Parcell, A. C., Kooyman, D. L., and Winder, W. W. (2010) Skeletal muscle dysfunction in muscle-specific LKB1 knockout mice. *J. Appl. Physiol.* **108**, 1775–1785
437. Egawa, T., Goto, A., Ohno, Y., Yokoyama, S., Ikuta, A., Suzuki, M., Sugiura, T., Ohira, Y., Yoshioka, T., Hayashi, T., and Goto, K. (2015) Involvement of AMPK in regulating slow-twitch muscle atrophy during hindlimb unloading in mice. *Am. J. Physiol. Endocrinol. Metab.* **309**, E651–E662
438. Saclier, M., Yacoub-Youssef, H., Mackey, A. L., Arnold, L., Ardjoune, H., Magnan, M., Sailhan, F., Chelly, J., Pavlath, G. K., Mounier, R., Kjaer, M., and Chazaud, B. (2013) Differentially activated macrophages orchestrate myogenic precursor cell fate during human skeletal muscle regeneration. *Stem Cells* **31**, 384–396
439. Arnold, L., Henry, A., Poron, F., Baba-Amer, Y., van Rooijen, N., Plonquet, A., Gherardi, R. K., and Chazaud, B. (2007) Inflammatory monocytes recruited after skeletal muscle injury switch into antiinflammatory macrophages to support myogenesis. *J. Exp. Med.* **204**, 1057–1069
440. Varga, T., Mounier, R., Horvath, A., Cuvelier, S., Dumont, F., Poliska, S., Ardjoune, H., Juban, G., Nagy, L., and Chazaud, B. (2016) Highly dynamic transcriptional signature of distinct macrophage subsets during sterile inflammation, resolution, and tissue repair. *J. Immunol.* **196**, 4771–4782
441. Ardite, E., Perdiguer, E., Vidal, B., Gutarra, S., Serrano, A. L., and Muñoz-Cánoves, P. (2012) PAI-1-regulated miR-21 defines a novel age-associated fibrogenic pathway in muscular dystrophy. *J. Cell Biol.* **196**, 163–175
442. Perdiguer, E., Kharraz, Y., Serrano, A. L., and Muñoz-Cánoves, P. (2012) MKP-1 coordinates ordered macrophage-phenotype transitions essential for stem cell-dependent tissue repair. *Cell Cycle* **11**, 877–886
443. Perdiguer, E., Sousa-Victor, P., Ruiz-Bonilla, V., Jardí, M., Caelles, C., Serrano, A. L., and Muñoz-Cánoves, P. (2011) p38/MKP-1-regulated AKT coordinates macrophage transitions and resolution of inflammation during tissue repair. *J. Cell Biol.* **195**, 307–322
444. Juban, G., and Chazaud, B. (2017) Metabolic regulation of macrophages during tissue repair: insights from skeletal muscle regeneration. *FEBS Lett.* **591**, 3007–3021
445. Zhao, X., Zmijewski, J. W., Lorne, E., Liu, G., Park, Y.-J., Tsuruta, Y., and Abraham, E. (2008) Activation of AMPK attenuates neutrophil proinflammatory activity and decreases the severity of acute lung injury. *Am. J. Physiol. Lung Cell. Mol. Physiol.* **295**, L497–L504
446. Sag, D., Carling, D., Stout, R. D., and Suttles, J. (2008) Adenosine 5'-monophosphate-activated protein kinase promotes macrophage polarization to an anti-inflammatory functional phenotype. *J. Immunol.* **181**, 8633–8641
447. Jeong, H. W., Hsu, K. C., Lee, J.-W., Ham, M., Huh, J. Y., Shin, H. J., Kim, W. S., and Kim, J. B. (2009) Berberine suppresses proinflammatory responses through AMPK activation in macrophages. *Am. J. Physiol. Endocrinol. Metab.* **296**, E955–E964
448. Mounier, R., Thérêt, M., Arnold, L., Cuvelier, S., Bultot, L., Göransson, O., Sanz, N., Ferry, A., Sakamoto, K., Foretz, M., Violette, B., and Chazaud, B. (2013) AMPK $\alpha 1$ regulates macrophage skewing at the time of resolution of inflammation during skeletal muscle regeneration. *Cell Metab.* **18**, 251–264
449. Varga, T., Mounier, R., Gogolak, P., Poliska, S., Chazaud, B., and Nagy, L. (2013) Tissue LyC6- macrophages are generated in the absence of circulating LyC6- monocytes and Nur77 in a model of muscle regeneration. *J. Immunol.* **191**, 5695–5701
450. Shan, T., Zhang, P., Liang, X., Bi, P., Yue, F., and Kuang, S. (2014) Lkb1 is indispensable for skeletal muscle development, regeneration, and satellite cell homeostasis. *Stem Cells* **32**, 2893–2907
451. Gopinath, S. D., Webb, A. E., Brunet, A., and Rando, T. A. (2014) FOXO3 promotes quiescence in adult muscle stem cells during the process of self-renewal. *Stem Cell Reports* **2**, 414–426
452. Simsek, T., Kocbas, F., Zheng, J., Debernardinis, R. J., Mahmoud, A. I., Olson, E. N., Schneider, J. W., Zhang, C. C., and Sadek, H. A. (2010) The distinct metabolic profile of hematopoietic stem cells reflects their location in a hypoxic niche. *Cell Stem Cell* **7**, 380–390
453. Rafalski, V. A., Mancini, E., and Brunet, A. (2012) Energy metabolism and energy-sensing pathways in mammalian embryonic and adult stem cell fate. *J. Cell Sci.* **125**, 5597–5608
454. Ryall, J. G., Dell'Orso, S., Derfoul, A., Juan, A., Zare, H., Feng, X., Clermont, D., Koulis, M., Gutierrez-Cruz, G., Fulco, M., and Sartorelli, V. (2015) The NAD(+)-dependent SIRT1 deacetylase translates a metabolic switch into regulatory epigenetics in skeletal muscle stem cells. *Cell Stem Cell* **16**, 171–183
455. Theret, M., Gsaier, L., Schaffer, B., Juban, G., Ben Larbi, S., Weiss-Gayet, M., Bultot, L., Collodet, C., Foretz, M., Desplanches, D., Sanz, P., Zang, Z., Yang, L., Vial, G., Viollet, B., Sakamoto, K., Brunet, A., Chazaud, B., and Mounier, R. (2017) AMPK $\alpha 1$ -LDH pathway regulates muscle stem cell self-renewal by controlling metabolic homeostasis. *EMBO J.* **36**, 1946–1962
456. Jahnke, V. E., Van Der Meulen, J. H., Johnston, H. K., Ghimbovski, S., Partridge, T., Hoffman, E. P., and Nagaraju, K. (2012) Metabolic remodeling agents show beneficial effects in the dystrophin-deficient mdx mouse model. *Skelet. Muscle* **2**, 16
457. Webster, C., Silberstein, L., Hays, A. P., and Blau, H. M. (1988) Fast muscle fibers are preferentially affected in Duchenne muscular dystrophy. *Cell* **52**, 503–513
458. Ervasti, J. M., and Campbell, K. P. (1993) Dystrophin and the membrane skeleton. *Curr. Opin. Cell Biol.* **5**, 82–87
459. Davies, K. E., and Nowak, K. J. (2006) Molecular mechanisms of muscular dystrophies: old and new players. *Nat. Rev. Mol. Cell Biol.* **7**, 762–773
460. Deconinck, A. E., Rafael, J. A., Skinner, J. A., Brown, S. C., Potter, A. C., Metzinger, L., Watt, D. J., Dickson, J. G., Tinsley, J. M., and Davies, K. E. (1997) Utrophin-dystrophin-deficient mice as a model for Duchenne muscular dystrophy. *Cell* **90**, 717–727
461. Rafael, J. A., Tinsley, J. M., Potter, A. C., Deconinck, A. E., and Davies, K. E. (1998) Skeletal muscle-specific expression of a utrophin transgene rescues utrophin-dystrophin deficient mice. *Nat. Genet.* **19**, 79–82
462. Chakkalakal, J. V., Stocksley, M. A., Harrison, M.-A., Angus, L. M., Deschenes-Furry, J., St-Pierre, S., Megeney, L. A., Chin, E. R., Michel, R. N., and Jasmin, B. J. (2003) Expression of utrophin A mRNA correlates with the oxidative capacity of skeletal muscle fiber types and is regulated by calcineurin/NFAT signaling. *Proc. Natl. Acad. Sci. USA* **100**, 7791–7796
463. Ljubicic, V., Burt, M., Lunde, J. A., and Jasmin, B. J. (2014) Resveratrol induces expression of the slow, oxidative phenotype in mdx mouse muscle together with enhanced activity of the SIRT1-PGC-1 α axis. *Am. J. Physiol. Cell Physiol.* **307**, C66–C82
464. Ljubicic, V., and Jasmin, B. J. (2015) Metformin increases peroxisome proliferator-activated receptor γ Co-activator-1 α and utrophin A expression in dystrophic skeletal muscle. *Muscle Nerve* **52**, 139–142
465. Al-Rewashdy, H., Ljubicic, V., Lin, W., Renaud, J.-M., and Jasmin, B. J. (2015) Utrophin A is essential in mediating the functional adaptations of mdx mouse muscle following chronic AMPK activation. *Hum. Mol. Genet.* **24**, 1243–1255
466. Baltgalvis, K. A., Call, J. A., Cochrane, G. D., Laker, R. C., Yan, Z., and Lowe, D. A. (2012) Exercise training improves plantar flexor muscle function in mdx mice. *Med. Sci. Sports Exerc.* **44**, 1671–1679
467. Markert, C. D., Case, L. E., Carter, G. T., Furlong, P. A., and Grange, R. W. (2012) Exercise and Duchenne muscular dystrophy: where we have been and where we need to go. *Muscle Nerve* **45**, 746–751
468. Alemdaroglu, I., Karaduman, A., Yilmaz, Ö. T., and Topaloglu, H. (2015) Different types of upper extremity exercise training in

- Duchenne muscular dystrophy: effects on functional performance, strength, endurance, and ambulation. *Muscle Nerve* **51**, 697–705
469. Hu, F. B., Manson, J. E., Stampfer, M. J., Colditz, G., Liu, S., Solomon, C. G., and Willett, W. C. (2001) Diet, lifestyle, and the risk of type 2 diabetes mellitus in women. *N. Engl. J. Med.* **345**, 790–797
 470. Wing, R. R., Goldstein, M. G., Acton, K. J., Birch, L. L., Jakicic, J. M., Sallis, J. F., Jr., Smith-West, D., Jeffery, R. W., and Surwit, R. S. (2001) Behavioral science research in diabetes: lifestyle changes related to obesity, eating behavior, and physical activity. *Diabetes Care* **24**, 117–123
 471. Bultot, L., Jensen, T. E., Lai, Y.-C., Madsen, A. L., Collodet, C., Kviklyte, S., Deak, M., Yavari, A., Foretz, M., Ghaffari, S., Bellahcene, M., Ashrafian, H., Rider, M. H., Richter, E. A., and Sakamoto, K. (2016) Benzimidazole derivative small-molecule 991 enhances AMPK activity and glucose uptake induced by AICAR or contraction in skeletal muscle. *Am. J. Physiol. Endocrinol. Metab.* **311**, E706–E719
 472. Bernardo, B. C., Weeks, K. L., Pretorius, L., and McMullen, J. R. (2010) Molecular distinction between physiological and pathological cardiac hypertrophy: experimental findings and therapeutic strategies. *Pharmacol. Ther.* **128**, 191–227
 473. Bandyopadhyay, G. K., Yu, J. G., Ofrecio, J., and Olefsky, J. M. (2006) Increased malonyl-CoA levels in muscle from obese and type 2 diabetic subjects lead to decreased fatty acid oxidation and increased lipogenesis; thiazolidinedione treatment reverses these defects. *Diabetes* **55**, 2277–2285
 474. Højlund, K., Mustard, K. J., Staehr, P., Hardie, D. G., Beck-Nielsen, H., Richter, E. A., and Wojtaszewski, J. F. (2004) AMPK activity and isoform protein expression are similar in muscle of obese subjects with and without type 2 diabetes. *Am. J. Physiol. Endocrinol. Metab.* **286**, E239–E244
 475. Ruderman, N. B., Carling, D., Prentki, M., and Cacicado, J. M. (2013) AMPK, insulin resistance, and the metabolic syndrome. *J. Clin. Invest.* **123**, 2764–2772
 476. Pold, R., Jensen, L. S., Jessen, N., Buhl, E. S., Schmitz, O., Flyvbjerg, A., Fujii, N., Goodyear, L. J., Gottfredsen, C. F., Brand, C. L., and Lund, S. (2005) Long-term AICAR administration and exercise prevents diabetes in ZDF rats. *Diabetes* **54**, 928–934
 477. Buhl, E. S., Jessen, N., Pold, R., Ledet, T., Flyvbjerg, A., Pedersen, S. B., Pedersen, O., Schmitz, O., and Lund, S. (2002) Long-term AICAR administration reduces metabolic disturbances and lowers blood pressure in rats displaying features of the insulin resistance syndrome. *Diabetes* **51**, 2199–2206
 478. Iglesias, M. A., Ye, J. M., Frangioudakis, G., Saha, A. K., Tomas, E., Ruderman, N. B., Cooney, G. J., and Kraegen, E. W. (2002) AICAR administration causes an apparent enhancement of muscle and liver insulin action in insulin-resistant high-fat-fed rats. *Diabetes* **51**, 2886–2894
 479. Kelly, K. R., Abbott, M. J., and Turcotte, L. P. (2010) Short-term AMP-regulated protein kinase activation enhances insulin-sensitive fatty acid uptake and increases the effects of insulin on fatty acid oxidation in L6 muscle cells. *Exp. Biol. Med. (Maywood)* **235**, 514–521
 480. Bosselaar, M., Smits, P., van Loon, L. J., and Tack, C. J. (2011) Intravenous AICAR during hyperinsulinemia induces systemic hemodynamic changes but has no local metabolic effect. *J. Clin. Pharmacol.* **51**, 1449–1458
 481. Boon, H., Bosselaar, M., Praet, S. F., Blaak, E. E., Saris, W. H., Wagenmakers, A. J., McGee, S. L., Tack, C. J., Smits, P., Hargreaves, M., and van Loon, L. J. (2008) Intravenous AICAR administration reduces hepatic glucose output and inhibits whole body lipolysis in type 2 diabetic patients. *Diabetologia* **51**, 1893–1900
 482. Hasenour, C. M., Ridley, D. E., Hughey, C. C., James, F. D., Donahue, E. P., Shearer, J., Viollet, B., Foretz, M., and Wasserman, D. H. (2014) 5-Aminoimidazole-4-carboxamide-1- β -D-ribofuranoside (AICAR) effect on glucose production, but not energy metabolism, is independent of hepatic AMPK in vivo. *J. Biol. Chem.* **289**, 5950–5959
 483. Bergeron, R., Previs, S. F., Cline, G. W., Perret, P., Russell III, R. R., Young, L. H., and Shulman, G. I. (2001) Effect of 5-aminoimidazole-4-carboxamide-1-beta-D-ribofuranoside infusion on in vivo glucose and lipid metabolism in lean and obese Zucker rats. *Diabetes* **50**, 1076–1082
 484. Steinberg, G. R., Michell, B. J., van Denderen, B. J., Watt, M. J., Carey, A. L., Fam, B. C., Andrikopoulos, S., Proietto, J., Görgün, C. Z., Carling, D., Hotamisligil, G. S., Febbraio, M. A., Kay, T. W., and Kemp, B. E. (2006) Tumor necrosis factor alpha-induced skeletal muscle insulin resistance involves suppression of AMP-kinase signaling. *Cell Metab.* **4**, 465–474
 485. Gregor, M. F., and Hotamisligil, G. S. (2011) Inflammatory mechanisms in obesity. *Annu. Rev. Immunol.* **29**, 415–445
 486. Pilon, G., Dallaire, P., and Marette, A. (2004) Inhibition of inducible nitric-oxide synthase by activators of AMP-activated protein kinase: a new mechanism of action of insulin-sensitizing drugs. *J. Biol. Chem.* **279**, 20767–20774
 487. Kjøbsted, R., Wojtaszewski, J. F., and Treebak, J. T. (2016) Role of AMP-activated protein kinase for regulating post-exercise insulin sensitivity. *EXS* **107**, 81–126
 488. Miura, S., Kai, Y., Kamei, Y., Bruce, C. R., Kubota, N., Febbraio, M. A., Kadowaki, T., and Ezaki, O. (2009) Alpha2-AMPK activity is not essential for an increase in fatty acid oxidation during low-intensity exercise. *Am. J. Physiol. Endocrinol. Metab.* **296**, E47–E55
 489. Miura, S., Kai, Y., Tadaishi, M., Tokutake, Y., Sakamoto, K., Bruce, C. R., Febbraio, M. A., Kita, K., Chohnan, S., and Ezaki, O. (2013) Marked phenotypic differences of endurance performance and exercise-induced oxygen consumption between AMPK and LKB1 deficiency in mouse skeletal muscle: changes occurring in the diaphragm. *Am. J. Physiol. Endocrinol. Metab.* **305**, E213–E229
 490. Viollet, B., Athea, Y., Mounier, R., Guigas, B., Zarrinpashneh, E., Horman, S., Lantier, L., Hebrard, S., Devin-Leclerc, J., Beauloye, C., Foretz, M., Andreelli, F., Ventura-Clapier, R., and Bertrand, L. (2009) AMPK: lessons from transgenic and knockout animals. *Front. Biosci. (Landmark Ed.)* **14**, 19–44
 491. Ost, M., Werner, F., Dokas, J., Klaus, S., and Voigt, A. (2014) Activation of AMPK α 2 is not crucial for mitochondrial uncoupling-induced metabolic effects but required to maintain skeletal muscle integrity. *PLoS One* **9**, e94689
 492. Lee-Young, R. S., Ayala, J. E., Fueger, P. T., Mayes, W. H., Kang, L., and Wasserman, D. H. (2011) Obesity impairs skeletal muscle AMPK signaling during exercise: role of AMPK α 2 in the regulation of exercise capacity in vivo. *Int. J. Obes.* **35**, 982–989
 493. Fujii, N., Seifert, M. M., Kane, E. M., Peter, L. E., Ho, R. C., Winstead, S., Hirshman, M. F., and Goodyear, L. J. (2007) Role of AMP-activated protein kinase in exercise capacity, whole body glucose homeostasis, and glucose transport in skeletal muscle: insight from analysis of a transgenic mouse model. *Diabetes Res. Clin. Pract.* **77**(Suppl 1), S92–S98

Received for publication May 16, 2017.
Accepted for publication November 13, 2017.